



|    |          |                               |
|----|----------|-------------------------------|
| 1. | 02.09.23 | REVISED PER TENANT COMMENTS   |
| 2. | 02.21.23 | ISSUED FOR PRELIMINARY REVIEW |
| 3. | 03.09.23 | REVISED PER ILC COMMENTS      |
| 4. | 03.15.23 | REVISED PER TENANT COMMENTS   |
|    |          |                               |
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|    |          |                               |

## DOORS

**TAS SECTION 404.2.4.1 - MANEUVERING CLEARANCES AT MANUAL SWINGING DOORS AND GATES**  
SWINGING DOORS AND GATES SHALL HAVE MANEUVERING CLEARANCES COMPLYING WITH TABLE 404.2.4.1.

**TAS SECTION 404.2.3 - CLEAR WIDTH**  
DOOR OPENINGS SHALL PROVIDE A CLEAR WIDTH OF 32" MINIMUM. CLEAR OPENINGS OF DOORWAYS WITH SWINGING DOORS SHALL BE MEASURED BETWEEN THE FACE OF THE DOOR AND THE STOP, WITH THE DOOR OPEN 90 DEGREES. OPENINGS MORE THAN 24" IN DEPTH SHALL PROVIDE A CLEAR OPENING OF 36" MINIMUM. THERE SHALL BE NO PROJECTIONS INTO THE REQUIRED CLEAR OPENING WIDTH LOWER THAN 34" ABOVE THE FINISH FLOOR OR GROUND. PROJECTIONS INTO THE CLEAR OPENING WIDTH BETWEEN 34" AND 80" ABOVE THE FINISH FLOOR OR GROUND SHALL NOT EXCEED 4 INCHES.

**TAS SECTION 404.2.5 - THRESHOLDS**  
THRESHOLDS, IF PROVIDED AT DOORWAYS, SHALL BE 1/2" HIGH MAXIMUM. RAISED THRESHOLDS AND CHANGES IN LEVEL AT DOORWAYS SHALL COMPLY WITH 302 AND 303.

**TAS SECTION 404.2.7 - DOOR AND GATE HARDWARE**  
HANDLES, PULLS, LATCHES, LOCKS, AND OTHER OPERABLE PARTS ON DOORS AND GATES SHALL COMPLY WITH 309.4. OPERABLE PARTS OF SUCH HARDWARE SHALL BE 34" MINIMUM AND 48" MAXIMUM ABOVE THE FINISH FLOOR OR GROUND. WHERE SLIDING DOORS ARE IN THE FULLY OPEN POSITIONS, OPERATING HARDWARE SHALL BE EXPOSED AND USABLE FROM BOTH SIDES.

**TAS SECTION 404.2.8.1 - DOOR CLOSERS AND GATE CLOSERS**  
DOOR CLOSERS AND GATE CLOSERS SHALL BE ADJUSTED SO THAT FROM AN OPEN POSITION OF 90 DEGREES, THE TIME REQUIRED TO MOVE THE DOOR TO A POSITION OF 12 DEGREES FROM THE LATCH IS 5 SECONDS MINIMUM.

**TAS SECTION 404.2.9 - DOOR AND GATE OPENING FORCE**  
THE MAXIMUM FORCE PERTAINS TO THE CONTINUOUS APPLICATION OF FORCE NECESSARY TO FULLY OPEN A DOOR, NOT THE INITIAL FORCE NEEDED TO OVERCOME THE INERTIA OF THE DOOR. IT DOES NOT APPLY TO THE FORCE REQUIRED TO RETRACT BOLTS OR TO DISENGAGE OTHER DEVICES USED TO KEEP THE DOOR IN A CLOSED POSITION.

## TOILET AND SHOWER STALLS

**TAS SECTION 604.8.1.1 - TOILET STALLS**  
WHEELCHAIR ACCESSIBLE COMPARTMENTS SHALL BE 60 INCHES WIDE MINIMUM MEASURED PERPENDICULAR TO THE SIDE WALL, AND 56 INCHES DEEP MINIMUM FOR WALL HUNG WATER CLOSETS AND 59 MINIMUM FOR FLOOR MOUNTED WATER CLOSETS MEASURED PERPENDICULAR TO THE REAR WALL. WHEELCHAIR ACCESSIBLE COMPARTMENTS FOR CHILDREN'S USE SHALL BE 60 INCHES WIDE MINIMUM MEASURED PERPENDICULAR TO THE SIDE WALL, AND 59 INCHES DEEP MINIMUM FOR WALL HUNG AND FLOOR MOUNTED WATER CLOSETS MEASURED PERPENDICULAR TO THE REAR WALL.

**TAS SECTION 605.2 - URINALS HEIGHT AND DEPTH**  
URINALS SHALL BE THE STALL-TYPE OR THE WALL-HUNG TYPE WITH THE RIM 17 INCHES MAXIMUM ABOVE THE FINISH FLOOR OR GROUND. URINALS SHALL BE 13 1/2 INCHES DEEP MINIMUM MEASURED FROM THE OUTER FACE OF THE URINAL RIM TO THE BACK OF THE FIXTURE. FLUSH CONTROLS SHALL BE HAND OPERATED OR AUTOMATIC. HAND OPERATED FLUSH CONTROLS SHALL COMPLY WITH 309.

**TAS SECTION 606.2 & 606.3 - LAVATORIES AND SINKS**  
LAVATORIES AND SINKS SHALL COMPLY WITH 606. A CLEAR FLOOR SPACE COMPLYING WITH 305, POSITIONED FOR A FORWARD APPROACH, AND KNEE AND TOE CLEARANCE COMPLYING WITH 306 SHALL BE PROVIDED. WATER SUPPLY AND DRAIN PIPES UNDER LAVATORIES AND SINKS SHALL BE INSULATED OR OTHERWISE CONFIGURED TO PROTECT AGAINST CONTACT. THERE SHALL BE NO SHARP OR ABRASIVE SURFACES UNDER LAVATORIES AND SINKS.

**TAS SECTION 608.5.1 - TRANSFER TYPE SHOWER COMPARTMENTS**  
IN TRANSFER TYPE SHOWER COMPARTMENTS, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE INSTALLED ON THE SIDE WALL OPPOSITE THE SEAT 38 INCHES MINIMUM AND 48 INCHES MAXIMUM ABOVE THE SHOWER FLOOR AND SHALL BE LOCATED ON THE CONTROL WALL 15 INCHES MAXIMUM FROM THE CENTERLINE OF THE SEAT TOWARD THE SHOWER OPENING.

**TAS SECTION 608.5.2 - STANDARD ROLL-IN TYPE SHOWER COMPARTMENTS**  
IN STANDARD ROLL-IN TYPE SHOWER COMPARTMENTS, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE LOCATED ABOVE THE GRAB BAR, BUT NO HIGHER THAN 48 INCHES ABOVE THE SHOWER FLOOR. WHERE A SEAT IS PROVIDED, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE INSTALLED ON THE BACK WALL ADJACENT TO THE SEAT WALL AND SHALL BE LOCATED 27 INCHES MAXIMUM FROM THE SEAT WALL.

**TAS SECTION 608.5.3 - ALTERNATE ROLL-IN TYPE SHOWER COMPARTMENTS**  
IN ALTERNATE ROLL-IN TYPE SHOWER COMPARTMENTS, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE LOCATED ABOVE THE GRAB BAR, BUT NO HIGHER THAN 48 INCHES ABOVE THE SHOWER FLOOR. WHERE A SEAT IS PROVIDED, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE LOCATED ON THE SIDE WALL ADJACENT TO THE SEAT 27 INCHES MAXIMUM FROM THE SIDE WALL BEHIND THE SEAT OR SHALL BE LOCATED ON THE BACK WALL OPPOSITE THE SEAT 15 INCHES MAXIMUM, LEFT OR RIGHT, OF THE CENTERLINE OF THE SEAT. WHERE A SEAT IS NOT PROVIDED, THE CONTROLS, FAUCETS, AND SHOWER SPRAY UNIT SHALL BE INSTALLED ON THE SIDE WALL FARTHEST FROM THE COMPARTMENT ENTRY.

## SIGNAGE

**TAS SECTION 216.1 - BUILDING SIGNS**  
SIGNS WHICH DESIGNATE PERMANENT ROOMS AND SPACES SHALL COMPLY WITH ALL SECTIONS OF SECTIONS 216 AND 703. ELEMENTS AND SPACES OF ACCESSIBLE FACILITIES WHICH SHALL BE IDENTIFIED BY THE INTERNATIONAL SYMBOL OF ACCESSIBILITY ARE PARKING SPACES AS RESERVED FOR INDIVIDUALS WITH DISABILITIES, ACCESSIBLE PASSENGER LOADING ZONES, ACCESSIBLE ENTRANCES WHEN NOT ALL ARE ACCESSIBLE, AND ACCESSIBLE TOILET AND BATHING FACILITIES WHEN NOT ALL ARE ACCESSIBLE.

**TAS SECTION 216.2 & 216.3 - BUILDING SIGNS**  
SIGNS WHICH DESIGNATE PERMANENT ROOMS AND SPACES AND OTHER SIGNS WHICH PROVIDE DIRECTION TO OR INFORMATION ABOUT FUNCTIONAL SPACES OF THE BUILDING SHALL COMPLY WITH ALL SECTIONS OF 703.5. (BUILDING DIRECTORIES, MENUS, AND ALL OTHER SIGNS WHICH ARE TEMPORARY ARE NOT REQUIRED TO COMPLY.)

**TAS SECTION 216.6 - ENTRANCES**  
WHERE NOT ALL ENTRANCES COMPLY WITH 404, ENTRANCES COMPLYING WITH 404 SHALL BE IDENTIFIED BY THE INTERNATIONAL SYMBOL OF ACCESSIBILITY COMPLYING WITH 703.7.2.1. DIRECTIONAL SIGNS COMPLYING WITH 703.5 THAT INDICATE THE LOCATION OF THE NEAREST ENTRANCE COMPLYING WITH 404 SHALL BE PROVIDED AT ENTRANCES THAT DO NOT COMPLY WITH 404.

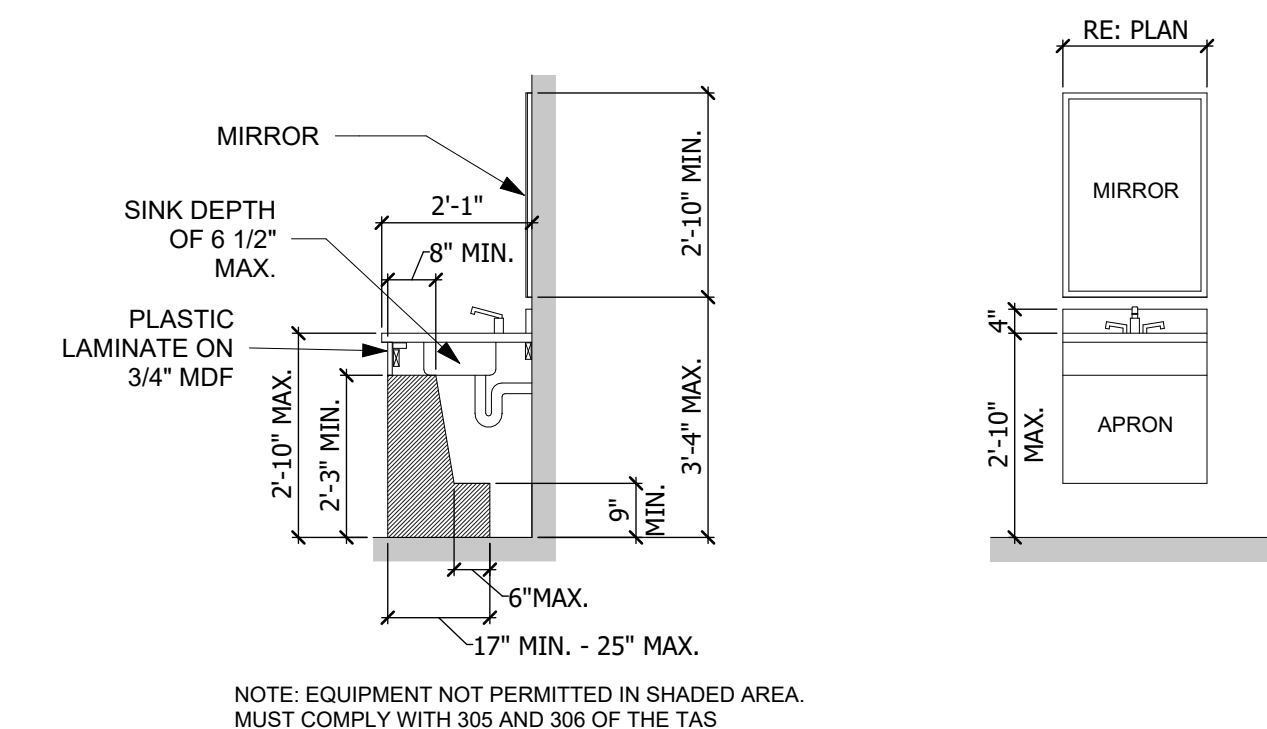
**TAS SECTION 703.1 - SIGNS**  
SIGNS SHALL COMPLY WITH 703. WHERE BOTH VISUAL AND TACTILE CHARACTERS ARE REQUIRED, EITHER ONE SIGN WITH BOTH VISUAL AND TACTILE CHARACTERS, OR TWO SEPARATE SIGNS, ONE WITH VISUAL, AND ONE WITH TACTILE CHARACTERS, SHALL BE PROVIDED.

**TAS SECTION 703.2.1 & 703.3 - RAISED AND BRAILLED CHARACTERS**  
RAISED CHARACTERS SHALL BE 1/32" MINIMUM ABOVE THEIR BACKGROUND. CHARACTERS SHALL BE UPPERCASE, SANS SERIF, AND SHALL NOT BE ITALIC, OBLIQUE, SCRIPT, HIGHLY DECORATIVE, OR OF OTHER UNUSUAL FORMS. BRAILLE CHARACTERS SHALL BE CONTRACTED (GRADE 2). CHARACTER HEIGHT MEASURED VERTICALLY FROM THE BASELINE OF THE CHARACTER SHALL BE 5/8" MINIMUM AND 2" MAXIMUM BASED ON THE HEIGHT OF THE UPPERCASE LETTER "T".

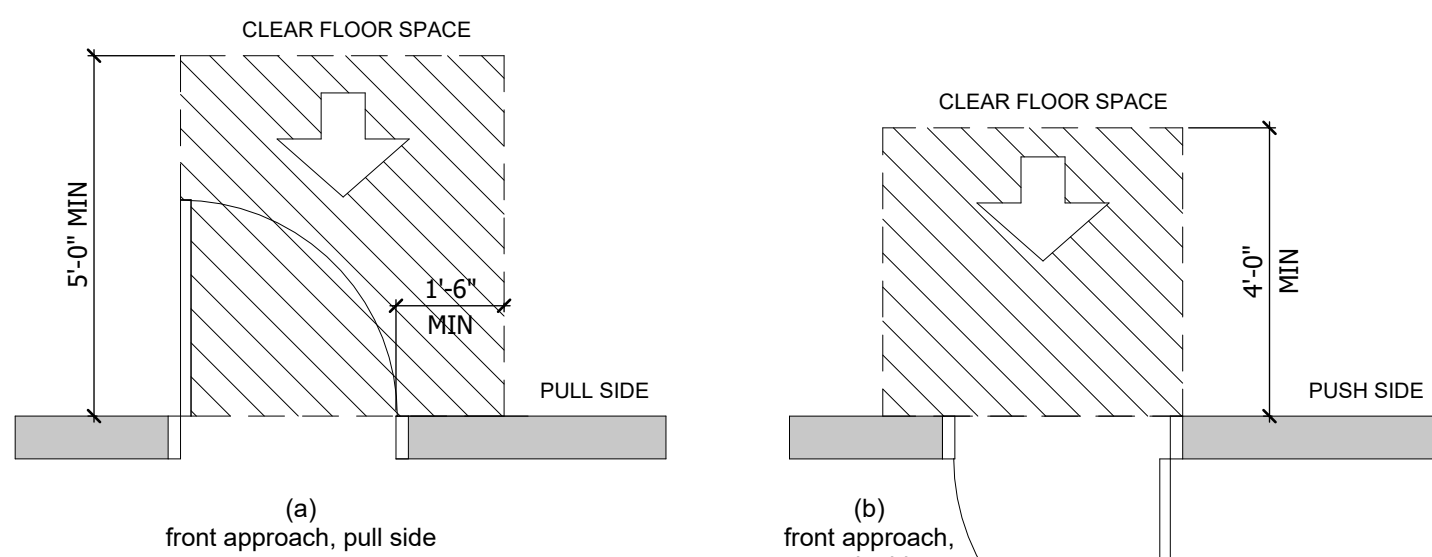
**TAS SECTION 703.2.4 - CHARACTER PROPORTION**  
CHARACTERS SHALL BE SELECTED FROM FONTS WHERE THE WIDTH OF THE UPPERCASE LETTER "O" IS 55 PERCENT MINIMUM AND 110 PERCENT MAXIMUM OF THE HEIGHT OF THE UPPERCASE LETTER "T".

**TAS SECTION 703.5.1 - FINISH AND CONTRAST**  
CHARACTERS AND THEIR BACKGROUND SHALL HAVE A NON-GLARE FINISH. CHARACTERS SHALL CONTRAST WITH THEIR BACKGROUND WITH EITHER LIGHT CHARACTERS ON A DARK BACKGROUND OR DARK CHARACTERS ON A LIGHT BACKGROUND

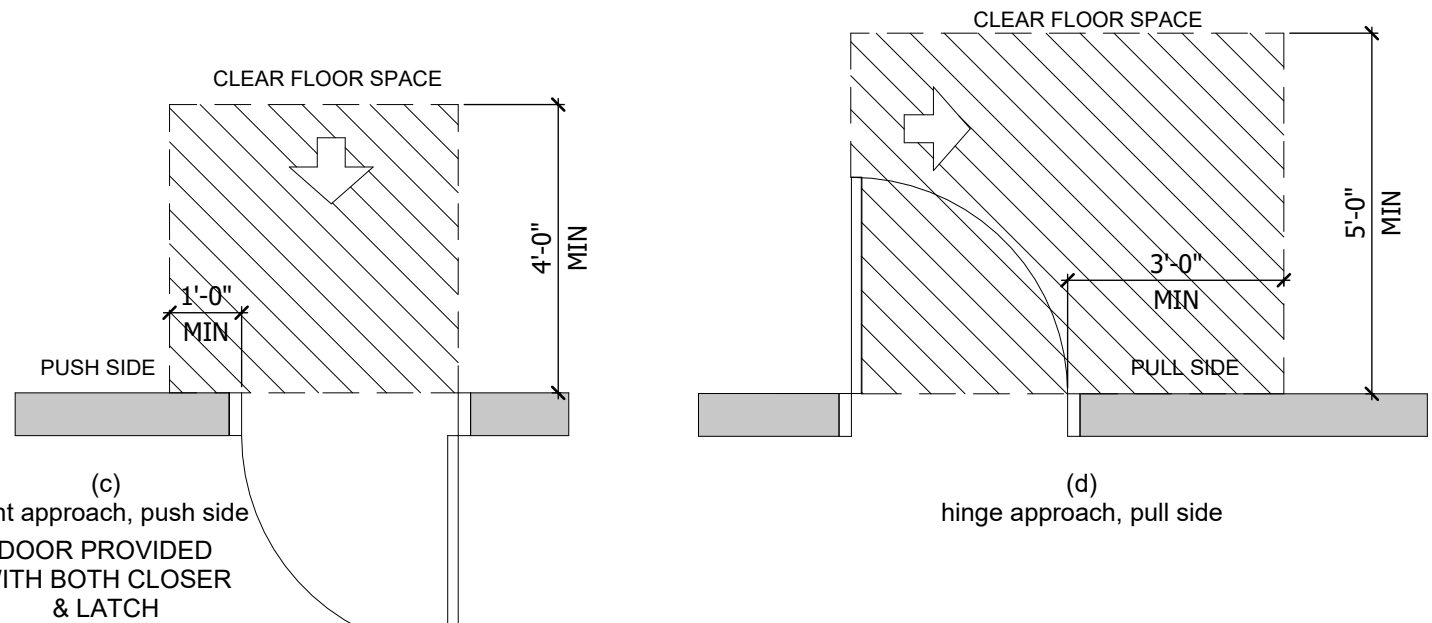
**NOTE: THESE DRAWINGS ARE FOR GENERAL MOUNTING CLEARANCE REFERENCE. SEE PLANS FOR CONFIGURATIONS. ALL DIMENSIONS SHALL CONFORM TO TEXAS ACCESSIBILITY STANDARDS.**



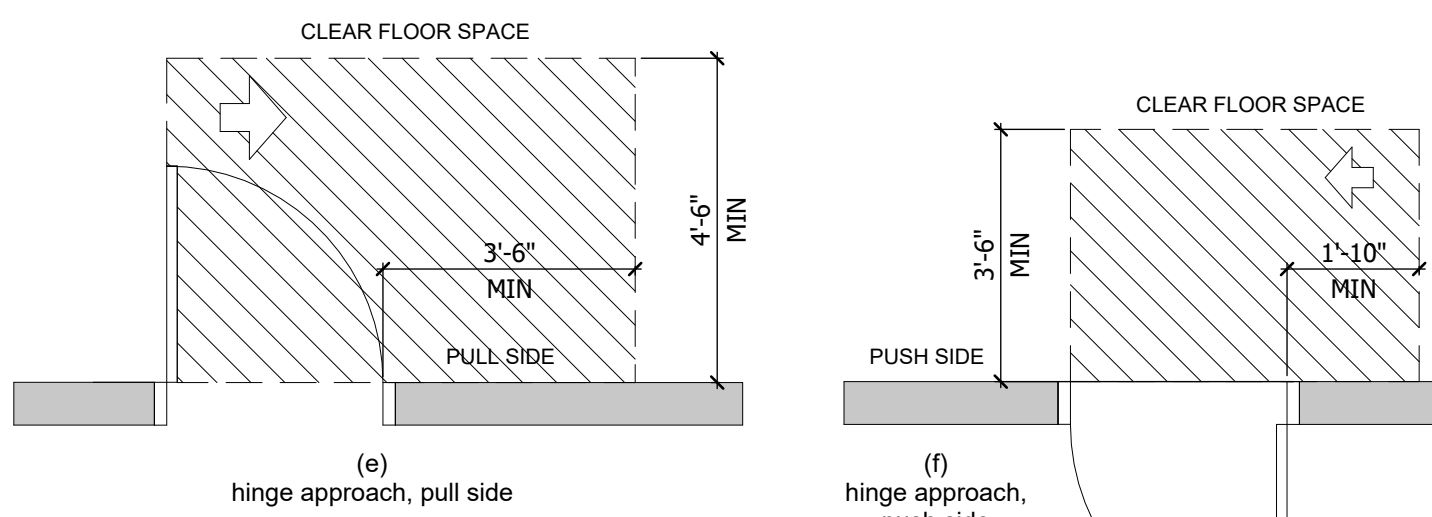
## LAVATORY WITH COUNTERTOP - TAS SECTION 606.2 - 606.5



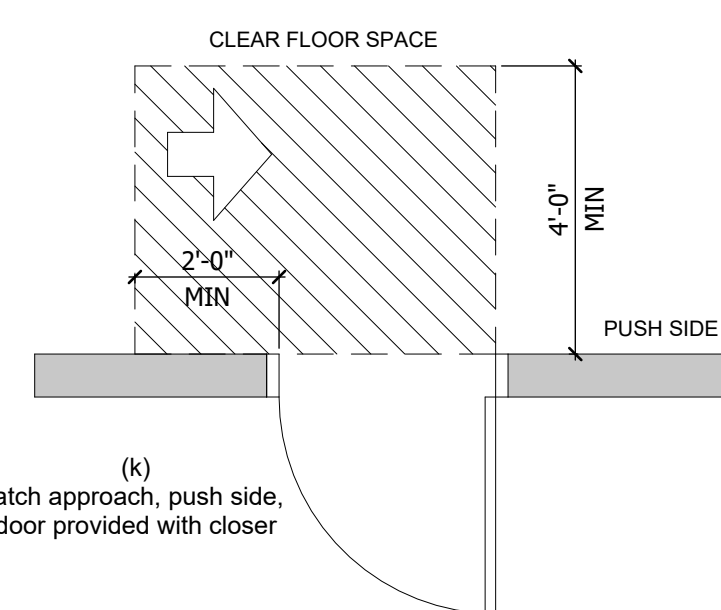
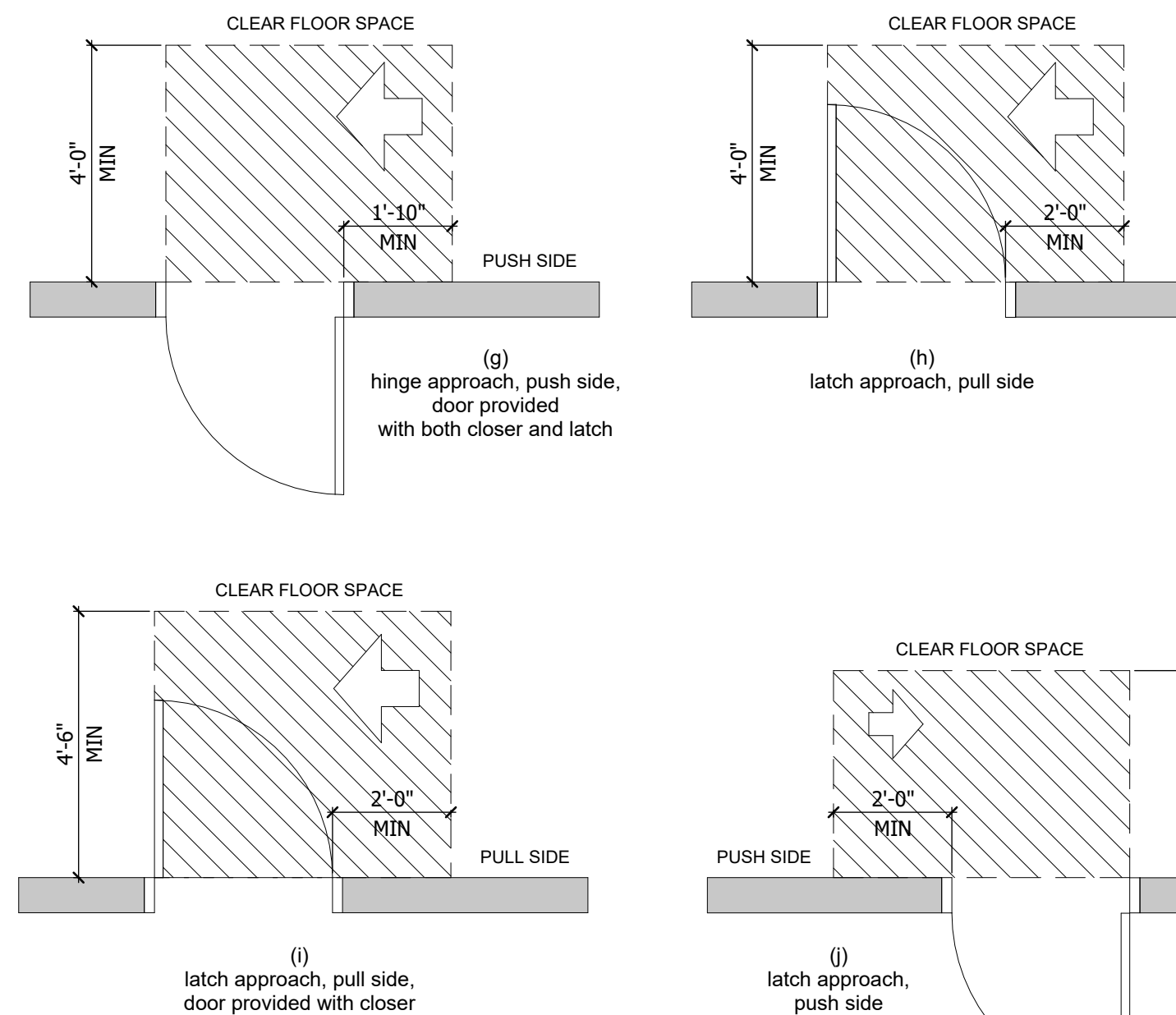
## FRONT APPROACH - SWINGING DOORS



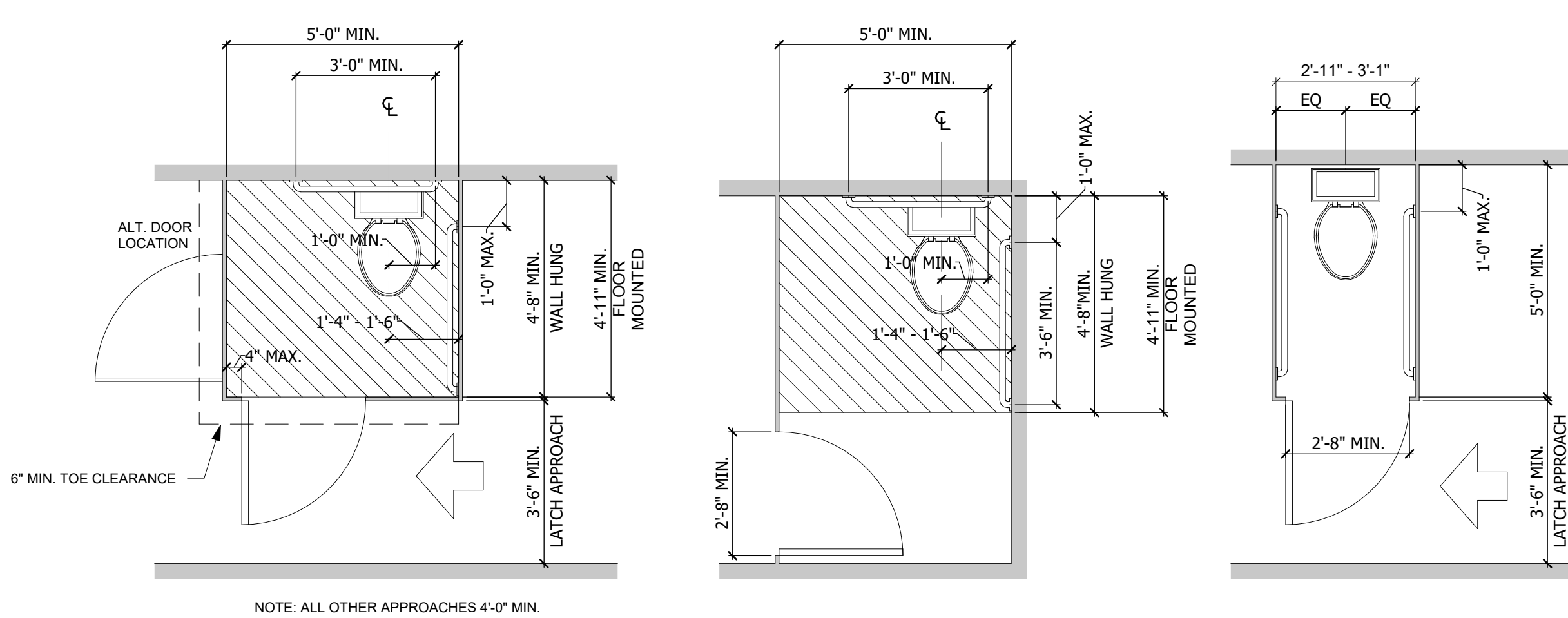
## LATCH SIDE APPROACH - SWINGING DOORS



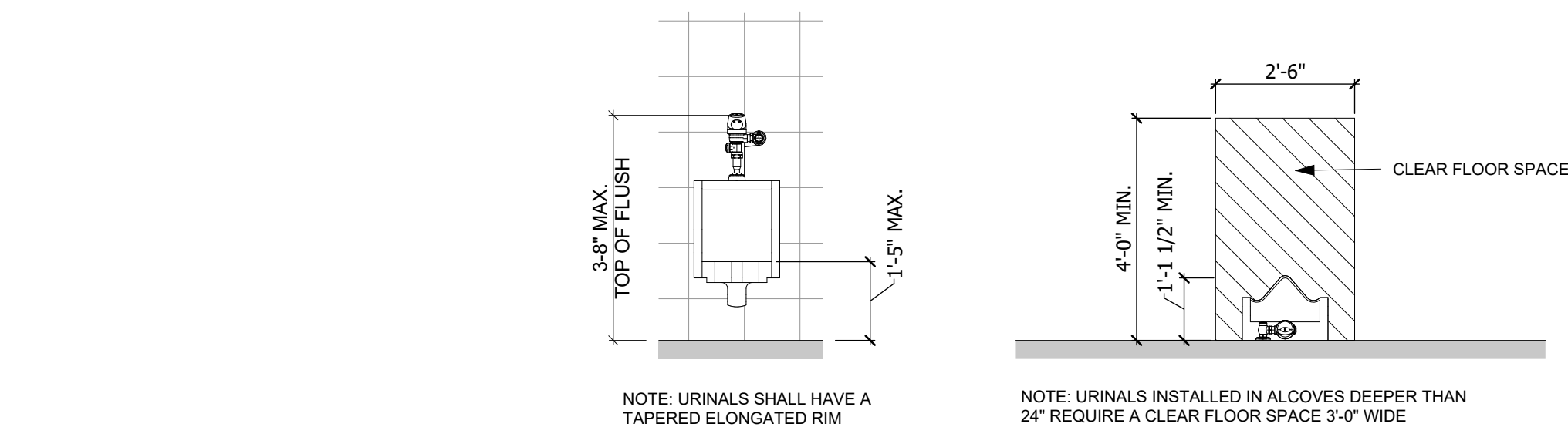
## HINGE SIDE APPROACH - SWINGING DOORS



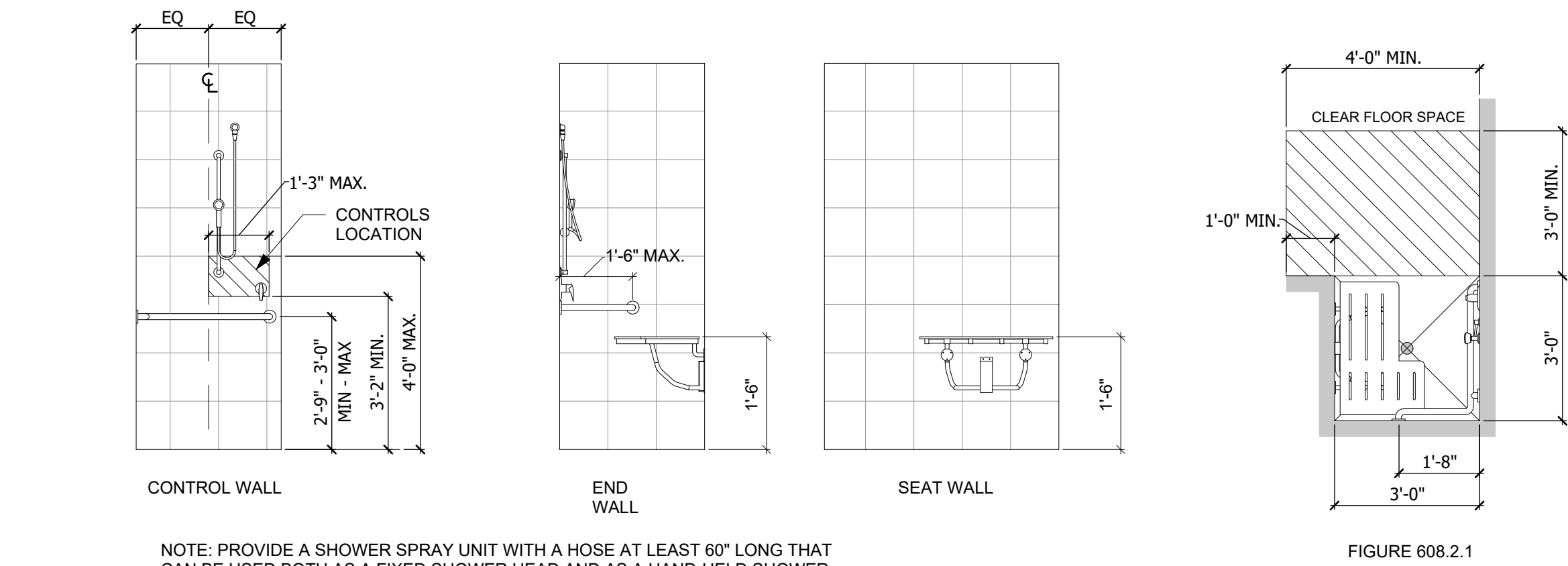
## SWINGING DOORS AND GATES- TAS SECTION 404.2.4.1



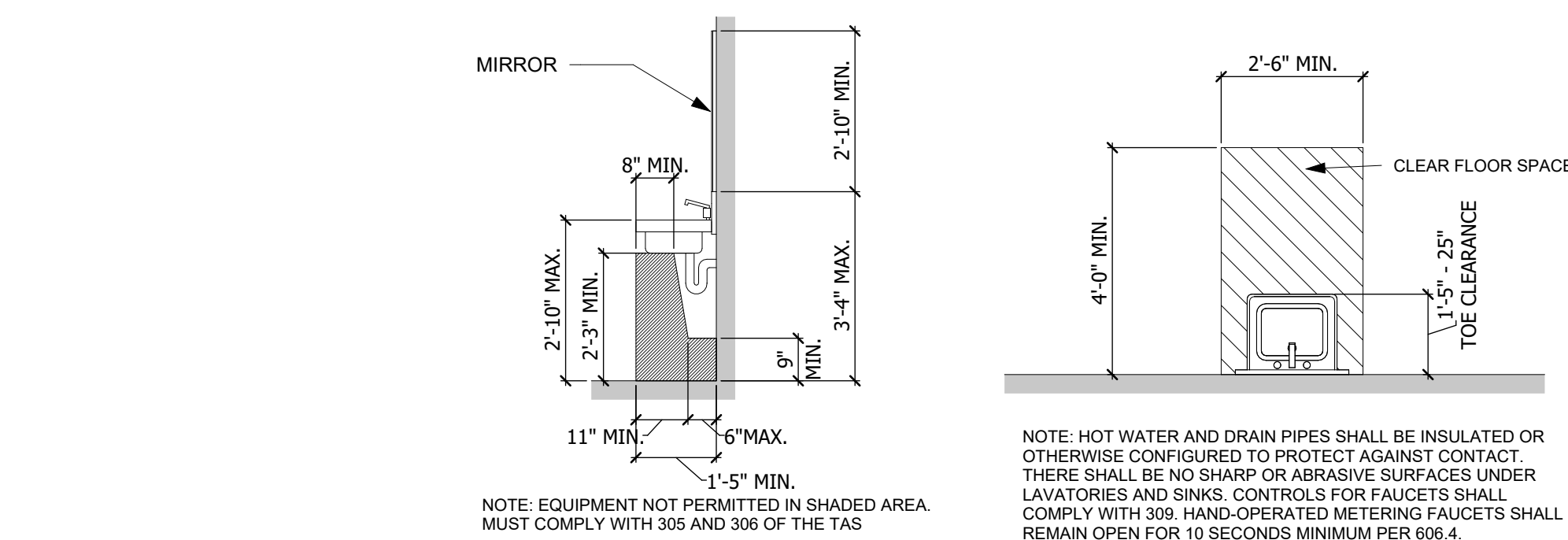
## SIZE - TAS SECTION 604.8.1.1



## URINALS - TAS SECTION 605.2



## TRANSFER TYPE SHOWER COMPARTMENTS - TAS SECTION 608.2.1, 608.5.1, 608.5.2 & 608.5.3



## CLEAR FLOOR SPACE & HEIGHT - TAS SECTION 606.2 - 606.5



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|   |          |                               |

FIRE EXTINGUISHER LOCATION

⦿ DENOTES FIRE EXTINGUISHER LOCATIONS - MAXIMUM TRAVEL DISTANCE TO EXTINGUISHER IS 75 FEET

EMERGENCY EGRESS LIGHTING LEGEND

⊠ NEW LED ILLUMINATED EXIT SIGN  
□ NEW EXTERIOR EGRESS LIGHTING

EGRESS LEGEND

4

APPLICABLE CODES - CITY OF HOUSTON

2015 INTERNATIONAL BUILDING CODE W/ CITY OF HOUSTON AMENDMENTS  
2020 NATIONAL ELECTRICAL CODE  
2015 UNIFORM MECHANICAL CODE W/ CITY OF HOUSTON AMENDMENTS  
2015 UNIFORM PLUMBING CODE W/ CITY OF HOUSTON AMENDMENTS  
2015 INTERNATIONAL FIRE CODE W/ CITY OF HOUSTON AMENDMENTS  
2015 INTERNATIONAL ENERGY CONSERVATION CODE - COMMERCIAL W/ CITY OF HOUSTON AMENDMENTS

USE AND OCCUPANCY

304.1 BUSINESS GROUP M: RETAIL

THE AREAS LABELED AS SHOWROOM WILL SERVE A GROUP M RETAIL OCCUPANCY WHERE THE TENANTS SPECIFIC USE WILL BE DISPLAY AND SALE OF MERCHANDISE.

2,311 S.F. / 100 OCCUPANT FACTOR (1004.2) = 24 OCCUPANTS

TOTAL: 24 OCCUPANTS

ALLOWABLE AREA - NON SEPARATED OCCUPANCIES

OCCUPANCIES:  
M - RETAIL  
CONSTRUCTION TYPE: IIB  
SPRINKLER: YES

BASE AREA: 50,000 S.F.  
SPRINKLER INCREASE: 37,500 S.F.  
ALLOWABLE AREA: 87,500 S.F.  
ACTUAL BUILDING: 2,311 S.F.

MOST RESTRICTIVE PROVISIONS:  
SINGLE FIRE AREA IS CREATED, BOUNDED BY THE EXTERIOR WALLS OF THE BUILDING - S1 OCCUPANCY, 2 STORY HEIGHT LIMIT, TYPE IIB CONSTRUCTION AND AREA LIMITATIONS AS NOTED BELOW:

PER SECTION 507.3, THE AREA OF A GROUP B, F, M, OR S BUILDING NO MORE THAN ONE STORY ABOVE GRADE PLANE OF ANY CONSTRUCTION TYPE, SHALL NOT BE LIMITED WHERE THE BUILDING IS PROVIDED WITH AN AUTOMATIC SPRINKLER SYSTEM THROUGHOUT IN ACCORDANCE WITH SECTION 903.3.1.1 AND IS SURROUNDED AND ADJOINED BY PUBLIC WAYS OR YARDS NOT LESS THAN 60 FEET IN WIDTH.

PER SECTIONS 508.3.1, THE TOTAL AREA (2,311 S.F.) IS LESS THAN THE ALLOWABLE AREA LIMITATIONS FOR THE MOST RESTRICTIVE OCCUPANCY TYPE (M, 87,500 SF ) AND THEREFORE QUALIFIES FOR NON-SEPARATED USE.

LAVATORY CALCULATIONS - IBC TABLE 2902.1

SHOWROOM (M) : 1 LAV / 750 OCCUPANTS

TOTAL LAVATORY CALC: 24 OCCUPANTS / 750 = .032 LAVATORIES

TOTAL LAVATORIES REQUIRED: .032  
TOTAL LAVATORIES PROVIDED: 1

WATER CLOSET CALCULATIONS - IBC TABLE 2902.1

SHOWROOM (M) : 1 WATER CLOSET / 500 OCCUPANTS

TOTAL WATER CLOSET CALC: 24 OCCUPANTS / 500 = .048 WC

TOTAL WATER CLOSETS REQUIRED: .048  
TOTAL WATER CLOSETS PROVIDED: 1

EGRESS NOTES

MINIMUM OF 44" EGRESS WIDTH PROVIDED TO ALL EXITS.

DOORS LABELED AS REQUIRED EXITS TO REMAIN UNLOCKED DURING BUSINESS HOURS AND HAVE APPLICABLE SIGNAGE.

ALL EGRESS DOORS SHALL BE READILY OPENABLE FROM THE EGRESS SIDE WITHOUT THE USE OF A KEY OR SPECIAL KNOWLEDGE.

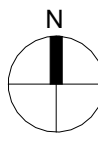
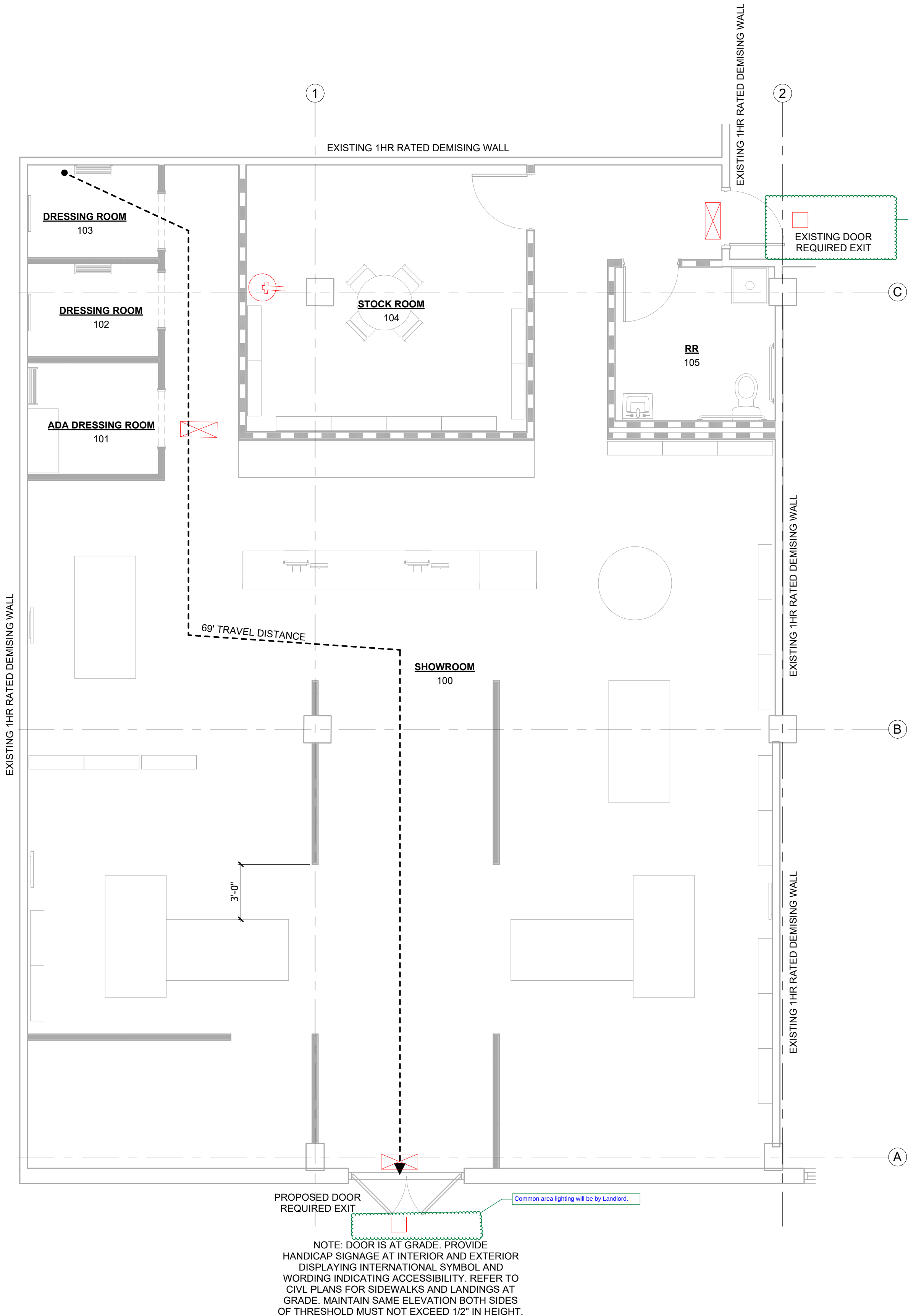
EGRESS PLAN

5

1/4" = 1'-0"

NOTES

1



PROJECT #

23.1002

DESIGN TEAM

Loe Ortega Architecture PLLC  
346.708.7733  
2339 Commerce St.  
Suite 165  
Houston, TX 77002  
www.LOAarchitecture.com



SEAL

This drawing is for preliminary review or bidding and NOT for regulatory approval, permitting, or construction.

PROJECT

**LUMINARY**  
**RICE VILLAGE**  
2556 UNIVERSITY BLVD., SUITE A-103  
HOUSTON, TX 77005

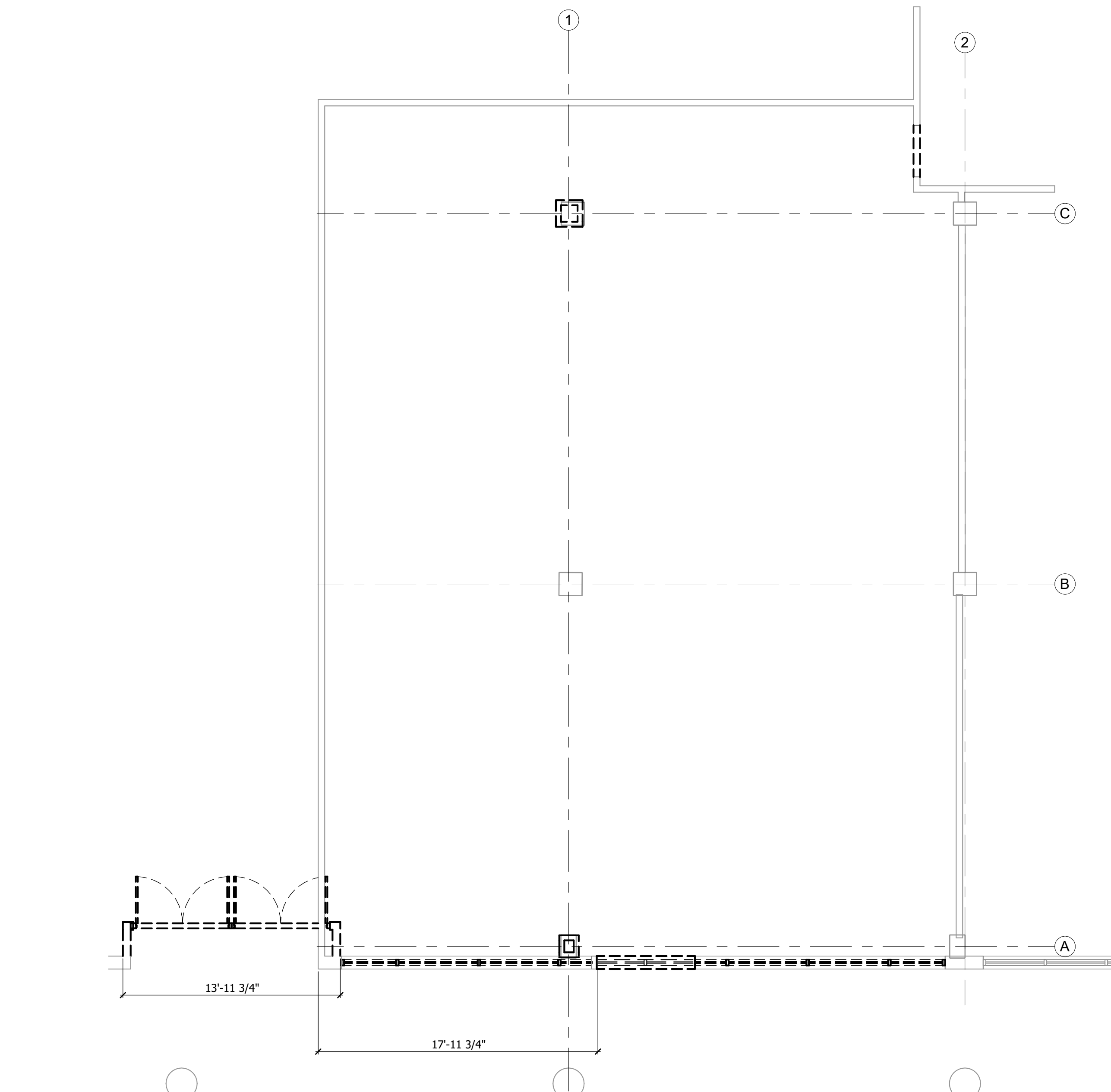
SET  
CONSTRUCTION  
DOCUMENTS

ISSUE DATE

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|   |          |                               |

SHEET  
DEMO PLAN  
AND NOTES

A1.00



#### DEMO NOTES

PROTECT ALL EXISTING CEILINGS, LIGHTING, FIXTURES, MILLWORK, FINISHES, AND STRUCTURE DURING DEMOLITION.

CONTRACTOR SHALL CONSULT WITH STRUCTURAL ENGINEER PRIOR TO MODIFICATION OF ANY MASONRY WALLS, INCLUDING BUT NOT LIMITED TO DEMOLITION FOR NEW OPENINGS, DOORS, OR WINDOWS.

CONTRACTOR IS RESPONSIBLE FOR THE SAFETY AND INTEGRITY OF THE STRUCTURE AND JOB SITE DURING DEMOLITION. MEANS AND METHODS OF DEMOLITION AND CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR.

#### DEMO WALL LEGEND

————— EXISTING WALL  
- - - - - DEMO WALL

DEMOLITION PLAN  
3/16" = 1'-0"

5

DEMO LEGEND & NOTES

1

## DIMENSIONS

ALL INTERIOR DIMENSIONS ARE TO FACE OF FINISHED WALL U.N.O.

ALL DOOR OPENINGS TO BE 6" FROM ADJACENT WALLS ON HINGE SIDE OF DOOR UNLESS OTHERWISE NOTED.

## BLOCKING

PROVIDE METAL STUD OR STEEL BLOCKING (AND/OR FIRE-RETARDANT 2X WOOD BLOCKING WHERE PERMITTED BY CODE) ADEQUATE TO SUPPORT GRAB BARS, HANDRAILS, TRIM, MOULDINGS, WALL MOUNTED EQUIPMENT AND FIXTURES AS SCHEDULED OR NOTED ELSEWHERE. ALL BLOCKING MUST PROVIDE ADEQUATE STRUCTURAL SUPPORT TO MEET ALL APPLICABLE CODES RELATED TO SUCH ITEMS.

## MEP NOTES

PROVIDE A MINIMUM OF 1 FOOT-CANDLE IN ALL AREAS ON A BATTERY BACK-UP TO PROPERLY ILLUMINATE PATH OF EGRESS IN THE EVENT OF POWER FAILURE.

## RESTROOM NOTES

PER IBC SECTION 1210.2.1 IN OTHER THAN DWELLING UNITS, BATHING AND SHOWER ROOM FLOORS SHALL HAVE A SMOOTH, HARD, NONABSORBENT SURFACE. THE INTERSECTIONS OF SUCH FLOORS WITH WALLS SHALL HAVE A SMOOTH, HARD, NONABSORBENT VERTICAL BASE THAT EXTENDS UPWARD ONTO THE WALLS NOT LESS THAN 4 INCHES

PER THE IBC, WALLS WITHIN TWO (2) FEET OF URINALS AND WATER CLOSETS SHALL HAVE A SMOOTH, HARD, NONABSORBENT SURFACE TO A HEIGHT OF FOUR (4) FEET A.F.F., AND EXCEPT FOR STRUCTURAL ELEMENTS, THE MATERIALS USED IN SUCH WALLS SHALL BE OF A TYPE THAT IS NOT ADVERSELY AFFECTED BY MOISTURE. ACCESSORIES SUCH AS GRAB BARS, TOWEL BARS, PAPER DISPENSERS AND SOAP DISHES PROVIDED ON OR WITHIN SUCH WALLS SHALL BE BE INSTALLED AND SEALED TO PROTECT STRUCTURAL ELEMENTS FROM MOISTURE

WHEN GYPSUM BOARD IS USED AS A BASE FOR TILE OR WALL PANELS FOR TUBS, SHOWER OR WATER CLOSET COMPARTMENT WALLS, WATER-RESISTANT GYPSUM BACKING BOARD SHALL BE USED AS A SUBSTRATE

PROVIDE FRP TO 48" A.F.F. ON ALL WET WALLS, WALLS ADJACENT TO WET WALLS AND WALLS WITHIN TWO (2) FEET OF URINALS AND WATER CLOSETS IN REMODELED RESTROOMS

PROVIDE STANDARD RESTROOM ACCESSORIES INCLUDING MIRRORS, PAPER TOWEL DISPENSER, TOILET PAPER DISPENSERS, AND ADA GRAB BARS IN REMODELED RESTROOMS

## GENERAL

ANY CONSTRUCTION INDICATED AS NEW MUST CONFORM TO ALL APPLICABLE CODES, ORDINANCES AND REGULATIONS.

CONTRACTOR TO VERIFY WITH OWNER AND ARCHITECT THE SCOPE OF WORK TO BE PERFORMED PRIOR TO COMMENCING ANY WORK. CONTRACTOR IS RESPONSIBLE FOR CHECKING ALL CONTRACT DOCUMENTS, FIELD CONDITIONS, AND DIMENSIONS FOR ACCURACY AND CONFIRMING THAT THE WORK IS CONSTRUCTABLE AS SHOWN AND MEETS ALL APPLICABLE CODES BEFORE PROCEEDING WITH CONSTRUCTION. IF THERE ARE ANY QUESTIONS REGARDING THESE OR OTHER COORDINATION ISSUES, THE CONTRACTOR IS RESPONSIBLE FOR OBTAINING A CLARIFICATION FROM THE ARCHITECT BEFORE PROCEEDING WITH THE WORK IN QUESTION.

CONTRACTOR WILL BE RESPONSIBLE FOR CONSTRUCTION MEAN, METHODS OF CONSTRUCTION, AND THE SEQUENCE OF CONSTRUCTION AND PROCEDURES FOR SAFETY PRECAUTIONS AND PROGRAMS OF THE PROPOSED PROJECT.

## GENERAL NOTES

## WALL LEGEND

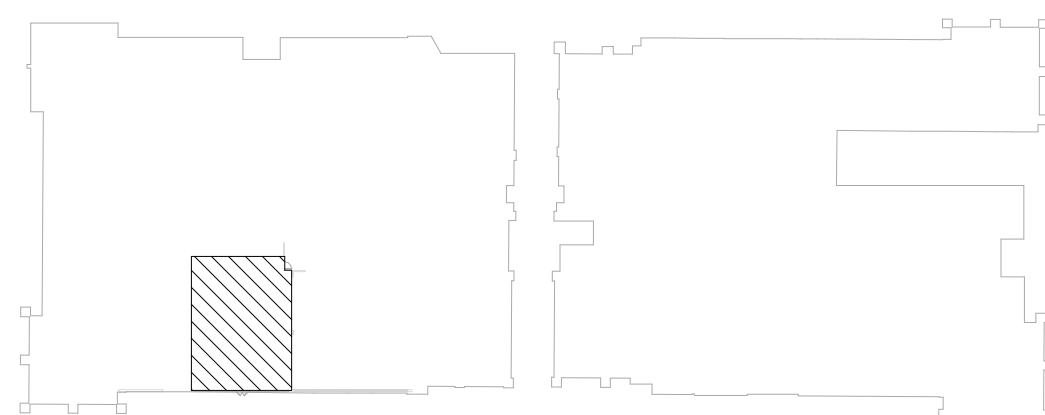
|  |                          |
|--|--------------------------|
|  | EXISTING WALL            |
|  | NEW WALL                 |
|  | NEW WALL TO 8'-0" A.F.F. |
|  | NEW INSULATED WALL       |

## SYMBOL LEGEND

|  |                      |
|--|----------------------|
|  | SWITCH               |
|  | DUPLEX OUTLET        |
|  | DEDICATED GFI DUPLEX |
|  | QUAD OUTLET          |
|  | DATA                 |
|  | FLOOR OUTLET         |
|  | FLOOR DATA           |

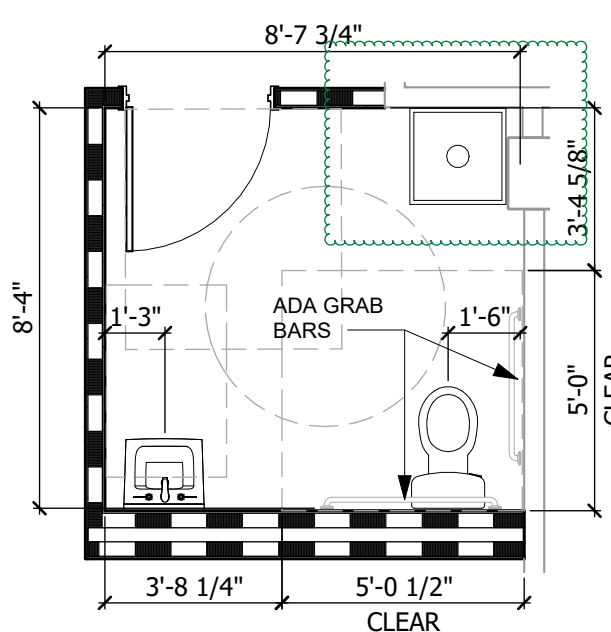
+X" - DENOTES HEIGHT WHERE NOTED

## LEGENDS



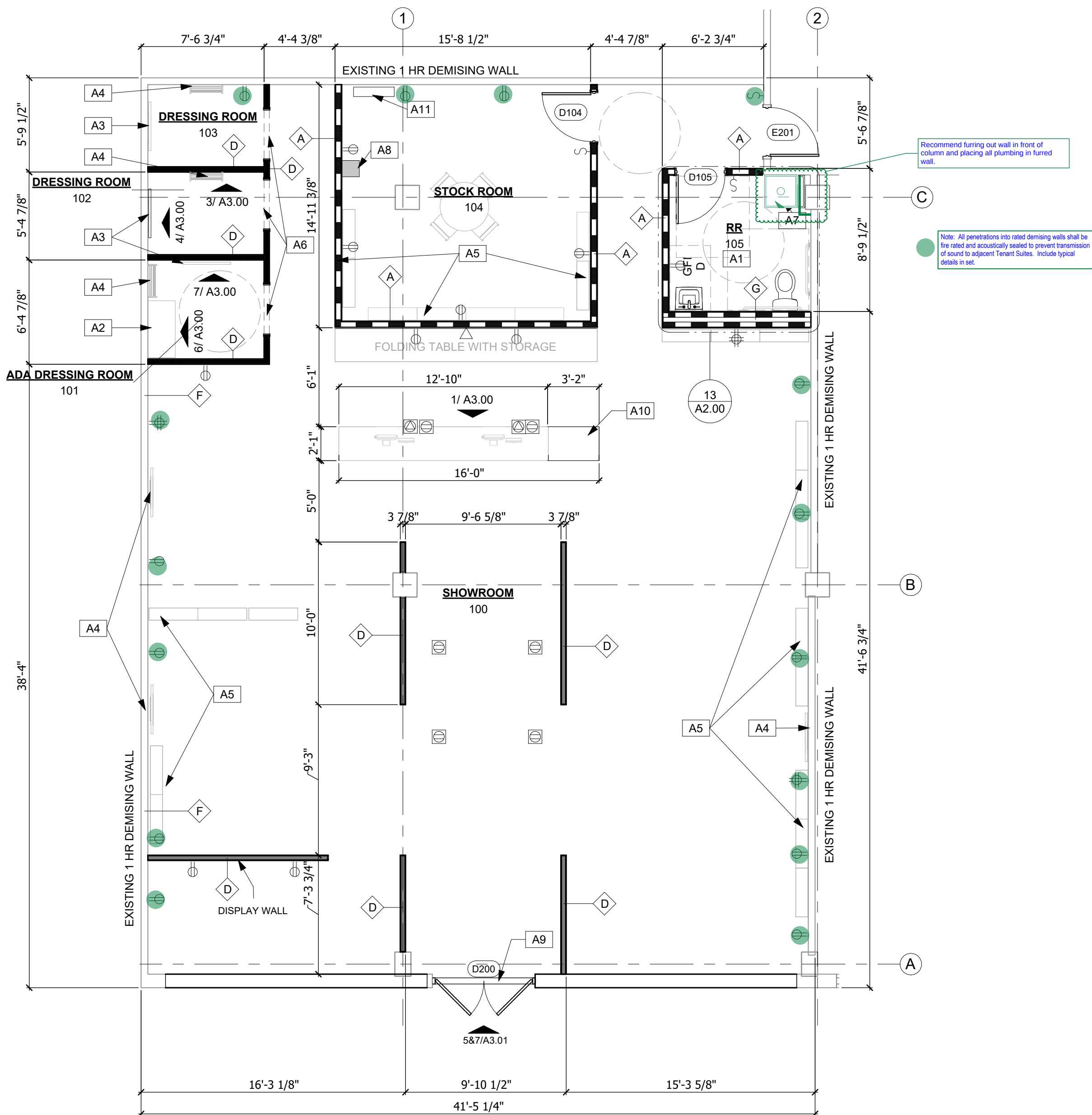
TOTAL RETAIL: 2,311 S.F.

## LOCATION PLAN



## ENLARGED RESTROOM PLAN

1/4" = 1'-0"



## OVERALL FLOOR PLAN

3/16" = 1'-0"

|   |          |                               |
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DESIGN TEAM

Loe Ortega Architecture PLLC  
344.708.7733  
2339 Commerce St.  
Suite 165  
Houston, TX 77002  
www.LOAarchitecture.com



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PROJECT

**LUMINARY**  
**RICE VILLAGE**  
2556 UNIVERSITY BLVD., SUITE A-103  
HOUSTON, TX 77005

SET  
CONSTRUCTION  
DOCUMENTS

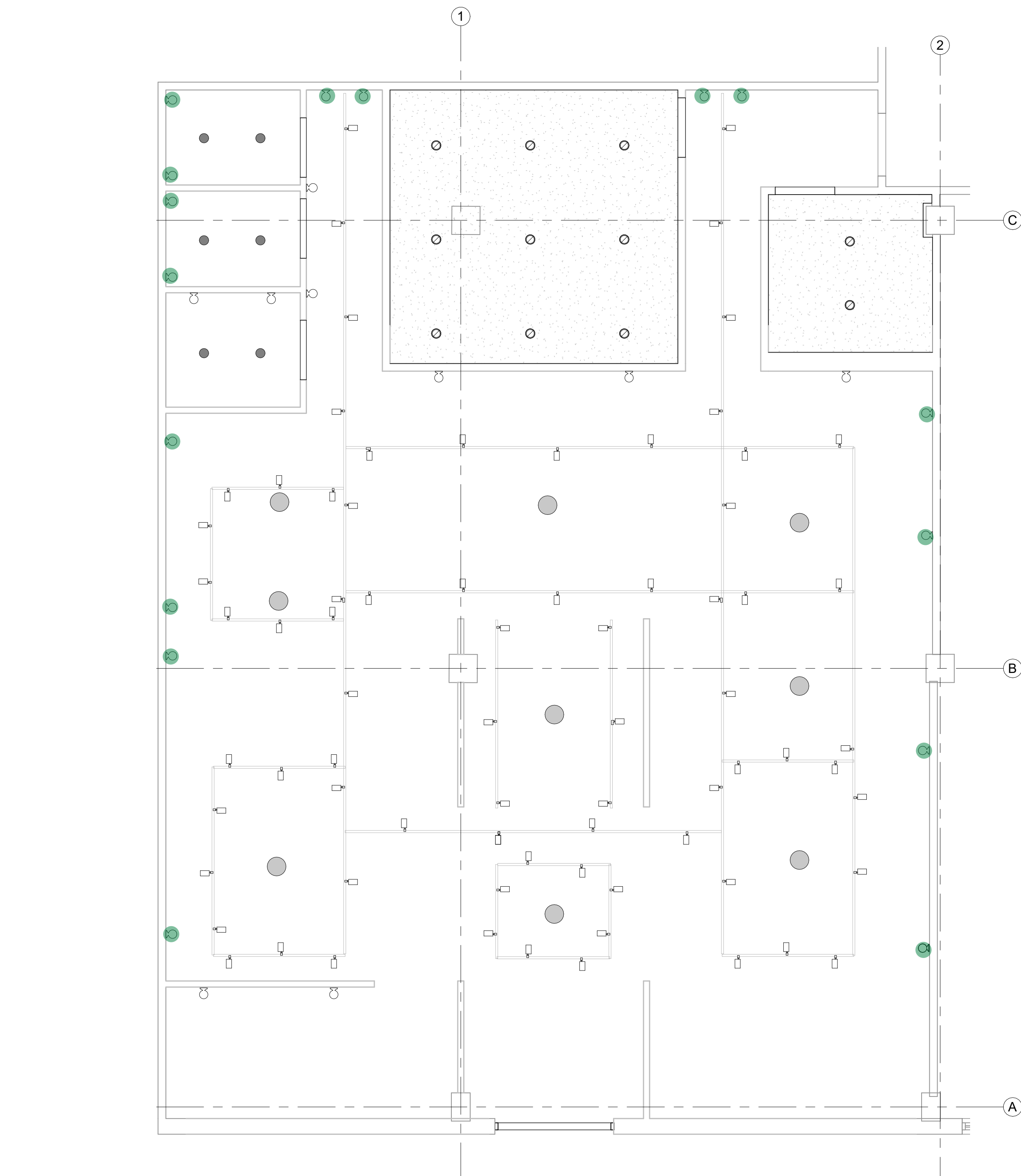
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SHEET  
**REFLECTED  
CEILING PLAN  
AND NOTES**

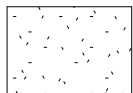
**A2.20**

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Note: All penetrations into rated demising walls shall be fire rated and acoustically sealed to prevent transmission of sound to adjacent Tenant Suites. Include typical details to set.

LIGHTING LEGEND



NEW 2'x4' ACOUSTICAL CEILING TILE GRID @ 9'-0" A.F.F.



NEW LED TRACK LIGHT FIXTURE



NEW PENDANT LIGHT FIXTURE



NEW 6" PENDANT LIGHT FIXTURE



NEW 6" RECESSED CAN LIGHT FIXTURE



NEW WALL SCONCE AT 60" A.F.F.

REFLECTED CEILING PLAN

5

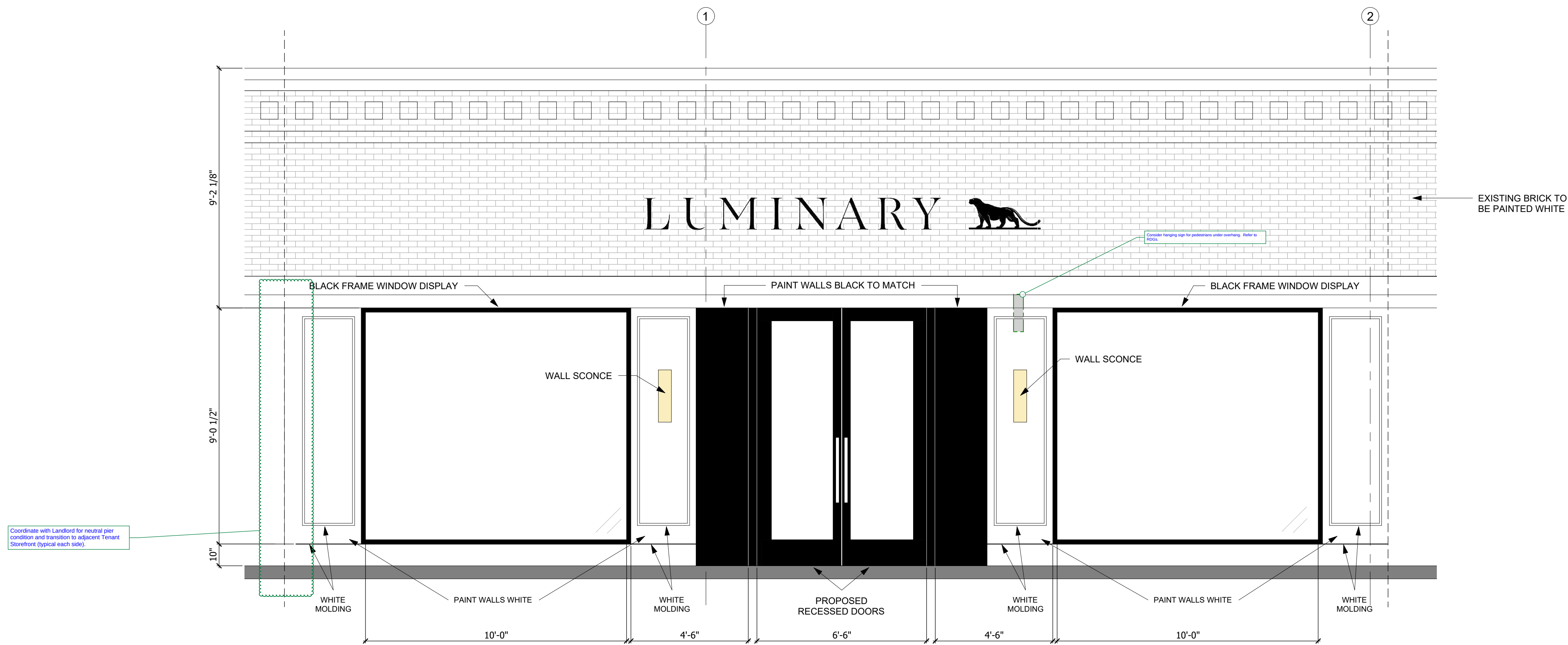
1/4" = 1'-0"

RCP LEGEND

1

PROJECT #  
23.1002



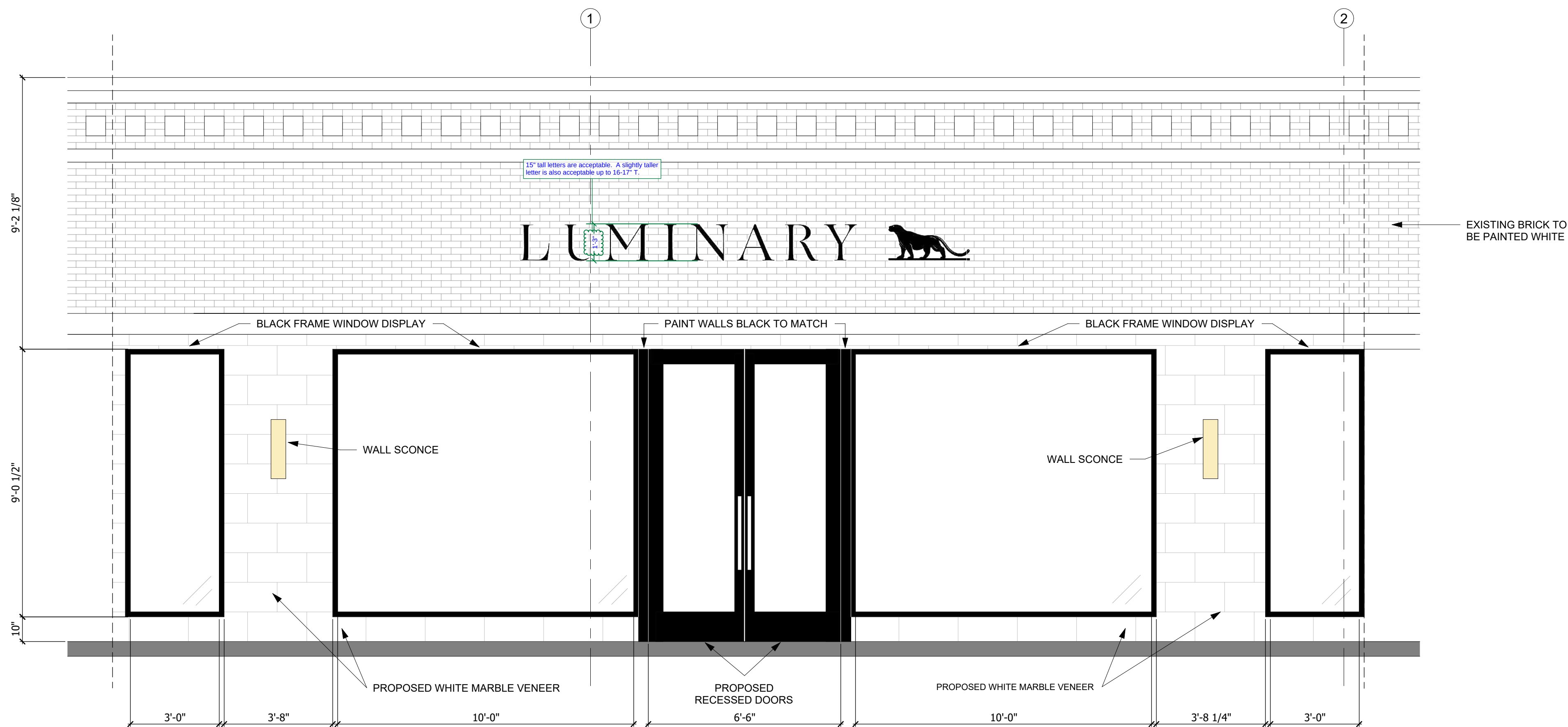


EXTERIOR ELEVATION - OPT A

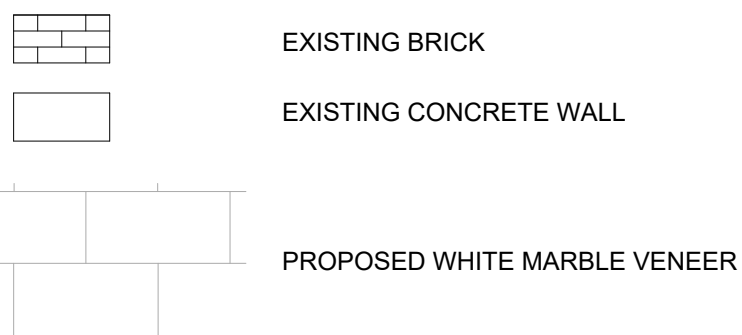
7

3/8" = 1'-0"

- 1) We recommend Option A layout as the show windows are better configured with interior.
- 2) Provide 3D rendering of proposed storefront with all materials accurately depicted. Consider ceiling condition immediately adjacent to storefront.
- 3) Provide material samples / specifications for all proposed materials. High-quality composite wood materials shall be used for paneled areas.
- 4) Storefront base shall be stone or highly durable material.



MATERIAL LEGEND



EXTERIOR ELEVATION - OPT B

5

3/8" = 1'-0"

MATERIAL LEGEND

1

|   |          |                               |
|---|----------|-------------------------------|
| 1 | 02.09.23 | REVISED PER TENANT COMMENTS   |
| 2 | 02.21.23 | ISSUED FOR PRELIMINARY REVIEW |
| 3 | 03.09.23 | REVISED PER LLO COMMENTS      |
| 4 | 03.15.23 | REVISED PER TENANT COMMENTS   |
|   |          |                               |
|   |          |                               |
|   |          |                               |
|   |          |                               |
|   |          |                               |



PROJECT #

23.1002

DESIGN TEAM

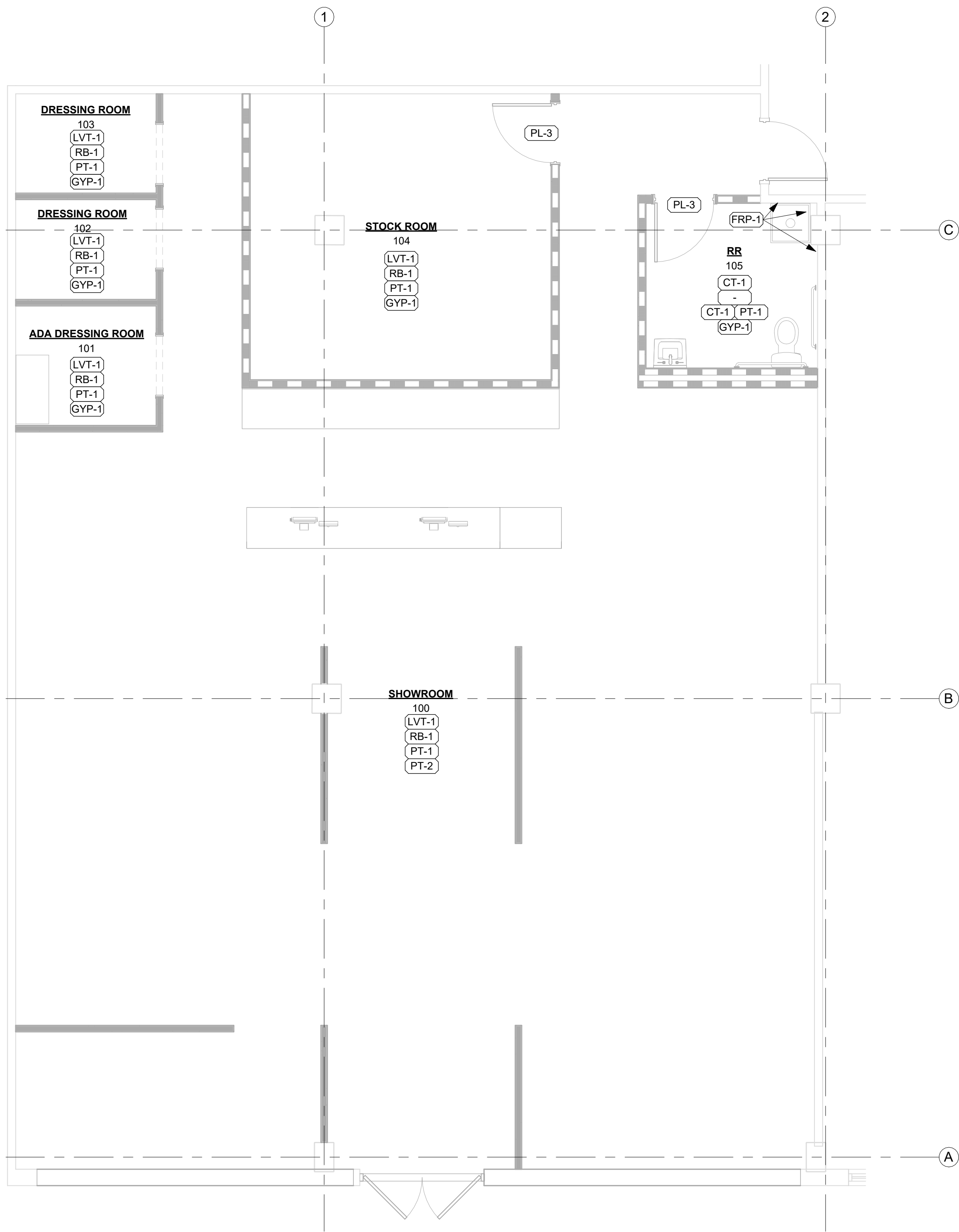
Loe Ortega Architecture PLLC  
346.708.7733  
2339 Commerce St.  
Suite 165  
Houston, TX 77002  
www.LOAarchitecture.com



SEAL

This drawing is for preliminary review or bidding and NOT for regulatory approval, permitting, or construction.

PROJECT



INTERIOR DESIGN CONCEPT PROVIDED BY TENANT AND APPROVED



| FINISH SCHEDULE |                              |              |       |       |                                    |
|-----------------|------------------------------|--------------|-------|-------|------------------------------------|
| SYMBOL          | DESCRIPTION                  | MANUFACTURER | STYLE | COLOR | REMARKS                            |
| FLOOR           |                              |              |       |       |                                    |
| CT-1            | CERAMIC TILE                 | TBD          | TBD   | TBD   | WHERE NOTED                        |
| LVT-1           | LUXURY VINYL TILE            | TBD          | TBD   | TBD   | WHERE NOTED                        |
| BASE            |                              |              |       |       |                                    |
| -               | RUBBER BASE                  | -            | -     | -     | WHERE NOTED                        |
| RB-1            | RUBBER BASE                  | TBD          | TBD   | TBD   | WHERE NOTED                        |
| WALL            |                              |              |       |       |                                    |
| CT-1            | CERAMIC TILE                 | TBD          | TBD   | TBD   | TO 48" ON ALL RESTROOM WALLS       |
| FRP-1           | FIBERGLASS REINFORCED PANELS | TBD          | TBD   | TBD   | AT 10'-0" A.F.F. AND ABOVE TO DECK |
| PT-1            | PAINT                        | TBD          | TBD   | TBD   | WHERE NOTED                        |
| CEILING         |                              |              |       |       |                                    |
| ACT-1           | ACOUSTICAL CEILING TILE      | TBD          | TBD   | TBD   | AT 10'-0" A.F.F.                   |
| GYP-1           | GYP BOARD                    | -            | -     | PT-2  | AT 10'-0" A.F.F.                   |
| PT-2            | PAINT                        | TBD          | TBD   | TBD   | AT 10'-0" A.F.F. AND ABOVE TO DECK |
| MILLWORK        |                              |              |       |       |                                    |
| PL-1            | PLASTIC LAMINATE             | TBD          | TBD   | TBD   | VERTICAL APPLICATIONS              |
| PL-2            | PLASTIC LAMINATE             | TBD          | TBD   | TBD   | HORIZONTAL APPLICATIONS            |
| PL-3            | PLASTIC LAMINATE             | TBD          | TBD   | TBD   | ALL NEW DOORS                      |

| INTERIOR FINISHES - CLASSIFICATION CHART |                                   |           |                           |
|------------------------------------------|-----------------------------------|-----------|---------------------------|
| OCCUPANCY GROUP                          | EXIT ENCLOSURES, EXIT PASSAGEWAYS | CORRIDORS | ROOMS AND ENCLOSED SPACES |
| M - SPRINKLERED                          | B                                 | C         | C                         |

- XXX-# FLOOR
- XXX-# BASE
- XXX-# WALL
- XXX-# CEILING

FINISH NOTES

PAINT ALL OFFICE WALLS INCLUDING WAREHOUSE SIDE OF OFFICE PERIMETER WALL.

PROVIDE CERAMIC TILE TO 48" ON ALL WALLS IN RESTROOM. PAINT WALL 48" AND ABOVE WITH ENAMEL PAINT LATEX BASE.

PAINT ALL WALLS THROUGHOUT SHOWROOM WITH EGG-SHELL PAINT LATEX BASE.

PROVIDE 4" VINYL BASE THROUGHOUT OFFICE AREA.

LUMINARY  
RICE VILLAGE  
2556 UNIVERSITY BLVD, SUITE A-103  
HOUSTON, TX 77005

SET

CONSTRUCTION  
DOCUMENTS

ISSUE DATE

|   |          |                               |
|---|----------|-------------------------------|
| 1 | 02.09.23 | REVISED PER TENANT COMMENTS   |
| 2 | 02.21.23 | ISSUED FOR PRELIMINARY REVIEW |
| 3 | 03.09.23 | REVISED PER L.C. COMMENTS     |
| 4 | 03.15.23 | REVISED PER TENANT COMMENTS   |
|   |          |                               |
|   |          |                               |
|   |          |                               |
|   |          |                               |
|   |          |                               |
|   |          |                               |

SHEET

FINISH PLAN  
AND NOTES

A8.00



| DIFFUSER NECK SIZES     |                            |                                |                     |
|-------------------------|----------------------------|--------------------------------|---------------------|
| SUPPLY CFM<br>(MIN-MAX) | RETURN<br>CFM<br>(MIN-MAX) | RECTANGULAR NECK<br>SIZE (IN.) | ROUND NECK SIZE (ø) |
| 0 - 100                 | 0-90                       | 6x6                            | 6"                  |
| 101 - 200               | 91-200                     | 8x8                            | 8"                  |
| 201 - 400               | 201-300                    | 10x10                          | 10"                 |
| 401 - 600               | 301-450                    | 12x12                          | 12"                 |
| 601 - 900               | 451-700                    | 14x14                          | 14"                 |
| --                      | 701-2000                   | 22 x 22                        | --                  |

| AIR DEVICE SCHEDULE                                                                                                                                                                                                                              |              |        |          |         |                    |                    |                                                      |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------|----------|---------|--------------------|--------------------|------------------------------------------------------|-------|
| DESIGNATION                                                                                                                                                                                                                                      | MANUFACTURER | MODEL  | MATERIAL | SERVICE | FACE<br>DIMENSIONS | NECK<br>DIMENSIONS | DESCRIPTION                                          | NOTES |
| A                                                                                                                                                                                                                                                | TITUS        | S300FL | STEEL    | SUPPLY  | PER PLANS          | PER PLANS          | 3/4" BLADE SPACING, 0° DEFLECTION, WITH SCOOP DAMPER | 1,2   |
| B                                                                                                                                                                                                                                                | TITUS        | TMR    | STEEL    | SUPPLY  | PER PLANS          | PER PLANS          | ROUND LOUVERED DIFFUSER                              | 1,2   |
| C                                                                                                                                                                                                                                                | TITUS        | OMNI   | STEEL    | SUPPLY  | 24X24              | PER PLANS          | PLAQUE DIFFUSER                                      | 1,2,3 |
| D                                                                                                                                                                                                                                                | TITUS        | OMNI   | STEEL    | RETURN  | 24X24              | PER PLANS          | PLAQUE AIR GRILLE                                    | 2     |
| E                                                                                                                                                                                                                                                | TITUS        | 350    | STEEL    | RETURN  | PER PLANS          | PER PLANS          | 3/4" BLADE SPACING, 0° DEFLECTION                    | 2     |
| NOTES:<br>1. PROVIDE WITH BALANCING DAMPER AT BRANCH TAKEOFF.<br>2. COORDINATE BORDER TYPE AND COLOR WITH ARCHITECT PRIOR TO PURCHASE.<br>3. PROVIDE REMOTE CABLE DAMPER WHERE MANUAL BALANCING DAMPER WOULD BE LOCATED IN INACCESSIBLE CEILING. |              |        |          |         |                    |                    |                                                      |       |

| DUCT INSULATION SCHEDULE                                                                    |                       |     |       |     |           |        |         |             |        |                               |        |         |             |        |                                      |        |         |             |        |       |
|---------------------------------------------------------------------------------------------|-----------------------|-----|-------|-----|-----------|--------|---------|-------------|--------|-------------------------------|--------|---------|-------------|--------|--------------------------------------|--------|---------|-------------|--------|-------|
| DUCT INSULATION                                                                             |                       |     |       |     | CONCEALED |        |         |             |        | EXPOSED IN NON-SERVICE SPACES |        |         |             |        | EXPOSED IN SERVICE/ MECHANICAL ROOMS |        |         |             |        | NOTES |
|                                                                                             |                       |     |       |     | SUPPLY    | RETURN | EXHAUST | OUTSIDE AIR | JACKET | SUPPLY                        | RETURN | EXHAUST | OUTSIDE AIR | JACKET | SUPPLY                               | RETURN | EXHAUST | OUTSIDE AIR | JACKET |       |
| INDOOR<br>ROUND/ FLAT-OVAL/<br>RECTANGULAR DUCT                                             | MINERAL FIBER BLANKET | R-6 | 2     | 3/4 | X         | X      |         | X           | FSK    |                               |        |         |             |        | X                                    | X      |         | X           | FSK    | 1     |
|                                                                                             | MINERAL FIBER BOARD   | R-6 | 1-1/2 | 2   |           |        |         |             |        |                               |        |         |             |        |                                      |        |         |             |        |       |
| NOTES:                                                                                      |                       |     |       |     |           |        |         |             |        |                               |        |         |             |        |                                      |        |         |             |        |       |
| 1.       REFER TO SPECIFICATIONS FOR ADDITIONAL INSULATION REQUIREMENTS AND JACKET DETAILS. |                       |     |       |     |           |        |         |             |        |                               |        |         |             |        |                                      |        |         |             |        |       |

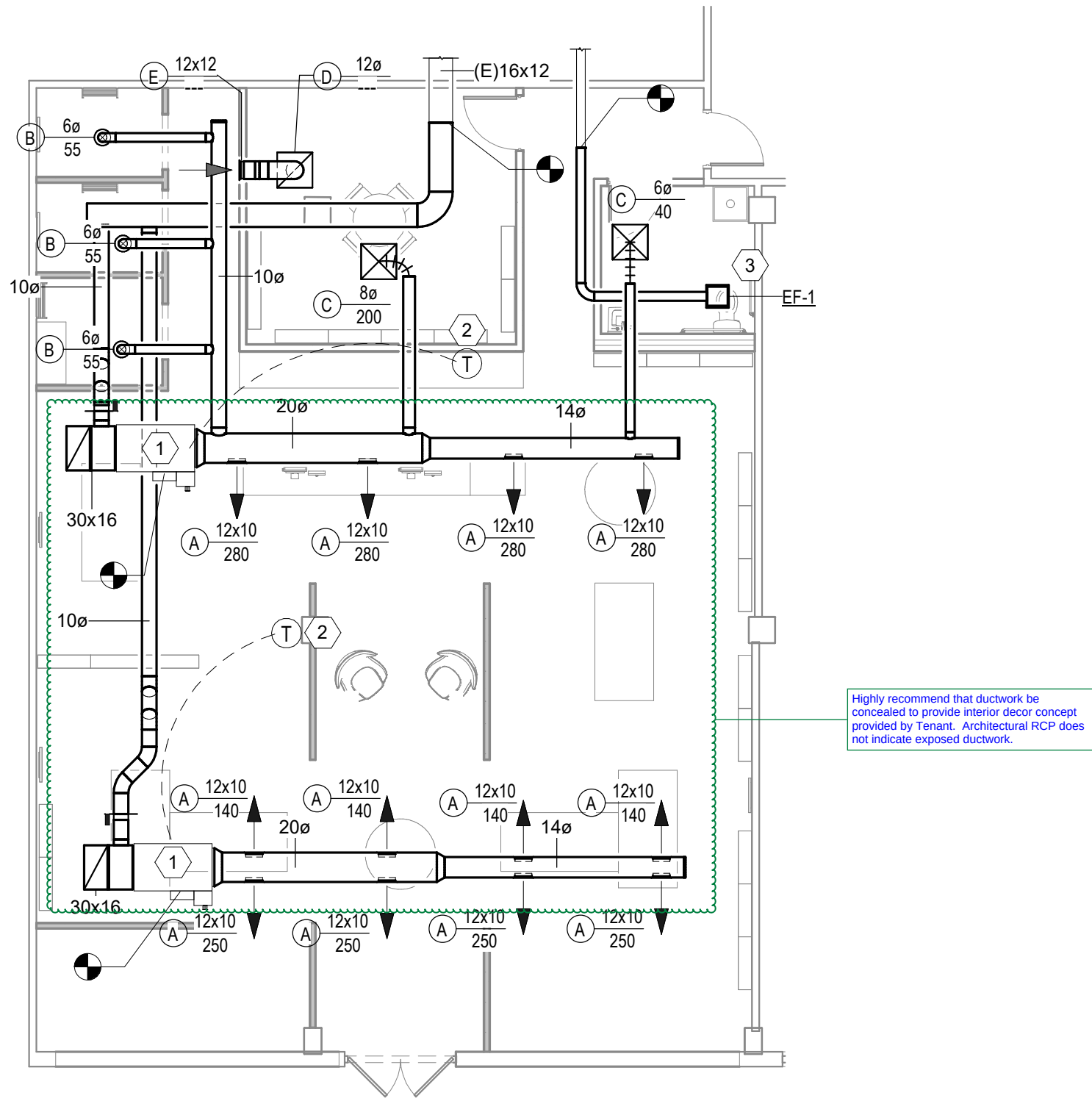
| FAN SCHEDULE                                                        |        |          |              |        |     |                     |       |      |                 |       |      |                 |                |                           |       |
|---------------------------------------------------------------------|--------|----------|--------------|--------|-----|---------------------|-------|------|-----------------|-------|------|-----------------|----------------|---------------------------|-------|
| EQUIPMENT                                                           |        | LOCATION | UNIT         |        | CFM | EXT. S.P. (IN.W.G.) | MOTOR |      | ELECTRICAL DATA |       |      | EMERGENCY POWER | VARIABLE SPEED | OPERATIONAL WEIGHT (LBS.) | NOTES |
| TYPE                                                                | NUMBER |          | MANUFACTURER | MODEL  |     |                     | RPM   | BHP  | VOLTAGE         | PHASE | FLA  |                 |                |                           |       |
| EF                                                                  | 1      | RESTROOM | GREENHECK    | SP-A90 | 50  | 0.3                 | 900   | 0.01 | 115             | 1     | 0.17 | NO              | NO             | 15                        | 1,2,3 |
| NOTES:                                                              |        |          |              |        |     |                     |       |      |                 |       |      |                 |                |                           |       |
| 1. PROVIDE WITH FACTORY MOUNTED ELECTRICAL DISCONNECT AND EC MOTOR. |        |          |              |        |     |                     |       |      |                 |       |      |                 |                |                           |       |
| 2. FAN SHALL BE CONTROLLED BY LIGHT SWITCH.                         |        |          |              |        |     |                     |       |      |                 |       |      |                 |                |                           |       |
| 3. PROVIDE WITH CEILING MOUNTING KIT.                               |        |          |              |        |     |                     |       |      |                 |       |      |                 |                |                           |       |

| OUTSIDE AIR CALCULATION                                       |        |                               |                      |              |                             |                    |                                     |                                           |                        |
|---------------------------------------------------------------|--------|-------------------------------|----------------------|--------------|-----------------------------|--------------------|-------------------------------------|-------------------------------------------|------------------------|
| ROOM TYPE                                                     | PEOPLE | PEOPLE<br>OUTSIDE<br>AIR RATE | OCCUPANT<br>AIR RATE | ROOM<br>AREA | AREA<br>OUTSIDE<br>AIR RATE | AREA AIR<br>RATE 2 | BREATHING<br>ZONE<br>OUTDOOR<br>AIR | ZONE AIR<br>DISTRIBUTION<br>EFFECTIVENESS | ZONE<br>OUTDOOR<br>AIR |
|                                                               | Pz     | Rp                            | Pz*Rp                | Ra           | Az                          | Rz*Az              | Vbz = RpPz +<br>RaAz                | Ez                                        | Voz = Vbz/<br>Ez       |
| STORAGE                                                       | 6      | 5                             | 30                   | 224          | 0.06                        | 13                 | 43                                  | 0.8                                       | 54.3                   |
| RETAIL SALES                                                  | 25     | 5                             | 125                  | 1716         | 0.12                        | 206                | 331                                 | 0.8                                       | 413.65                 |
| DRESSING ROOM                                                 | 3      | 5                             | 15                   | 119          | 0.12                        | 14                 | 29                                  | 0.8                                       | 36.6                   |
| TOTAL:                                                        |        |                               |                      |              |                             |                    | 404 CFM                             |                                           | 505 CFM                |
| BALANCE EXISTING OUTSIDE AIR DAMPER SERVING FLOOR TO: 510 CFM |        |                               |                      |              |                             |                    |                                     |                                           |                        |



| GENERAL NOTES |                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A             | CONTRACTOR SHALL VERIFY THERMOSTATS SERVE THEIR RESPECTIVE ZONES.                                                                                                                                                                                                                                                                       |
| B             | CONTRACTOR SHALL CONDUCT FIELD SURVEY TO VERIFY ALL EXISTING CONDITIONS ASSOCIATED WITH SCOPE OF WORK PRIOR TO SUBMITTING BIDS. ANY ADDITIONAL WORK REQUIRED ASSOCIATED WITH FAILURE TO PERFORM A FIELD SURVEY PRIOR TO SUBMITTING BIDS SHALL BE AT NO ADDITIONAL COST TO THE OWNER OR ENGINEER.                                        |
| C             | CONTRACTOR SHALL REPORT ANY DISCREPANCIES, OMISSIONS, OR INCONSISTENCIES ON THE DRAWINGS TO THE ENGINEER FOR VERIFICATION BEFORE STARTING CONSTRUCTION. OWNER AND ENGINEER ARE NOT RESPONSIBLE FOR ANY ERRORS IN CONSTRUCTION WHERE SUCH DISCREPANCIES OMISSIONS OR INCONSISTENCIES HAVE NOT BEEN PROPERLY REPORTED IN A TIMELY MANNER. |
| D             | CONTRACTOR SHALL PERFORM A PRE-BALANCE. MECHANICAL EQUIPMENT INCLUDING BUT NOT LIMITED TO AIR TERMINAL UNITS AND ASSOCIATED AIR QUANTITIES SHALL BE IDENTIFIED AND SUBMITTED TO MECHANICAL ENGINEER FOR REVIEW. PRE-BALANCE SHALL ALSO INCLUDE MIXING BOX CAPACITY.                                                                     |
| E             | RELOCATE ANY AND ALL MECHANICAL, ELECTRICAL, AND PLUMBING EQUIPMENT LOCATED ABOVE WALLS TO DECK OR AS INDICATED ON MECHANICAL, ELECTRICAL, OR PLUMBING DRAWINGS. ALL CODE REQUIRED CLEARANCES SHALL BE MET.                                                                                                                             |

| KEYNOTES |                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KEYNOTE  | DESCRIPTION                                                                                                                                                                                                                                                          |
| 1        | EXISTING AHU TO REMAIN. FIELD VERIFY EXACT LOCATION. BALANCE TO CFM OF ALL CONNECTED AIR DEVICES. EXISTING CONDENSATE PIPING TO REMAIN. BALANCE OUTSIDE AIR DAMPER TO 205 CFM.                                                                                       |
| 2        | FIELD VERIFY EXACT LOCATION OF EXISTING THERMOSTAT AND RELOCATE TO APPROXIMATE LOCATION SHOWN. EXTEND CONTROL WIRING AS NEEDED. CONFIRM THAT IT SERVES ZONE INDICATED. PROVIDE LOCK BOX ON THERMOSTAT. COORDINATE EXACT LOCATION AND MOUNTING HEIGHT WITH ARCHITECT. |
| 3        | NEW CEILING MOUNTED EXHAUST FAN SHALL BE LOCATED IN APPROXIMATE LOCATION SHOWN. TIE EXHAUST DUCT TO EXISTING EXHAUST DUCT. RE- SCHEDULES AND DETAILS.                                                                                                                |



**1 MECHANICAL PLAN**  
1/8" = 1'-0"

NAHRA DESIGN GROUP  
Landlord Review Comments - Architectural  
03/13/2023  
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PROJECT #  
23.1002

DESIGN TEAM  
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Houston, TX 77007  
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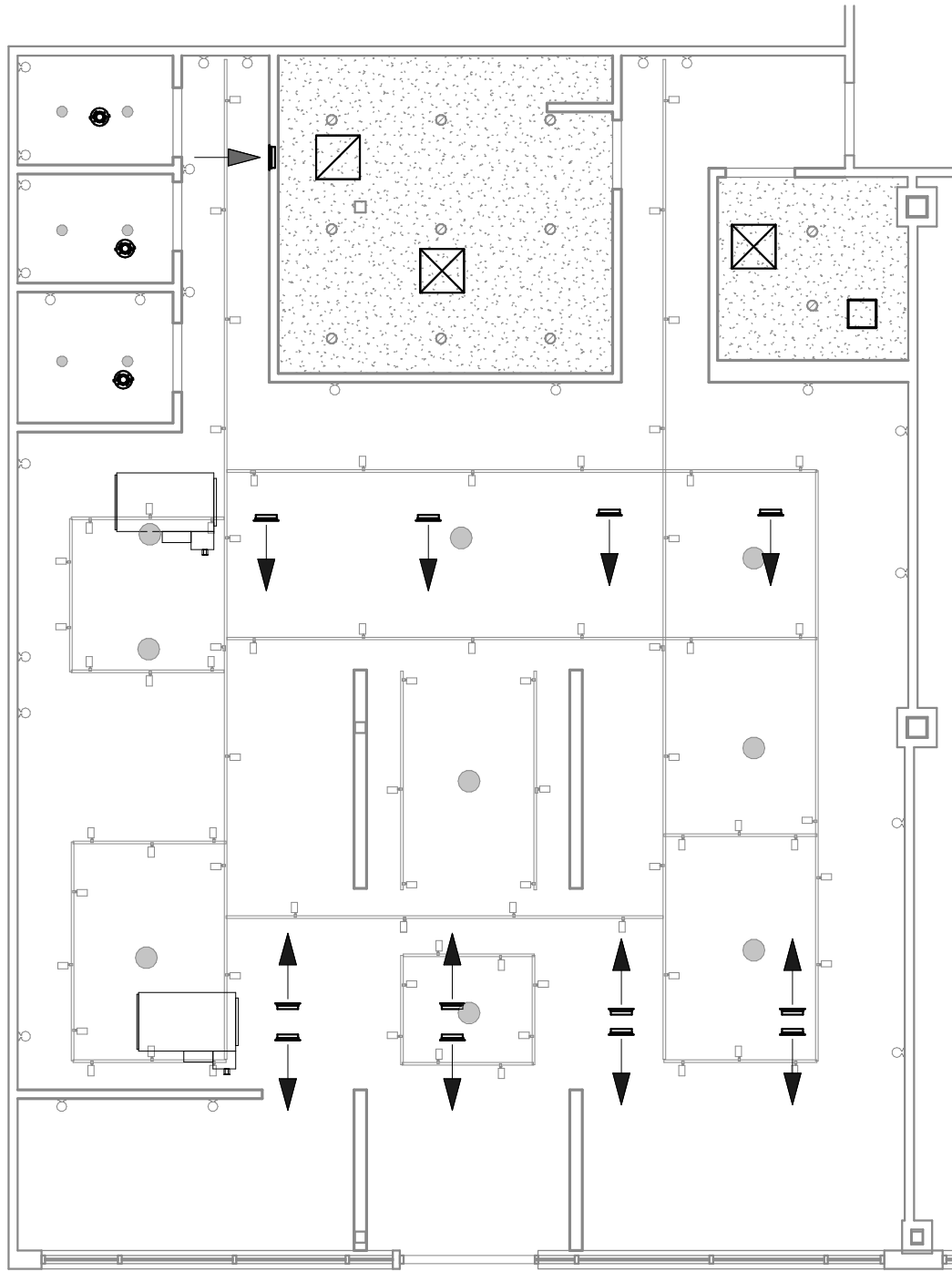
SEAL  
STATE OF TEXAS  
JONATHAN D. MILLER  
110124  
LICENSED PROFESSIONAL ENGINEER  
03/13/2023

PROJECT

LUMINARY  
RICE VILLAGE  
2556 UNIVERSITY BLVD, SUITE A-103  
HOUSTON, TEXAS 77005

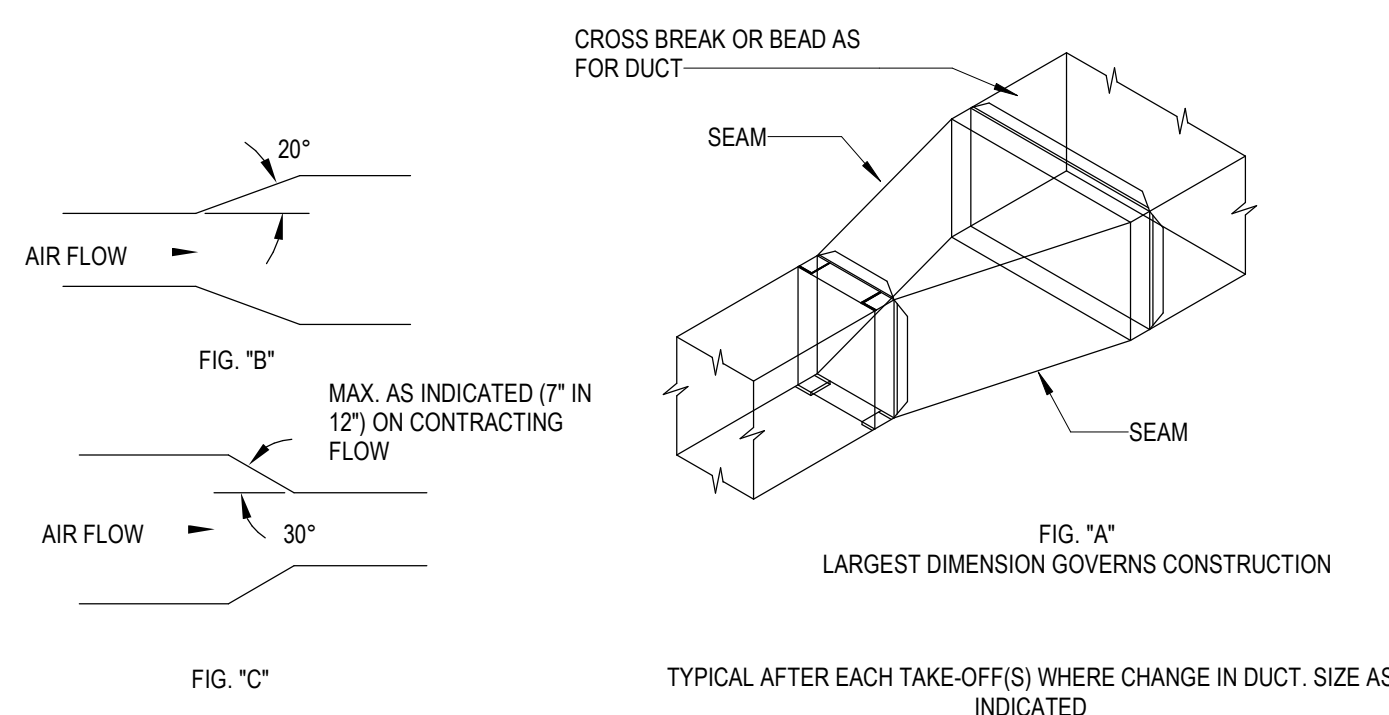
SET  
Construction Drawings  
ISSUE DATE 04/18/18  
03/13/23  
ISSUE FOR PERMIT  
SHEET  
MECHANICAL  
PLAN  
M2.01

| GENERAL NOTES |                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A             | CONTRACTOR SHALL VERIFY THERMOSTATS SERVE THEIR RESPECTIVE ZONES.                                                                                                                                                                                                                                                                       |
| B             | CONTRACTOR SHALL CONDUCT FIELD SURVEY TO VERIFY ALL EXISTING CONDITIONS ASSOCIATED WITH SCOPE OF WORK PRIOR TO SUBMITTING BIDS. ANY ADDITIONAL WORK REQUIRED ASSOCIATED WITH FAILURE TO PERFORM A FIELD SURVEY PRIOR TO SUBMITTING BIDS SHALL BE AT NO ADDITIONAL COST TO THE OWNER OR ENGINEER.                                        |
| C             | CONTRACTOR SHALL REPORT ANY DISCREPANCIES, OMISSIONS, OR INCONSISTENCIES ON THE DRAWINGS TO THE ENGINEER FOR VERIFICATION BEFORE STARTING CONSTRUCTION. OWNER AND ENGINEER ARE NOT RESPONSIBLE FOR ANY ERRORS IN CONSTRUCTION WHERE SUCH DISCREPANCIES OMISSIONS OR INCONSISTENCIES HAVE NOT BEEN PROPERLY REPORTED IN A TIMELY MANNER. |
| D             | CONTRACTOR SHALL PERFORM A PRE-BALANCE. MECHANICAL EQUIPMENT INCLUDING BUT NOT LIMITED TO AIR TERMINAL UNITS AND ASSOCIATED AIR QUANTITIES SHALL BE IDENTIFIED AND SUBMITTED TO MECHANICAL ENGINEER FOR REVIEW. PRE-BALANCE SHALL ALSO INCLUDE MIXING BOX CAPACITY.                                                                     |
| E             | RELOCATE ANY AND ALL MECHANICAL, ELECTRICAL, AND PLUMBING EQUIPMENT LOCATED ABOVE WALLS TO DECK OR AS INDICATED ON MECHANICAL, ELECTRICAL, OR PLUMBING DRAWINGS. ALL CODE REQUIRED CLEARANCES SHALL BE MET.                                                                                                                             |

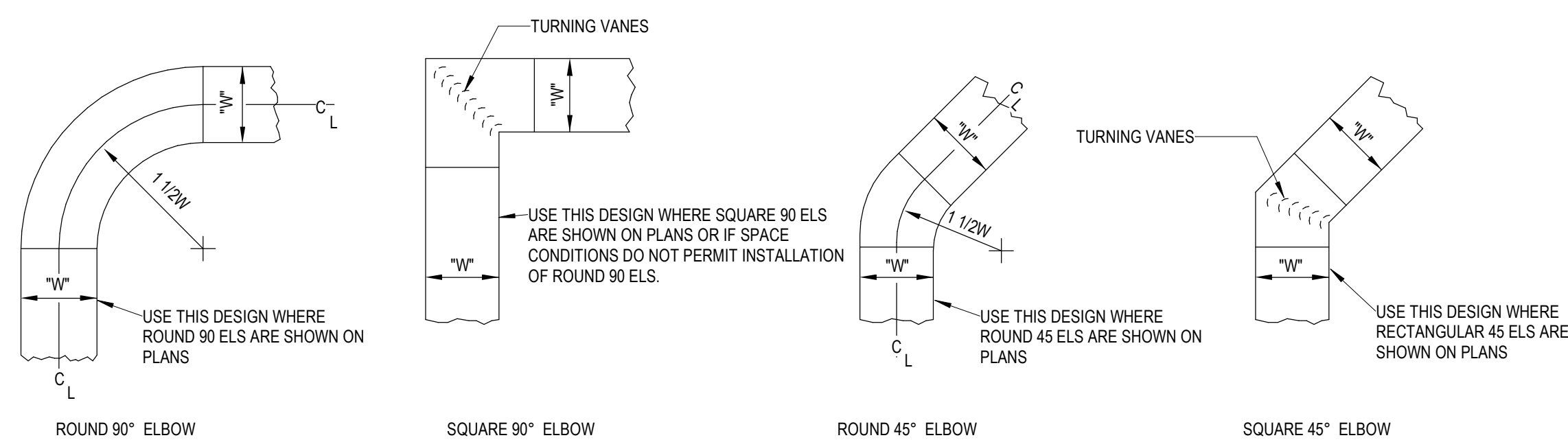


**1 MECHANICAL CEILING COORDINATION PLAN**  
1/8" = 1'-0"

|          |                  |
|----------|------------------|
| 03/13/23 | ISSUE FOR PERMIT |
|          |                  |
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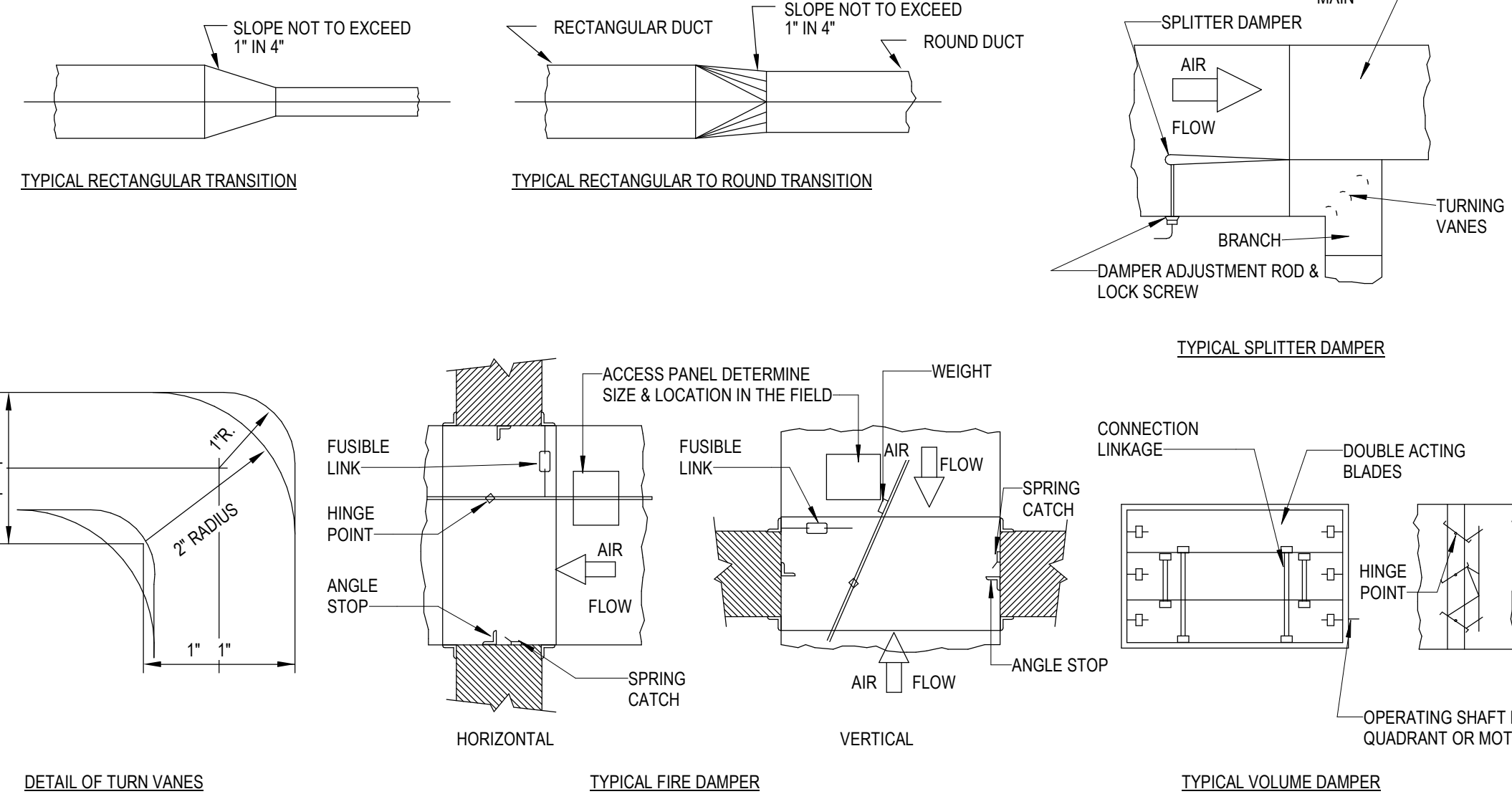


## 4 DUCT TRANSITION DETAIL

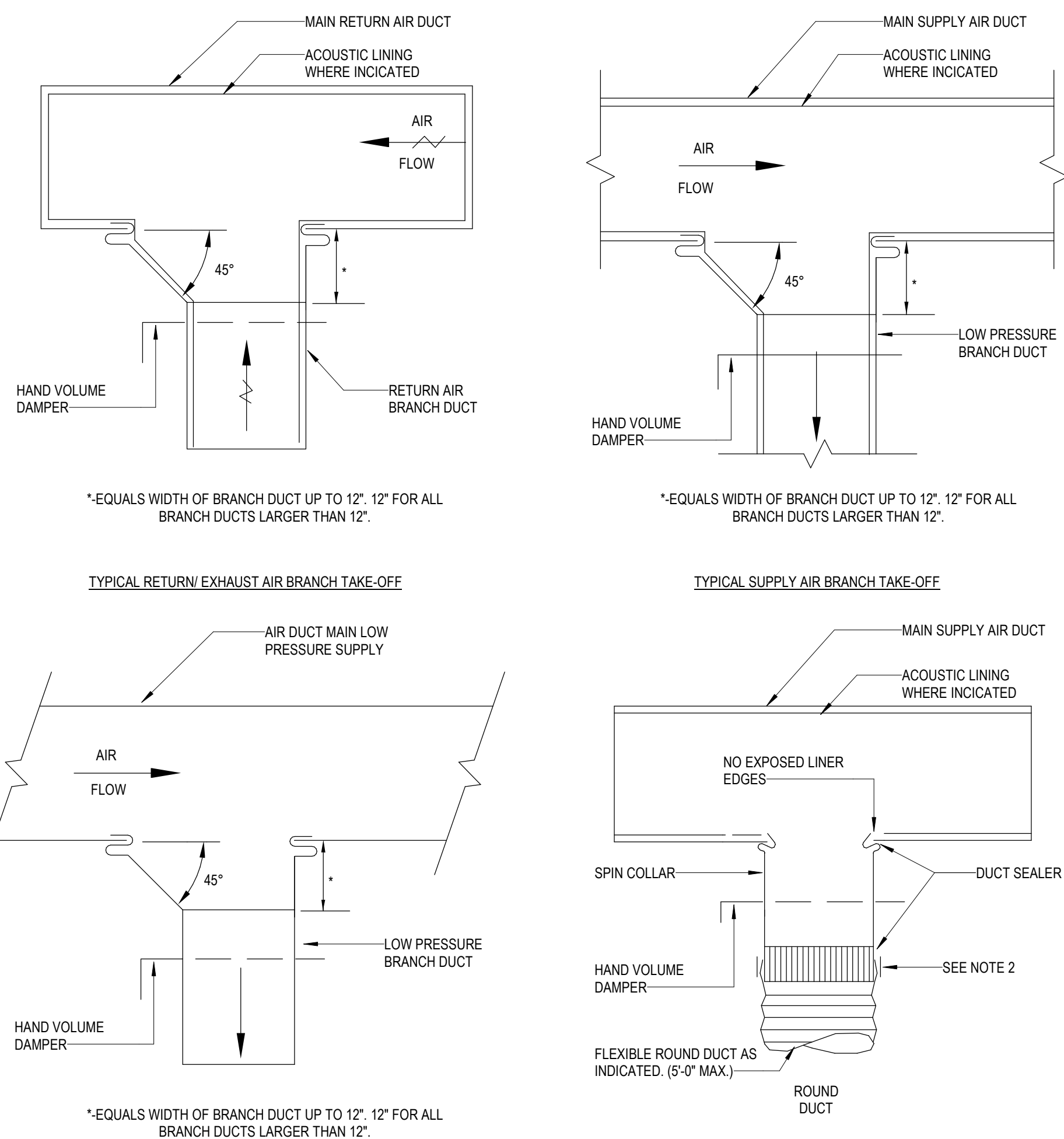


### CONSTRUCTION OF 90° ELBOWS

### CONSTRUCTION OF 45° ELBOWS



### 3 LOW VELOCITY DUCT DETAIL

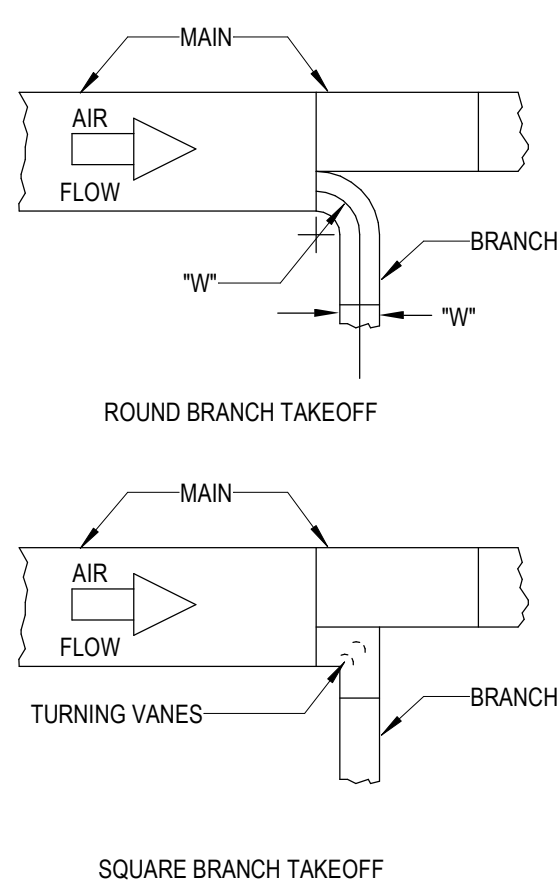


## NOTES:

1. PROVIDE BALANCING DAMPER AT BRANCH TAKEOFF.
2. SEAL FLEX DUCT TO METAL COLLAR WITH DUCT TAPE (NASHUA 398 OR EQUAL), INSTALL TIE-WRAP (ULINE JUMBO NATURAL NYLON CABLE TIES) OVER DUCT TAPE. INSTALL DUCT INSULATION OVER METAL COLLAR AND SECURE WITH TIE-WRAP.

TYPICAL LOW PRESSURE BRANCH DUCT TAKE-OFF

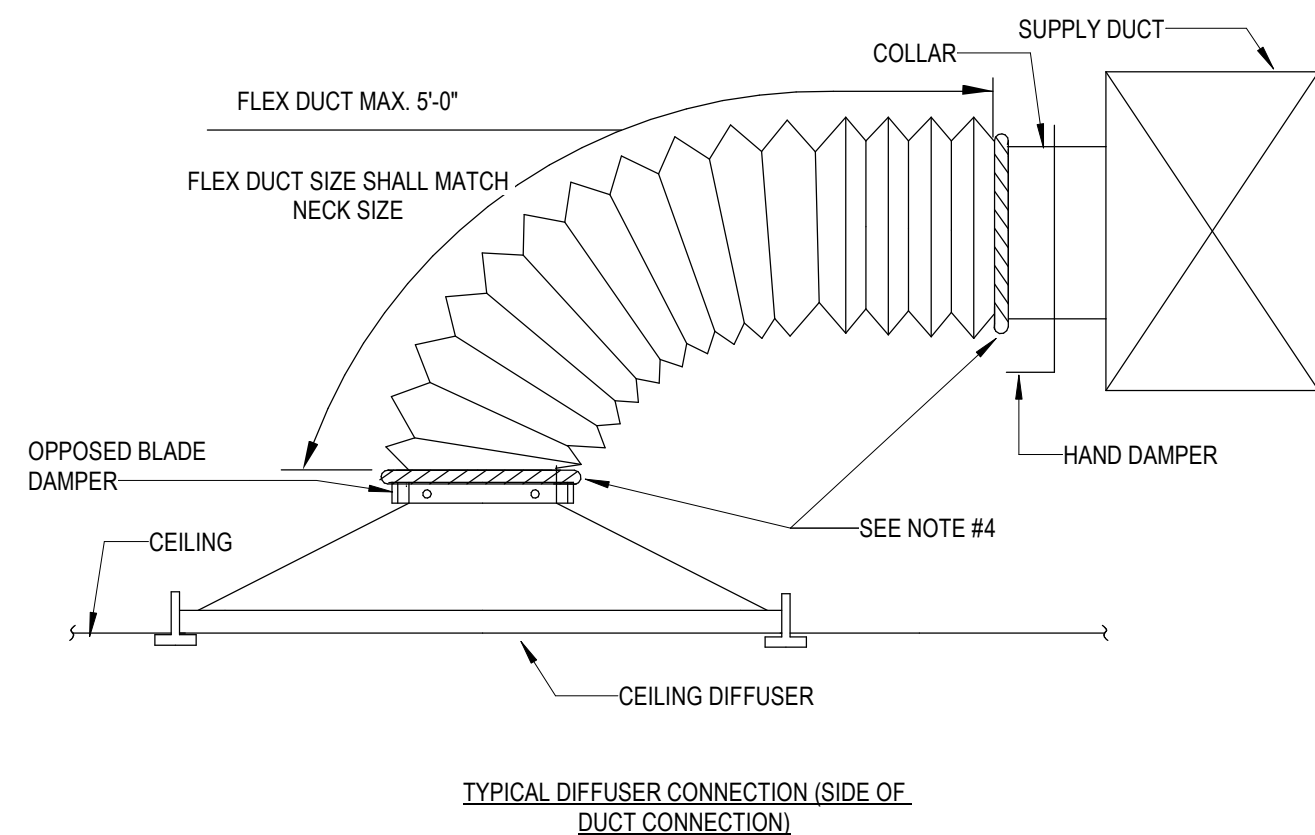
## 2 TYPICAL LOW PRESSURE BRANCH DUCT TAKE-OFF DETAIL



### CONSTRUCTION OF BRANCH TAKEOFFS FROM MAIN

INSTALLATION NOTES:

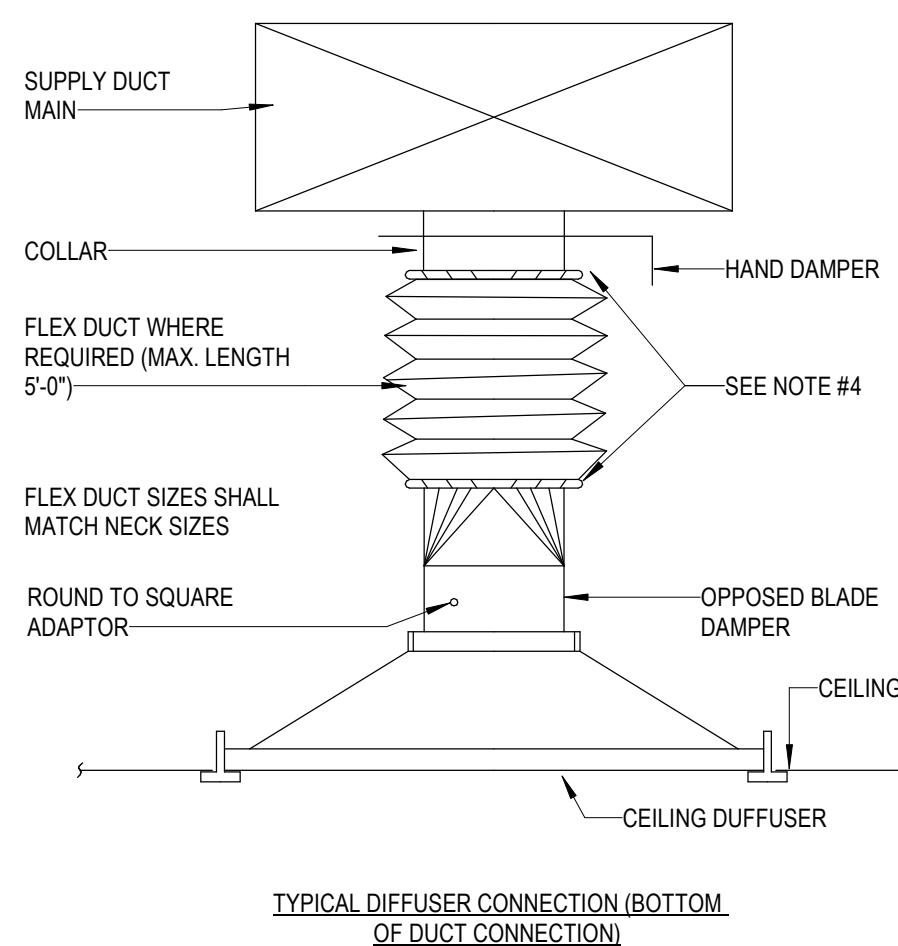
1. ALL DUCTS SHALL BE CONSTRUCTED AND ERECTED IN A NEAT AND WORKMANLIKE MANNER.
2. DUCTS SHALL BE CONSTRUCTED OF THE WEIGHTS, GAGES AND MATERIAL SHOWN IN THE SCHEDULE ON THESE DRAWINGS.
3. THE DIMENSION SHOWN FOR ALL DUCTS SHOWN IN PLAN GIVE THE WIDTH FIRST AND THEN THE HEIGHT.
4. DUCT RISERS SHOULD BE SUPPORTED BY ANGLES AT EVERY FLOOR.
5. AIR TURNING VANES SHALL BE INSTALLED IN ALL ABRUPT ELBOWS TO PREVENT TURBULENCE.
6. DUCTS SHALL BE SECURELY ATTACHED TO THE BUILDING CONSTRUCTION IN AN APPROVED MANNER.
7. DIVIDING TRANSITION PIECES SHALL BE MADE AS GRADUAL AS POSSIBLE.
8. INSTALL FIRE DAMPERS IN ACCORDANCE WITH UL 555.
9. ACCESS PANELS SHOULD BE PLACED BEFORE AND/OR AFTER EQUIPMENT INSTALLED IN THE DUCT.
10. DUCT AREA SHOULD NOT BE DECREASED MORE THAN 10% WHEN OBSTRUCTIONS CANNOT BE AVOIDED, AND THEN A STREAMLINE FITTING SHOULD BE USED.
11. JOINTS AND SEAMS OF SUPPLY DUCTS SHALL BE FASTENED SECURELY AND MADE AIR TIGHT.



NOTES:

1. PROVIDE REMOTE CABLE DAMPERS FOR ALL GYP / NON ACCESSIBLE CEILINGS.
2. PROVIDE BALANCING DAMPER AT BRANCH TAKEOFF.
3. PROVIDE WITH EQUAL TO TITUS "TITUS FLEX RIGHT" OR HARD RIGHT ELBOW TO ENSURE ADEQUATE AIRFLOW.
4. SEAL FLEX DUCT TO METAL COLLAR WITH DUCT TAPE (NASHUA 398 OR EQUAL), INSTALL TIE-WRAP (ULINE JUMBO NATURAL NYLON CABLE TIES) OVER DUCT TAPE. INSTALL DUCT INSULATION OVER METAL COLLAR AND SECURE WITH TIE-WRAP.

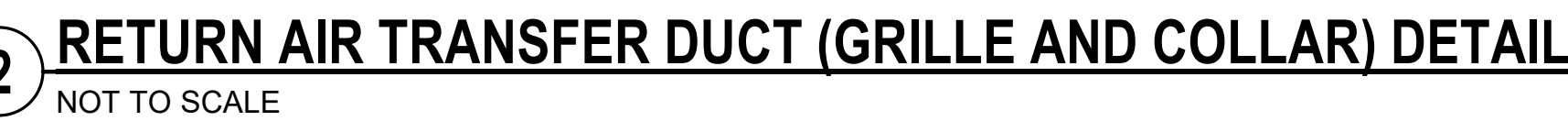
## 1 TYPICAL DIFFUSER CONNECTION DETAIL



TYPICAL DIFFUSER CONNECTION (BOTTOM  
OF DUCT CONNECTION)



SET  
Costruction Drawing



PLUMBING ABBREVIATIONS

|      |                                 |
|------|---------------------------------|
| (E)  | EXISTING                        |
| (N)  | NEW                             |
| AFF  | ABOVE FINISHED FLOOR            |
| AP   | ACCESS PANEL                    |
| BFF  | BELOW FINISHED FLOOR            |
| C    | CONDENSATE DRAIN                |
| CO   | CLEANOUT                        |
| CW   | DOMESTIC COLD WATER             |
| DF   | DRINKING FOUNTAIN               |
| DN   | DOWN                            |
| DR   | DRAIN                           |
| ELEV | ELEVATION                       |
| ET   | EXPANSION TANK                  |
| EW   | EYE WASH                        |
| EWB  | ELECTRIC WATER HEATER           |
| FCO  | FLOOR CLEANOUT                  |
| FD   | FLOOR DRAIN                     |
| FDC  | FIRE DEPARTMENT CONNECTION      |
| FP   | FIRE PUMP                       |
| FS   | FLOOR SINK                      |
| HB   | HOSE BIBB                       |
| HW   | HOT WATER                       |
| HWR  | HOT WATER RETURN                |
| IE   | INVERT ELEVATION                |
| IW   | INDIRECT WASTE                  |
| JP   | JOCKEY PUMP                     |
| L    | LAVATORY                        |
| MSB  | MOP SINK BASIN                  |
| NC   | NORMALLY CLOSED                 |
| NIC  | NOT IN CONTRACT                 |
| NO   | NORMALLY OPEN                   |
| OFD  | OVERFLOW DRAIN                  |
| OS&Y | OUTSIDE SCREW & YOKE GATE VALVE |
| POC  | POINT OF CONNECTION             |
| PRV  | PRESSURE REDUCING VALVE         |
| RD   | ROOF DRAIN                      |
| S    | SINK                            |
| SAN  | SANITARY WASTE                  |
| SH   | SHOWER                          |
| SOI  | SAND-OIL INTERCEPTOR            |
| SP   | SUMP PUMP                       |
| ST   | STORM PIPING                    |
| TD   | TRENCH DRAIN                    |
| TMV  | TEMPERATURE MIXING VALVE        |
| TP   | TRAP PRIMER                     |
| UR   | URINAL                          |
| V    | VENT                            |
| VTR  | VENT THROUGH ROOF               |
| WC   | WATER CLOSET                    |
| WCO  | WALL CLEANOUT                   |
| WH   | WALL HYDRANT                    |
| WHA  | WATER HAMMER ARRESTOR           |

PLUMBING LEGEND

|  |                                                                  |
|--|------------------------------------------------------------------|
|  | COMPRESSED AIR PIPING                                            |
|  | CONDENSATE DRAIN PIPING                                          |
|  | DOMESTIC COLD WATER PIPING                                       |
|  | DOMESTIC HOT WATER PIPING                                        |
|  | DOMESTIC HOT WATER RETURN PIPING                                 |
|  | NATURAL GAS PIPING                                               |
|  | OVERFLOW STORM PIPING                                            |
|  | SANITARY (ABOVE FLOOR)                                           |
|  | SANITARY (BELOW FLOOR)                                           |
|  | SPRINKLER PIPING                                                 |
|  | STORM DRAIN PIPING                                               |
|  | VENT PIPING                                                      |
|  | PIPING TO BE DEMOLISHED                                          |
|  | CLEANOUT/PLUG                                                    |
|  | PIPE DOWN                                                        |
|  | PIPE UP                                                          |
|  | PIPE CAP                                                         |
|  | CHANGE IN PIPE ELEVATION                                         |
|  | INSULATED AND HEAT TRACED PIPING                                 |
|  | ACCESS PANEL FOR TRAP PRIMER                                     |
|  | DOUBLE CHECK VALVE ASSEMBLY                                      |
|  | REDUCED PRESSURE BACKFLOW PREVENTER ASSEMBLY                     |
|  | BOTTOM PIPE CONNECTION                                           |
|  | CLEANOUT (TWO-WAY) (PROVIDE CONCRETE PAD OUTSIDE 18" X 24" X 4") |
|  | FLOOR CLEANOUT/GRADE CLEANOUT                                    |
|  | P-TRAP                                                           |
|  | TOP PIPE CONNECTION                                              |
|  | VALVE IN VERTICAL                                                |
|  | WALL HYDRANT                                                     |
|  | ANGLE RELIEF VALVE                                               |
|  | AUTOMATIC AIR VENT                                               |
|  | AUTOMATIC CONTROL VALVE                                          |
|  | AUTOMATIC CONTROL VALVE (3-WAY)                                  |
|  | BUTTERFLY VALVE MANUAL                                           |
|  | BALL VALVE                                                       |
|  | CALIBRATED BALANCING VALVE                                       |
|  | CHECK VALVE                                                      |
|  | CHECK VALVE WITH A.B.D.                                          |
|  | FLEXIBLE CONNECTION                                              |
|  | GAS COCK VALVE                                                   |
|  | GLOBE VALVE                                                      |
|  | OS&Y (OUTSIDE SCREW & YOKE) VALVE                                |
|  | PRESSURE REDUCING VALVE                                          |
|  | PRESSURE GAUGE AND COCK                                          |
|  | RELIEF SAFETY VALVE                                              |
|  | SOLENOID VALVE                                                   |
|  | THERMOMETER                                                      |
|  | TEMPERATURE MIXING VALVE                                         |
|  | WATER HAMMER ARRESTOR                                            |
|  | VACUUM BREAKER                                                   |
|  | VALVE WITH TAMPER SWITCH                                         |
|  | Y-STRAINER WITH BLOW OFF VALVE                                   |
|  | HOSE BIBB                                                        |
|  | METER                                                            |
|  | PUMP                                                             |
|  | FLOW METER                                                       |
|  | VENTURI METER                                                    |
|  | POINT OF CONNECTION (NEW TO EXISTING)                            |
|  | EQUIPMENT DESIGNATION                                            |

PLUMBING GENERAL NOTES

|    |                                                                                                                                                                                                                                                             |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | ALL SYSTEMS WILL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE LOCAL AND STATE CODES AND NATIONAL STANDARDS.                                                                                                                                               |
| 2  | NO PVC PIPING SHALL BE INSTALLED IN A RETURN AIR PLENUM.                                                                                                                                                                                                    |
| 3  | CONTRACTOR SHALL PROVIDE A FULLY FUNCTIONAL, CODE COMPLIANT SYSTEM.                                                                                                                                                                                         |
| 4  | CONTRACTOR SHALL TEST ALL SYSTEMS FOR FUNCTIONALITY BOTH BEFORE WORK IS STARTED AND ALSO AFTER CONTRACTED WORK IS FINISHED. REPORT ANY ISSUES TO THE ENGINEER AND TO THE BUILDING MANAGER.                                                                  |
| 5  | CONTRACTOR SHALL FIELD VERIFY ALL EXISTING CONDITIONS TO FAMILIARIZE THEMSELVES WITH THE PROJECT BEFORE BIDDING. ALL ISSUES SHALL BE TAKEN INTO ACCOUNT WITH THEIR BID. CONTRACTOR AGREES TO ACCEPT ALL EXISTING CONDITIONS AS PART OF THEIR BID.           |
| 6  | INFORMATION ON THE DRAWINGS IS DIAGRAMMATIC IN NATURE. NOT ALL OFFSETS ARE SHOWN FOR CLARITY PURPOSES. CONTRACTOR TO PROVIDE ALL OFFSETS AND FITTINGS TO MAKE A FULLY FUNCTIONAL SYSTEM.                                                                    |
| 7  | EXISTING SYSTEMS MAY NOT MATCH WHAT IS SHOWN ON THE DRAWINGS. CONTRACTOR TO ADJUST CONSTRUCTION AS REQUIRED TO COMPLETE WORK.                                                                                                                               |
| 8  | CONTRACTOR SHALL PROTECT ALL EXISTING BUILDING STRUCTURES, FLOORING, PIPING, FIXTURES, ETC. CONTRACTOR IS RESPONSIBLE FOR ANY DAMAGE THEY CAUSE TO THE SITE, PLUMBING FIXTURES OR PIPING, ETC.                                                              |
| 10 | COORDINATE ALL WORK WITH BUILDING OWNER'S OPERATING SCHEDULE.                                                                                                                                                                                               |
| 11 | ALL SYSTEMS SHALL REMAIN OPERATIONAL DURING CONSTRUCTION AS REQUIRED BY BUILDING OWNER.                                                                                                                                                                     |
| 12 | PROVIDE ANY AND ALL DEMOLITION WORK REQUIRED TO INSTALL NEW EQUIPMENT. RETURN SURROUNDING AREA TO ORIGINAL CONDITION AFTER INSTALLATION IS COMPLETE. MAINTAIN PLUMBING SYSTEMS IN ADJACENT SPACES.                                                          |
| 13 | CONTRACTOR TO COORDINATE ALL PLUMBING FIXTURE COMPATIBILITY WHEN MATCHING FIXTURES, FLUSH VALVES, FAUCETS, DRAINS, ETC.                                                                                                                                     |
| 14 | CONTRACTOR TO VERIFY ARCHITECTURAL REQUIREMENTS AND COORDINATION WITH SPECIFIED PLUMBING FIXTURES. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ARCHITECTS AND ENGINEER'S ATTENTION.                                                                           |
| 15 | CONTRACTOR SHALL PROVIDE ALL MATERIAL, EQUIPMENT, ACCESSORIES, AND LABOR REQUIRED FOR INSTALLATION OF A COMPLETE AND OPERABLE SYSTEMS.                                                                                                                      |
| 16 | CONTRACTOR SHALL COORDINATE ALL PIPE ROUTING AND EQUIPMENT LOCATIONS WITH OTHER TRADES AND EXISTING CONDITIONS. CONTRACTOR SHALL MODIFY EXISTING DESIGN CONDITIONS AS REQUIRED TO PROVIDE A FUNCTIONAL, CODE COMPLIANT INSTALLATION.                        |
| 17 | CONTRACTOR TO COORDINATE ALL WORK WITH ARCHITECTS AND ENGINEER'S PHASING PLANS. MAKE ALL NECESSARY ALLOCATIONS TO ACCOMMODATE PHASING AND PHASED CONSTRUCTION.                                                                                              |
| 18 | INSTALL ALL EQUIPMENT PER MANUFACTURER'S INSTRUCTIONS AND RECOMMENDATIONS. MAINTAIN ALL REQUIRED CLEARANCES FOR MAINTENANCE AND ACCESS.                                                                                                                     |
| 19 | ALL PLUMBING FIXTURES SHALL MEET TASADA REQUIREMENTS. FIXTURES SHALL BE INSTALLED TO MEET TASADA HEIGHT AND CLEARANCE REQUIREMENTS. COORDINATE WITH ARCHITECTURAL DRAWING DIMENSIONS.                                                                       |
| 20 | ROUTE ALL PIPING AS HIGH AS POSSIBLE AND TIGHT TO STRUCTURE. INSTALL PERPENDICULAR TO WALLS AND COLUMNS.                                                                                                                                                    |
| 21 | ALL EQUIPMENT SHALL BE INSTALLED ON 4" CONCRETE HOUSEKEEPING PADS.                                                                                                                                                                                          |
| 22 | PROVIDE DIELECTRIC FITTINGS OR UNIONS BETWEEN PIPING OF DISSIMILAR METALS.                                                                                                                                                                                  |
| 23 | PROVIDE ACCESS PANELS AS REQUIRED TO SERVICE ALL VALVES AND EQUIPMENT. COORDINATE LOCATION OF ACCESS PANELS WITH ARCHITECTURAL CEILING AND WALL ELEVATION DRAWINGS.                                                                                         |
| 24 | DO NOT ROUTE PIPING THROUGH ELECTRICAL ROOMS/CLOSETS OR IT/DATA ROOMS. IF THE CONSTRUCTION SITUATION MAKES THIS UNAVOIDABLE, PROVIDE SECONDARY LEAK PROTECTION SLOPED DRAIN PAN UNDER PIPE. ROUTE DRAIN PAN LINE TO NEAREST FLOOR DRAIN OR SINK TAIL PIECE. |
| 25 | PROVIDE SHUT-OFF VALVES ON ALL PLUMBING BRANCH PIPING, EQUIPMENT, TOILET ROOM GROUPS, AND INDIVIDUAL FLOORS.                                                                                                                                                |
| 26 | SLOPE PIPING AS REQUIRED TO MEET CODE REQUIREMENTS, AVOID LOW POINTS, AND ESTABLISH HIGH POINTS FOR AIR REMOVAL.                                                                                                                                            |
| 27 | PROVIDE TRAP GUARD ON ALL FLOOR DRAINS, AIR DRAINS AND FLOOR SINKS OR OTHER APPROVED TRAP SEALING DEVICE.                                                                                                                                                   |
| 28 | ALL WORK SHALL BE COMPLETED TO THE APPROVAL OF THE BUILDING ENGINEER.                                                                                                                                                                                       |
| 29 | CONTRACTOR TO ACCOMMODATE ALL CLEARANCE AND DIMENSIONAL DIFFERENCES OF EQUIPMENT ACTUALLY PURCHASED FOR INSTALLATION.                                                                                                                                       |
| 30 | CONTRACTOR SHALL X-RAY SLAB AS REQUIRED TO VERIFY EXACT LOCATION OF EXISTING PIPING IN SLAB.                                                                                                                                                                |

AUTOMATIC SPRINKLER NOTES

|   |                                                                                                                                                                                                                                                                                                                                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | COMPLY WITH ALL NFPA CHAPTERS. EXACT NUMBERS OF HEADS REQUIRED SHALL BE AS DETERMINED BY THE CONTRACTOR. HOWEVER, SPECIFIC GUIDELINES AS REQUIRED BY OWNER, AND AS SPECIFIED PER SECTION IN THE SPECIFICATIONS SHALL BE MET.                                                                                                                             |
| 2 | CONTRACTOR SHALL REWORK EXISTING SPRINKLER HEADS AND PIPING IN REMODELED AREA. SYSTEM SHALL BE DESIGNED IN ACCORDANCE WITH NFPA 13 AND LOCAL CODES AND REQUIREMENTS. CONTRACTOR SHALL MATCH EXISTING PIPING MATERIALS AND SPRINKLER HEADS.                                                                                                               |
| 3 | EXISTING FIRE PROTECTION SYSTEMS SHALL REMAIN OPERATIONAL THROUGHOUT DURATION OF THE PROJECT.                                                                                                                                                                                                                                                            |
| 4 | DESIGN OF THE SPRINKLER SYSTEM SHALL BE UNDER THE DIRECT SUPERVISION OF A FIRE PROTECTION ENGINEER LICENSED IN THE STATE OF TEXAS. ALL DESIGN SUBMITTAL DOCUMENTS SHALL BEAR THE ENGINEER'S SIGNED AND DATED REGISTRATION NUMBER.                                                                                                                        |
| 5 | THE SYSTEM SHALL BE INSTALLED BY A FIRM HAVING A MINIMUM OF 5 YEARS EXPERIENCE REGULARLY ENGAGED IN THE DESIGN AND INSTALLATION OF AUTOMATIC FIRE PROTECTION SYSTEMS IN ACCORDANCE WITH REQUIREMENTS OF NFPA AND THE STATE OF TEXAS FIRE MARSHALL'S OFFICE. EVIDENCE TO SUPPORT THE ABOVE REQUIREMENTS SHALL BE SUBMITTED WITH SHOP DRAWINGS TO THE AHJ. |
| 6 | FURNISH, INSTALL, AND COMPLETE A 100% HYDRAULICALLY CALCULATED, AUTOMATIC WET PIPE SPRINKLER SYSTEM SERVING THE ENTIRE SPACE. THE SYSTEM SHALL BE INSTALLED TO MEET OWNERS INSURING GUIDELINES, LATEST ADOPTED EDITION OF NFPA 13, AND STATE AND LOCAL CODE REQUIREMENTS.                                                                                |
| 7 | FIRE PROTECTION CONTRACTOR SHALL MODIFY ALL EXISTING FIRE PROTECTION AND SPRINKLER PIPES AS REQUIRED TO MEET THE MAINTENANCE AND REMOVAL CLEARANCES OF ALL EXISTING MECHANICAL, ELECTRICAL AND PLUMBING EQUIPMENT.                                                                                                                                       |

PLUMBING SHEET LIST

| SHEET NUMBER | SHEET NAME               |
|--------------|--------------------------|
| P0.01        | PLUMBING COVER SHEET     |
| P0.02        | PLUMBING SCHEDULES       |
| P0.03        | PLUMBING SPECIFICATIONS  |
| P2.00        | PLUMBING UNDERFLOOR PLAN |
| P2.01        | PLUMBING FLOOR PLAN      |
| P5.01        | PLUMBING RISERS          |
| P6.01        | PLUMBING DETAILS         |

SHOCK ARRESTOR SCHEDULE

|   | MANUFACTURER & MODEL | FIXTURE UNITS | SIZE       | MODEL NUMBER |
|---|----------------------|---------------|------------|--------------|
| A | PRECISION PLUMBING   | 1-11          | 1/2" NPT   | SC-500       |
| B | PRECISION PLUMBING   | 12-32         | 3/4" NPT   | SC-750       |
| C | PRECISION PLUMBING   | 33-60         | 1" NPT     | SC-1000      |
| D | PRECISION PLUMBING   | 61-113        | 1 1/4" NPT | SC-1250      |
| E | PRECISION PLUMBING   | 114-154       | 1 1/2" NPT | SC-1500      |
| F | PRECISION PLUMBING   | 155-330       | 2" NPT     | SC-2000      |

NOTE:  
PROVIDE SHOCK ARRESTORS AT ENDS OF DCW AND DHW PIPING RUNS; AT ALL QUICK-CLOSING FIXTURES SUCH AS SHOWERS, FLUSHVALVES, SOLENOID VALVES, SINGLE-HANDED FAUCETS, AND SENSOR OPERATED FAUCETS; AND, FOR ALL GROUPS OF FIXTURES. SHOCK ARRESTORS SHALL BE PLACED AS CLOSE AS POSSIBLE TO THE LAST FIXTURE ON EACH PIPING RUN.

SPRINKLER FLOW DENSITIES

| HAZARD             | AREA (SF) | GPM/SF |
|--------------------|-----------|--------|
| LIGHT HAZARD       | 1500      | 0.10   |
| ORDINARY HAZARD I  | 1500      | 0.15   |
| ORDINARY HAZARD II | 1500      | 0.20   |

PIPING MATERIAL SCHEDULE

| SYSTEM                          | BELOW GRADE     | ABOVE GRADE        |
|---------------------------------|-----------------|--------------------|
| DOMESTIC COLD/ HOT WATER PIPING | -               | TYPE "L" COPPER    |
| SANITARY/ VENT PIPING           | SCHEDULE 40 PVC | NO - HUB CAST IRON |

CODE SUMMARY

|    |                                                                                                                                                                                                                                                                                                                                                                                                 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. | APPLICABLE CODES INCLUDE BUT ARE NOT LIMITED TO:<br>1. CITY OF HOUSTON PLUMBING CODE: 2015 UPC, WITH AMENDMENTS<br>2. CITY OF HOUSTON BUILDING CODE: 2015 IBC, WITH AMENDMENTS<br>3. CITY OF HOUSTON COMMERCIAL ENERGY CONSERVATION CODE: 2015 IECC, WITH AMENDMENTS<br>4. CITY OF HOUSTON FIRE CODE: 2015 IFC, WITH AMENDMENTS<br>5. CITY OF HOUSTON FUEL GAS CODE: 2012 IFGC, WITH AMENDMENTS |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

PROJECT #

23.1002

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SEAL



PROJECT

LUMINARY  
RICE VILLAGE  
2556 UNIVERSITY BLVD, SUITE A-103  
HOUSTON, TEXAS 77005

SHEET

Costruction Drawings

ISSUE DATE 03/15/18

|          |                  |  |  |  |  |  |  |  |  |
|----------|------------------|--|--|--|--|--|--|--|--|
| 03/13/23 | ISSUE FOR PERMIT |  |  |  |  |  |  |  |  |
|----------|------------------|--|--|--|--|--|--|--|--|

SHEET

PLUMBING  
COVER SHEET

P0.01

INFINITY  
MEP+S CONSULTANTS  
TBPE Registration Number 18865  
10260 Westheimer Rd., Suite 400  
Houston, Texas 77042  
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| PUMP SCHEDULE                                                                                                                  |        |           |                  |              |         |              |     |            |      |      |                 |       |       |                           |
|--------------------------------------------------------------------------------------------------------------------------------|--------|-----------|------------------|--------------|---------|--------------|-----|------------|------|------|-----------------|-------|-------|---------------------------|
| TYPE                                                                                                                           | NUMBER | SERVICE   | LOCATION         | UNIT         |         |              | GPM | HEAD (FT.) | RPM  | HP   | ELECTRICAL DATA |       |       | NOTES                     |
|                                                                                                                                |        |           |                  | MANUFACTURER | MODEL   | TYPE         |     |            |      |      | VOLTAGE         | PHASE | HERTZ | OPERATIONAL WEIGHT (LBS.) |
| CP                                                                                                                             | 1      | HOT WATER | RESTROOM CEILING | TACO         | 008-BC6 | INLINE CIRC. | 1   | 10         | 3250 | 1/25 | 120             | 1     | 60    | 8                         |
| NOTES:                                                                                                                         |        |           |                  |              |         |              |     |            |      |      |                 |       |       |                           |
| 1. TIME CLOCK - TORK MODEL EWZ120. TIME CLOCK SHALL BE PROVIDED BY PLUMBING CONTRACTOR AND INSTALLED BY ELECTRICAL CONTRACTOR. |        |           |                  |              |         |              |     |            |      |      |                 |       |       |                           |

| ELECTRIC WATER HEATER SCHEDULE                     |        |                    |                  |              |             |                 |                    |                          |                 |         |       |    |                           |       |
|----------------------------------------------------|--------|--------------------|------------------|--------------|-------------|-----------------|--------------------|--------------------------|-----------------|---------|-------|----|---------------------------|-------|
| EQUIPMENT                                          |        | SERVICE            | LOCATION         | UNIT TYPE    |             | STORAGE GALLONS | RECOVERY GPH@100°F | TEMPERATURE SETTING (°F) | ELECTRICAL DATA |         |       |    | OPERATIONAL WEIGHT (LBS.) | NOTES |
| TYPE                                               | NUMBER |                    |                  | MANUFACTURER | MODEL       |                 |                    |                          | KW              | VOLTAGE | PHASE | HZ |                           |       |
| EWH                                                | 1      | DOMESTIC HOT WATER | RESTROOM CEILING | STATES       | ES6-20-SOMT | 19.9            | 10.3               | 140                      | 2.5             | 277     | 1     | 60 | 228                       | 1     |
| NOTES:                                             |        |                    |                  |              |             |                 |                    |                          |                 |         |       |    |                           |       |
| 1. PROVIDE WITH THERM-X-TROL ST-5C EXPANSION TANK. |        |                    |                  |              |             |                 |                    |                          |                 |         |       |    |                           |       |

| PLUMBING FIXTURE CONNECTION SCHEDULE               |      |      |                          |    |       |
|----------------------------------------------------|------|------|--------------------------|----|-------|
| FIXTURE                                            | CW   | HW   | MIN.TRAP SIZE & TRAP ARM | V  | NOTES |
| WATER CLOSET (FLUSH TANK)                          | 3/4" | -    | 4"                       | 2" | 1,2   |
| LAV                                                | 3/4" | 3/4" | 1-1/2"                   | 2" | 2     |
| MOP SERVICE BASIN                                  | 3/4" | 3/4" | 3"                       | 2" | 2     |
| NOTES:                                             |      |      |                          |    |       |
| 1. 2" MINIMUM WASTE AND 2" VENT PIPE BELOW GRADE.  |      |      |                          |    |       |
| 2. TRAP SIZE TO BE CONSISTENT WITH FIXTURE OUTLET. |      |      |                          |    |       |

| PLUMBING FIXTURE SCHEDULE |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |  |
|---------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--|
| SYMBOL                    | MANUFACTURER & MODEL NUMBER       | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | NOTES         |  |
| WC-1                      | AMERICAN STANDARD MODEL 215AA-104 | VITREOUS CHINA FLUSH TANK WATER CLOSET 1.28 GPF WITH OPEN FRONT LESS COVER SEAT. SUPPLY WITH STOP. MCGUIRE # 216BLK LOOSE KEY SUPPLY AND STOP KIT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |  |
| LV-1                      | AMERICAN STANDARD MODEL 0355-012  | ADA, VITREOUS CHINA WALL MOUNTED LAVATORY, 3 DECK HOLE, AND RECTANGULAR BASIN. FAUCETS: CHICAGO FAUCET # 802-VE2805-317ABCP- MANUAL FAUCET, 0.5 GPM AERATOR AND 4" VANDAL PROOF WRIST BLADE HANDLES. SUPPLY WITH STOP: MCGUIRE # BV2165CC LOOSE KEY SUPPLY AND STOP KIT. TRAP: MCGUIRE # 8090 CAST P-TRAP WITH CLEANOUT. INSULATION: TRUEBRO # LAV GUARD ON TRAP, DRAIN, AND BOTH HOT AND COLD SUPPLY. REFER TO ARCHITECTURAL FOR MOUNTING HEIGHT & LOCATION.                                                                                                                                                                       | PROVIDE TMV-1 |  |
| MS-1                      | FIAT TERRAZO TSB3000 SQUARE       | MOP SINK 24 X 24 X 10 INCH HIGH. RECEPTOR COMPOSED OF PEARL GREY MARBLE CHIPS AND WHITE PORTLAND CEMENT GROUND SMOOTH. GROUTED AND SEALED TO RESIST STAINS. FLOOR MOUNTED, WITH 1-1/4 INCH WIDE SHOULDERS, VINYL BUMPER GUARD, STAINLESS STEEL DOME STRAINER. FAUCET: FIAT 830-AA CHROME PLATED WITH VACUUM BREAKER, INTEGRAL STOPS. ADJUSTABLE WALL BRACE, PAIL HOOK, AND 3/4" HOSE THREAD ON SPOUT. BODY INLETS 8" CENTER TO CENTER. PROVIDE 3" QUICK DRAIN CONNECTOR, MODEL 832-AA 30" LONG HOSE AND BRACKET MODEL 889-CC MOP HANGER, MODEL 1453-BB STAINLESS STEEL STRAINER DRAIN AND WALL GUARD, (MOUNT FAUCET AT 36" A.F.F.). |               |  |
| FD-1                      | JAY R. SMITH MODEL 2010           | 6" DIAMETER NICKEL BRONZE STRAINER, ROUND FLOOR DRAIN-CAST IRON BODY WITH FLANGE, ADJUSTABLE NICKEL BRONZE STRAINER. TRAP GUARD AND NO-HUB OUTLET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |  |
| FS-1                      | JAY R. SMITH 3101-13              | SQUARE NICKEL BRONZE TOP, 8 1/2" (215) TOP - MEDIUM RECEPTORS, SANI-RECEPTOR ACID RESISTANT COATED, 3/4" - GRATE, TRAP GUARD WITH SEDIMENT BUCKET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |  |
| WCO-1                     | JAY R. SMITH MODEL 4710-05        | WALL CLEANOUT-CAST IRON BODY WITH SPIGOT INLET/OUTLET AND THREADED BRASS RAISED HEAD, DRILLED AND TAPPED FOR 1/20 SCREW. PROVIDE WITH 8480R, ROUND, STAINLESS STEEL ACCESS COVER HAVING 1/4-20X3-1/2 CENTER SCREW.                                                                                                                                                                                                                                                                                                                                                                                                                  |               |  |
| TMV-1                     | ACORN ST70-38                     | TEMPERING VALVE SIMILAR LEAD-FREE CERTIFIED DZR BRASS BODY, ASSE 1070 AT 0.25 GPM AND MAXIMUM SET POINT 115 DEGREE FAHRENHEIT, FACTOR SET AT 105 DEGREE FAHRENHEIT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |  |

NAHRA DESIGN GROUP  
Landlord Review Comments - Architectural  
03/13/2023

PROJECT #  
23.1002

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03/13/2023

PROJECT

LUMINARY  
RICE VILLAGE

2556 UNIVERSITY BLVD, SUITE A-103  
HOUSTON, TEXAS 77005

|                     |                       |
|---------------------|-----------------------|
| SET                 | Construction Drawings |
| ISSUE DATE 03/16/18 |                       |
| 03/13/23            | ISSUE FOR PERMIT      |
|                     |                       |
|                     |                       |
|                     |                       |
|                     |                       |
|                     |                       |

SHEET

PLUMBING  
SCHEDULES

SECTION 22 05 00 - COMMON WORK RESULTS FOR PLUMBING

1. THE PLUMBING SYSTEMS WILL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH THE INCLUDING ANY STATE OR LOCAL AMENDMENTS.

2. ALL PERMITS (FEES BY CONTRACTOR), AND INSPECTIONS WILL BE OBTAINED AS REQUIRED BY ALL LEGAL AUTHORITIES OF WORK INCLUDED IN THESE DOCUMENTS. MATERIAL WHICH ARE SPECIFIED BY REFERENCE TO FEDERAL, STATE, OR OWNER INSURANCE PROVIDERS, SPECIFICATION (I.E. ASTM, ASME, ANSI, TIA, FM GLOBAL, FGI OR AWWA SPECIFICATIONS), FEDERAL STANDARDS, OR OTHER STANDARD SPECIFICATIONS MUST COMPLY WITH THE LATEST EDITIONS UNLESS OTHERWISE NOTED WITHIN THE CONTRACT DOCUMENTS OR OTHER SPECIFICATION SECTIONS.

3. ALL MATERIALS AND EQUIPMENT WILL BE NEW AND WILL BE INSTALLED IN ACCORDANCE WITH INDUSTRY STANDARDS.

4. SUBMITTALS WILL BE PREPARED FOR ALL MATERIALS AND EQUIPMENT INDICATING PERFORMANCE DATA, CATALOG INFORMATION, INSTALLATION DETAILS, ETC. IN ADDITION, SOME AREAS WILL BE DETAILED ON SHOP DRAWINGS AND USED FOR COORDINATION. ALL SUBMITTALS AND SHOP DRAWINGS WILL BE REVIEWED AND APPROVED BY THE DESIGN ENGINEER.

5. AT THE COMPLETION OF THE WORK, BOUND SETS OF OPERATIONS AND MAINTENANCE MANUALS WILL BE PROVIDED. THESE MANUALS WILL INCLUDE EQUIPMENT PERFORMANCE DATA, INSTALLATION DETAILS, MAINTENANCE AND SERVICE INSTRUCTIONS, PARTS LISTS, WIRING AND CONTROLS DIAGRAMS.

6. A SET OF PLUMBING DRAWINGS WILL BE MAINTAINED AT THE JOB SITE A SEPARATE SET OF CLEAN CONTRACT DRAWINGS FOR THE SOLE PURPOSE OF RECORDING THE "AS-BUILT" CHANGES AND DIAGRAMS OF THOSE PORTIONS OF WORK IN WHICH ACTUAL CONSTRUCTION IS SIGNIFICANTLY DIFFERENT FROM THE CONTRACT DOCUMENTS. AT THE END OF THE PROJECT, OBTAIN WITHOUT COST TO THE OWNER A SET OF THE ORIGINAL PLUMBING DRAWINGS/CAD FILES INDICATING ALL CHANGES FROM THE CONTRACT DOCUMENTS FOR THE OWNER'S RECORDS. DELIVERY OF THE AS-BUILT PRINTS AND CAD FILES IS A CONDITION OF FINAL ACCEPTANCE OF THE WORK COMPLETED.

7. COORDINATION WILL TAKE PLACE WITH OTHER TRADES DURING CONSTRUCTION. ANY INTERFERENCES FOUND DUE TO OTHER TRADES WILL BE BROUGHT TO THE ATTENTION OF THE GENERAL CONTRACTOR OR CONSTRUCTION MANAGER.

8. UPON COMPLETION OF THE WORK, AND AT TIMES DURING THE INSTALLATION, RUBBISH AND DEBRIS RESULTING FROM THE SCOPE OF THIS WORK WILL BE REMOVED TO A LOCATION ON SITE PROVIDED BY THE GENERAL CONTRACTOR AND THE AREA WILL BE LEFT IN A NEAT, ALL FLOOR DRAINS AND PLUMBING FIXTURE/EQUIPMENT SHALL BE CLEANED TO LIKE NEW CONDITIONS.

9. ALL EQUIPMENT AND MATERIALS FURNISHED AND INSTALLED UNDER THIS SCOPE OF WORK WILL BE GUARANTEED TO BE FREE FROM DEFECTS OF MATERIALS AND WORKMANSHIP FOR A PERIOD OF ONE YEAR AFTER THE DATE OF EQUIPMENT START-UP.

10. ANY OPENINGS REQUIRED THROUGH STRUCTURAL WALLS, FLOORS AND ROOFS WILL BE IDENTIFIED ON BLOCK OUT DRAWINGS AND PROVIDED TO THE GENERAL CONTRACTOR. ALL FORMING, CUTTING AND FRAMING TO THESE OPENINGS WILL BE BY THE GENERAL CONTRACTOR.

11. ARCHITECTURAL ACCESS DOORS REQUIRED IN CEILINGS AND WALLS WILL BE PROVIDED AND INSTALLED BY THE GENERAL CONTRACTOR.

12. PIPE SLEEVES WILL BE PROVIDED WHERE PIPING PASSES THROUGH WALLS AND FLOORS. SLEEVES WILL BE 18 GAUGE GALVANIZED STEEL FOR WALLS AND STEEL PIPE FOR FLOORS. FIRE STOP WILL BE PROVIDED WHERE THE FLOOR OR WALLS ARE FIRE RATED. SLEEVES WILL BE EXTENDED 2 INCHES ABOVE THE FLOOR IN "WET" AREAS.

EXECUTION

A. INSTALLATION OF PIPING, VALVES AND EQUIPMENT

a. PITCH: HORIZONTAL SANITARY AND DRAIN PIPING WILL BE RUN AT AN UNIFORM GRADE OF 1/4" PER FOOT FOR 2-1/2" AND LESS PIPE SIZES. 1/2" PER FOOT FOR 3" AND ABOVE PIPE SIZE. CONTRACTOR SHALL INSTALL PLUMBING SYSTEM SLOPE PER CODE AND AUTHORITIES HAVING JURISDICTION.

b. WATER PIPING WITHIN WALLS AND ROUGH-INS FOR FIXTURES AND EQUIPMENT: COPPER PLATED STEEL SUPPORT SYSTEM SOLDERED TO PIPING AND SECURED TO BUILDING CONSTRUCTION SO THAT PIPES CANNOT BE DISPLACED. (HOLDRITE OR EQUIVALENT). HOT WATER PIPING INSULATION WITH STANDARD JACKETS, WITH OR WITHOUT VAPOR BARRIER, FACTORY APPLIED OR FIELD APPLIED. FITTINGS, JOINTS, AND VALVES WILL BE INSULATED WITH LIKE MATERIAL AND THICKNESS AS ADJOINING PIPE. PROVIDE REMOVABLE INSULATION AND COVER FOR ALL VALVES AND UNIONS.

c. WASTE AND VENT PIPING WITHIN WALLS AND ROUGH-INS FOR FIXTURES AND EQUIPMENT: COPPER PLATED STEEL SUPPORT SYSTEM FOR COPPER DWV PIPING OR GALVANIZED STEEL SUPPORT SYSTEM FOR CAST IRON OR GALVANIZED PIPING. SUPPORTS TO PIPING AND BUILDING CONSTRUCTION WILL BE SECURED SO THAT PIPES CANNOT BE DISPLACED. FELT STRIP INSULATION WILL BE PROVIDED BETWEEN DISSIMILAR METALS. (HOLDRITE OR EQUIVALENT)

d. UNDERGROUND PIPING: NO-HUB SOIL PIPE NOT PERMITTED.

e. UNION AND FLANGES: ON PIPING TO INLET AND OUTLET OF ALL APPARATUS AND EQUIPMENT TO FACILITATE REMOVAL OF EQUIPMENT, AND DOWNSTREAM OF ALL SHUTOFF VALVES.

f. WATER HAMMER ARRESTERS: WATER HAMMER ARRESTERS WILL BE INSTALLED AT ALL QUICK CLOSING VALVES SUCH AS FLUSH VALVES, FOOT CONTROL VALVES, FLOAT VALVES, SOLENOID VALVES, ETC. WATER HAMMER ARRESTERS WILL BE SIZED AND LOCATED AS RECOMMENDED BY PDI MANUAL WH 201.

B. CLEANOUTS WILL BE PROVIDED AND INSTALLED PER CPC SECTION 707.0 AND 719.0.

C. LABELS AND IDENTIFICATION: VALVE TAGS, PIPING SYSTEMS AND EQUIPMENT IDENTIFICATIONS AS SPECIFIED IN PLUMBING IDENTIFICATION.

D. PIPING SYSTEM LEAK TESTS:

a. POTABLE WATER PIPING WILL BE HYDROSTATICALLY PRESSURE TESTED FOR A DURATION OF FOUR HOURS AT A TEST PRESSURE OF 120 PSIG.

b. SANITARY DRAINAGE PIPING: ALL ABOVE GROUND PIPING WILL BE TESTED HYDROSTATICALLY BY CLOSING ALL OPENINGS IN THE PIPING SYSTEM, EXCEPT THE HIGHEST OPENING ABOVE THE ROOF, AND BY FILLING THE SYSTEM TO THE POINT OF OVERFLOWING. THE PRESSURE EXERTED ON THE SYSTEM WILL BE NO LESS THAN 10 FEET OF HEAD.

SECTION 23 05 93 – TESTING, ADJUSTING, AND BALANCING FOR HVAC EXECUTION

1. GENERAL PROCEDURES FOR TESTING AND BALANCING

A. PERFORM TESTING AND BALANCING PROCEDURES ON EACH SYSTEM ACCORDING TO THE PROCEDURES CONTAINED IN AABC (ASSOCIATED AIR BALANCE COUNCIL) NATIONAL STANDARDS FOR TOTAL SYSTEM BALANCE; NEEB'S "PROCEDURAL STANDARDS FOR TESTING, ADJUSTING, AND BALANCING OF ENVIRONMENTAL SYSTEMS"; CUT INSULATION, PIPES, AND EQUIPMENT CABINETS FOR INSTALLATION OF TEST PRESSURES TO THE MINIMUM EXTENT NECESSARY FOR TAB PROCEDURES. MARK EQUIPMENT AND BALANCING DEVICES AND VALVE POSITION INDICATORS WITH PAINT OR OTHER SUITABLE, PERMANENT IDENTIFICATION MATERIAL TO SHOW FINAL SETTINGS.

C. TAKE AND REPORT TESTING AND BALANCING MEASUREMENTS IN INCH-POUND (IP)UNITS.

D. VERIFY FINAL SYSTEM CONDITIONS.

SECTION 22 05 13 - COMMON MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT

1. TEST MOTORS IN ACCORDANCE WITH NEMA MG 1, INCLUDING WINDING RESISTANCE, NO-LOAD SPEED AND CURRENT, LOCKED ROTOR CURRENT, INSULATION HIGH-POTENTIAL TEST, AND MECHANICAL ALIGNMENT TESTS. INSTALL SECURELY ON FIRM FOUNDATION. MOUNT BALL BEARING MOTORS WITH SHAFT IN ANY POSITION. INSTALL ENGRAVED PLASTIC NAMEPLATES. GROUND AND BOND MOTORS.

2. SINGLE-PHASE MOTORS: PERMANENT SPLIT-CAPACITOR TYPE, WHERE AVAILABLE; OTHERWISE, USE SPLIT-PHASE START-CAPACITOR RUN CAPACITOR START-CAPACITOR RUN MOTOR. TERMINAL LUGS TO MATCH BRANCH CIRCUIT CONDUCTOR QUANTITIES, SIZES AND MATERIALS.

3. THREE-PHASE MOTORS: NEMA MG 1, DESIGN B, PREMIUM-EFFICIENCY SQUIRREL-CAGE INDUCTION MOTOR, WITH WINDINGS TO ACCOMPLISH STARTING METHODS AND NUMBER OF SPEEDS INDICATED. SERVICE FACTOR: 1.15 UNLESS OTHERWISE INDICATED ON DRAWINGS. ENCLOSURE: MEET CONDITIONS OF INSTALLATION UNLESS SPECIFIC ENCLOSURE IS SPECIFIED OR INDICATED. DESIGN FOR CONTINUOUS OPERATION IN 40 DEGREES C ENVIRONMENT, WITHIN INSULATION CLASS B. INSULATION CLASS B. INSULATION CLASS B. INSULATION CLASS B. SERVICE FACTOR, AND MOTOR ENCLOSURE TYPE. INSULATION SYSTEM, NEMA CLASS F. MOTOR FRAMES, NEMA STANDARD T-FRAMES OF STEEL, ALUMINUM, OR CAST IRON WITH END BRACKETS OF CAST IRON OR ALUMINUM WITH STEEL INSERTS. THERMISTOR SYSTEM (MOTOR FRAME SIZES 254T AND LARGER); THREE PTC THERMISTORS EMBEDDED IN MOTOR WINDINGS AND EPOXY ENCAPSULATED SOLID STATE CONTROL RELAY WITH WIRING TO TERMINAL BOX. BEARINGS: GREASE LUBRICATED ANTI-FRICTION BALL BEARINGS WITH HOUSINGS EQUIPPED WITH PLUGGED PROVISION FOR RELUBRICATION. RATED FOR MINIMUM 80% A, 1.0 LIFE OF 200,000 HOURS. CALCULATE BEARING LOAD WITH NEMA MINIMUM V-BELT PULLEY WITH BELT CENTER LINE AT END OF NEMA STANDARD SHAFT EXTENSION. STAMP BEARING SIZES ON NAMEPLATE. SOUND POWER LEVELS: CONFORM TO NEMA MG 1. TERMINAL LUGS TO MATCH BRANCH CIRCUIT CONDUCTOR QUANTITIES, SIZES AND MATERIALS.

SECTION 22 05 23.12 – BALL VALVES FOR PLUMBING PIPING

1. BRASS:

A. PRODUCTS: TWO-PIECE BRASS BALL VALVES WITH FULL PORT AND STAINLESS-STEEL TRIM.

B. MANUFACTURERS: NIBCO, KITZ CORPORATION, MILWAUKEE VALVE COMPANY.

C. DESCRIPTION: STANDARD: MSS SP-110, SWP RATING: 150 PSIG, CWP RATING: 800 PSIG, BODY DESIGN: TWO-PIECE, BODY MATERIAL: FORGED BRASS, ENDS: THREADED, SEATS: PTFE, STEM: STAINLESS STEEL, BALL: STAINLESS STEEL, VENTED, PORT: FULL.

2. BRONZE:

A. PRODUCTS: TWO-PIECE LEAD FREE BRONZE BALL VALVES WITH FULL PORT AND STAINLESS-STEEL TRIM.

B. MANUFACTURERS: NIBCO, KITZ CORPORATION, WATTS.

C. DESCRIPTION: STANDARD: MSS SP-110, SWP RATING: 150 PSIG, CWP RATING: 800 PSIG, BODY DESIGN: TWO-PIECE, BODY MATERIAL: FORGED BRONZE, ENDS: THREADED, SEATS: PTFE, STEM: STAINLESS STEEL, BALL: STAINLESS STEEL, VENTED, PORT: FULL.

3. SELECT VALVES WITH THE FOLLOWING END CONNECTIONS:

A. FOR COPPER TUBING, NPS 2 AND SMALLER: THREADED ENDS EXCEPT WHERE SOLDER-JOINT VALVE-END OPTION IS INDICATED IN VALVE SCHEDULES BELOW.

B. FOR STEEL PIPING, NPS 2 AND SMALLER: THREADED ENDS.

SECTION 22 05 23.14 – CHECK VALVES FOR PLUMBING PIPING

1. SOURCE LIMITATIONS FOR VALVES: OBTAIN EACH TYPE OF VALVE FROM SINGLE SOURCE FROM SINGLE MANUFACTURER.

2. COMPLY WITH ASME, COMPLY WITH AWWA C606 FOR GROOVED-END CONNECTIONS. NSF 61 ANNEX G AND NSF 372 FOR VALVE MATERIALS FOR POTABLE-WATER SERVICE.

3. BRONZE VALVES SHALL BE MADE WITH DEZINCIFICATION-RESISTANT MATERIAL.S. BRONZE VALVES MADE WITH COPPER ALLOY (BRASS) CONTAINING MORE THAN 15 PERCENT ZINC ARE NOT PERMITTED. VALVE PRESSURE-TEMPERATURE RATINGS: NOT LESS THAN INDICATED AND AS REQUIRED FOR SYSTEM PRESSURES AND TEMPERATURES. VALVE SIZES: SAME AS UPSTREAM PIPING UNLESS OTHERWISE INDICATED. VALVE BYPASS AND DRAIN CONNECTIONS: MSS SP-45.

4. BRONZE SWING CHECK VALVES

A. CLASS 125, BRONZE, SWING CHECK VALVES WITH BRONZE DISC.

B. MANUFACTURERS: NIBCO INC., WATTS.

C. DESCRIPTION: STANDARD: MSS SP-80, TYPE 3, CWP RATING: 300 PSIG., BODY DESIGN: HORIZONTAL FLOW, BODY MATERIAL: ASTM B 62, BRONZE, ENDS: THREADED OR SOLDERED. SEE VALVE SCHEDULE ARTICLES., DISC, BRONZE.

SECTION 22 05 29 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

1. SINGLE PIPE HANGERS WILL BE MALLEABLE IRON OR CARBON STEEL, ADJUSTABLE SWIVEL, SPLIT RING FOR PIPE SIZES UP TO 1 INCHES AND CARBON STEEL, ADJUSTABLE CLEVIS HANGERS FOR PIPE SIZES 2 INCHES AND OVER.

2. PROVIDE COPPER HANGERS OR WONDER TAPE AT ALL HANGERS FOR DISSIMILAR METALS.

3. MULTIPLE OR TRAPEZE HANGERS WILL BE UNISTRUT STEEL CHANNELS WITH UNISTRUT CLAMPS AND ALL-THREAD HANGER RODS AND DOUBLE NUTS. WHERE REQUIRED, UNISTRUT CHANNELS MAY BE REPLACED WITH STRUCTURAL STEEL CHANNELS OR I-BEAMS TO MEET STRUCTURAL LOADING.

4. WALL SUPPORTS WILL BE UNISTRUT BOLTED TO THE WALL WITH UNISTRUT PIPE CLAMPS FOR PIPE SIZES UP TO 3 INCHES AND WELDED STEEL BRACKETS AND STEEL CLAMPS FOR PIPE SIZES GREATER THAN 4 INCHES.

5. VERTICAL PIPE WILL BE SUPPORTED WITH STEEL RISER CLAMPS FOR STEEL PIPE AND COPPER RISER CLAMPS FOR COPPER PIPE.

6. GALVANIZED STEEL PIPE INSULATION SHIELDS WILL BE PROVIDED FOR PIPE SIZES UP TO 2 INCH. PRE-FABRICATED PIPE INSERTS AND SHIELDS WILL BE USED FOR PIPE SIZES OVER 2 INCHES.

7. HORIZONTAL PIPE WILL BE SUPPORTED WITH THE MAXIMUM SPACING:

| PIPE SIZE (INCH) | COPPER PIPE (FT.) | STEEL PIPE (FT.) |
|------------------|-------------------|------------------|
| 1/2"             | 6'                | 6'               |
| 3/4" TO 1"       | 6'                | 8'               |
| 1-1/4" TO 1-1/2" | 6'                | 10'              |
| 2"               | 10'               | 10'              |
| 2-1/2" TO 4"     | 10'               | 10'              |
| 6" TO 8"         | 10'               | 10'              |

8. ALL PIPING WILL BE BRONZED IN ACCORDANCE WITH NFPA 13, ASME B31.1, ASCE 705, AND ASCE 7-10 STANDARDS FOR THE APPROPRIATE SEISMIC HAZARD LEVEL.

9. INSERTS WILL BE USED TO SUPPORT PIPING FROM CONCRETE STRUCTURES. WHERE INSERTS ARE NOT USED DRILL-IN EXPANSION BOLTS WILL BE USED. LOW VELOCITY SHOT PIN FASTENERS WILL ALSO BE USED WHERE PERMITTED. APPROPRIATE DRILL-IN ANCHORS WILL BE USED ON POST TENSION SLABS.

SECTION 22 05 53 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

1. EQUIPMENT: PERMANENT LABEL (STENCIL, METAL TAG OR ENGRAVED PLASTIC) WITH UNIT TAG OR NAME AND AREA OR SPACE SERVED.

2. PIPING:

A. PIPING LABEL MANUFACTURERS: BRIMAR INDUSTRIES, INC., CRAFTMARK PIPE MARKERS., SETON IDENTIFICATION PRODUCTS.

B. GENERAL REQUIREMENTS FOR MANUFACTURED PIPE LABELS: PREPRINTED, COLOR-CODED, WITH LETTERING INDICATING SERVICE, AND SHOWING FLOW DIRECTION ACCORDING TO ASME A13.1. PROVIDE PIPE MARKERS EVERY 20 FEET. IDENTIFY SERVICE AND FLOW DIRECTION. INSTALL IN CLEAR VIEW AND ALIGN WITH AXIS OF PIPING.

C. PRE-INSULATED PIPE: PRECOILED, SEMIRIGID PLASTIC FORMED TO PARTIALLY COVER CIRCUMFERENCE OF PIPE AND TO ATTACH TO PIPE WITHOUT FASTENERS OR ADHESIVE.

D. SELF-ADHESIVE PIPE LABELS: PRINTED PLASTIC WITH CONTACT-TYPE, PERMANENT-ADHESIVE BACKING.

E. PIPE LABEL CONTENTS: INCLUDE IDENTIFICATION OF PIPING SERVICE USING SAME DESIGNATIONS OR ABBREVIATIONS AS USED ON DRAWINGS; ALSO INCLUDE PIPE SIZE AND AN ARROW INDICATING FLOW DIRECTION.

F. FLOW-DIRECTION ARROWS: INTEGRAL WITH PIPING SYSTEM LETTERING TO ACCOMMODATE BOTH DIRECTIONS OR AS SEPARATE UNIT ON EACH PIPE LABEL TO INDICATE FLOW DIRECTION, LETTERING SIZE: SIZE LETTERS ACCORDING TO ASME A13.1 FOR PIPING.

G. PIPE LABEL COLOR SCHEDULE:

a. DOMESTIC WATER PIPING:

• BACKGROUND: SAFETY GREEN

• LETTER COLORS: WHITE

b. SANITARY WASTE AND STORM DRAINAGE PIPING:

• BACKGROUND: SAFETY BLACK

• LETTER COLORS: WHITE

SECTION 22 07 19 - PLUMBING PIPING INSULATION

1. PIPE INSULATION WILL BE PROVIDED ON ALL COLD WATER, HOT WATER AND STEAM PIPING. THE INSULATION WILL BE GLASS FIBER WITH A K-FACTOR OF 0.24 AT 75°F, NON-COMBUSTIBLE, WITH A SERVICE TEMPERATURE RANGE OF -20°F TO 450°F.

2. AN ALL SERVICE JACKET WITH INTEGRAL VAPOR BARRIER CONSISTING OF WHITE KRAFT PAPER WITH FOIL FACE AND SEALED JOINTS WILL BE USED ON LOW TEMPERATURE PIPING. FITTING COVERS WILL BE PVC ONE-PIECE, PRE-MOLDED TYPE.

3. ALL INSULATED EXTERIOR PIPING WILL BE COVERED WITH 0.006 INCH THICK CORRUGATED ALUMINUM JACKET. PIPE INSULATION THICKNESS WILL BE AS FOLLOWS:

| INSULATION THICKNESS (IN.) FOR NOMINAL PIPE SIZES |                       |             |              |              |        |           |
|---------------------------------------------------|-----------------------|-------------|--------------|--------------|--------|-----------|
| SYSTEM                                            | TEMPERATURE RANGE (F) | 1" AND LESS | 1-1/4" TO 2" | 2-1/2" TO 4" | 6"     | 8" AND UP |
| DOMESTIC WATER                                    | ALL                   | 1"          | 1"           | 1"           | 1-1/2" | 1-1/2"    |

SECTION 22 11 16 - DOMESTIC WATER PIPING

1. PIPING

A. UNDERGROUND: NOT IN SCOPE.

B. ABOVE GROUND: ASTM B-88 TYPE I SEAMLESS COPPER TUBING, HARD TEMPER, COLD DRAWN, PEX-a, ASTM F 876/877, SDR 9, CTS, ENGL METHOD.

2. FITTINGS

A. COPPER TUBING: ANSI B16.22 WROUGHT COPPER SOLDER SWEAT TYPE.

B. CAST-COPPER, SOLDER-JOINT FITTINGS: PRESSURE FITTINGS.

C. WROUGHT-COPPER, SOLDER-JOINT FITTINGS: WROUGHT-COPPER PRESSURE FITTINGS.

D. BRASS: ANSI B16.15 85% RED BRASS, CAST IRON PATTERN. SCREWED TYPE MAY BE USED WHERE CLOSE FITTING IS REQUIRED.

E. ADAPTERS WILL BE USED WHERE COPPER TUBING CONNECTS TO IRON PIPE SIZE BRASS PIPING.

F. PEX-a: ENGINEERED POLYMER OR LEAD-FREE BRASS ASTM F 1960 COLD EXPANSION WITH PEX-a REINFORCING RING.

G. FLANGES: VICTAULIC MECHANICAL GROOVED COUPLINGS.

H. COPPER UNIONS:

a. CAST-COPPER-ALLOY, HEXAGONAL END BODY: BALL-AND-SOCKET, METAL-TO-METAL SEATING SURFACES, THREADED STODS OR PROGRESS.

b. COPPER PRESSURE-SEAL-JOINT FITTINGS:

a. MANUFACTURERS: VIEGA LLC.

b. FITTINGS FOR NPS 2 AND SMALLER: WROUGHT-COPPER FITTING WITH EPDM-RUBBER, O-RING SEAL IN EACH END.

c. FITTINGS FOR NPS 2-1/2 TO NPS 4: CAST-BRONZE OR WROUGHT-COPPER FITTING WITH EPDM-RUBBER, O-RING SEAL IN EACH END.

SECTION 22 11 19 - DOMESTIC WATER PIPING SPECIALTIES

1. DUAL-CHECK VALVE BACKFLOW PREVENTER

a. STANDARD: ASSE 1024.

b. OPERATION: CONTINUOUS PRESSURE APPLICATIONS.

c. SIZE: NPS 1/2 TO NPS 1-1/4.

d. BODY: BRONZE WITH UNION INLET.

2. OUTLET BOXES

A. ICEMAKER OUTLET BOXES

a. MOUNTED: RECESSED.

b. MATERIAL AND FINISH: (ENAMELED-STEEL OR EPOXY PAINTED STEEL) ( ENAMELED STEEL, EPOXY PAINTED STEEL OR PLASTIC) (PLASTIC) (STAINLESS STEEL) BOX AND FACEPLATE.

c. FAUCET: VALVED FITTING COMPLYING WITH ASME A112.18.1. INCLUDE NPS 1/2 OR SMALL COPPER TUBE OUTLET.

d. SUPPLY SHUTOFF FITTING: NPS 1/2 BALL VALVE AND NPS 1/2 COPPER WATER TUBING.

SECTION 22 13 16 - SANITARY WASTE AND VENT PIPING

1. MANUFACTURERS SHALL BE ISO 9001 ACCREDITED.

2. PIPE AND FITTINGS ABOVE GROUND: CISP 310 OR ASTM C 1277, CAST IRON NO-HUB SOIL PIPE AND FITTINGS WITH STAINLESS STEEL NO-HUB COUPLINGS. ALL CAST IRON SOIL PIPE AND FITTINGS SHALL BE MARKED WITH THE COLLECTIVE TRADEMARK OF THE CAST IRON SOIL PIPE INSTITUTE AND BE LISTED BY NSF INTERNATIONAL® INDIRECT DRAIN™ TYPE M, ASTM B-88, SEAMLESS HARD TEMPER, COLD DRAWN COPPER TUBING WITH ANSI B16.29 WROUGHT COPPER SOLDER SWEAT DRAINAGE FITTINGS.

4. CAST IRON SOIL PIPE NO-HUB COUPLINGS: CONFORMING TO ASTM C 1540. HEAVY DUTY, 24 GAUGE, TYPE 304 STAINLESS STEEL SHIELD AND CLAMP ASSEMBLY WITH ASTM C 564 NEOPRENE SEALING SLEEVES TORQUED TO A MINIMUM OF 100 INCH.LBS. COUPLING WILL BE FACTORY MUTUAL APPROVED PER STANDARD 1680, CLASS I. (HUSKY 4000 OR EQUIVALENT)

SECTION 22 33 30 - ELECTRIC DOMESTIC WATER HEATER

1. COMMERCIAL, LIGHT-DUTY, STORAGE, ELECTRICAL, DOMESTIC-WATER HEATERS:

A. STANDARD: UL 174.

B. STORAGE-TANK CONSTRUCTION: STEEL, VERTICAL ARRANGEMENT.

a. TAPPINGS: ASME B1.20.1 PIPE THREAD.

B. PRESSURE RATING: 150 PSIG

C. INTERIOR FINISH: COMPLY WITH NSF 61 ANNEX G BARRIER MATERIAL FOR POTABLE-WATER TANK LININGS, INCLUDING EXTENDING LINING MATERIAL INTO TAPPINGS.

2. FACTORY-INSTALLED STORAGE-TANK APPURTENANCES:

A. ANODE ROD: REPLACEABLE MAGNESIUM.

B. DIP TUBE: REQUIRED UNLESS COLD-WATER INTLET IS NEAR BOTTOM OF TANK.

C. DRAIN VALVE: ASSE 1005.

D. INSULATION: COMPLY WITH ASHRAE/IESNA 90.1.

E. JACKET: STEEL WITH ENAMELED FINISH.

F. HEAT TRAPPINGS: INLET TYPE IN COLD-WATER INLET AND OUTLET IN HOT-WATER OUTLET.

G. HEATING ELEMENTS: TWO: ELECTRIC, SCREW-IN IMMERSION TYPE, WIRED FOR SIMULTANEOUS OPERATION UNLESS OTHERWISE LIMITED. LIMITED TO 12 KW TOTAL.

H. TEMPERATURE CONTROL: ADJUSTABLE THERMOSTAT.

I. SAFETY CONTROL: HIGH-TEMPERATURE-LIMIT CUT-OFF DEVICE OR SYSTEM.

J. RELIEF VALVE: ASME RATED AND STAMPED FOR COMBINATION TEMPERATURE-AND-PRESSURE RELIEF VALVES. INCLUDE RELIEF CAPACITY AT LEAST AS GREAT AS HEAT INPUT, AND INCLUDE PRESSURE SETTING LESS THAN DOMESTIC-WATER HEATER WORKING-PRESSURE RATING. SELECT RELIEF VALVE WITH SENSING ELEMENT THAT EXTENDS INTO STORAGE TANK.

3. CAPACITY AND CHARACTERISTICS:

A. REFER TO PLUMBING ELECTRICAL WATER HEATER SCHEDULES FOR INFORMATION.

E. ELECTRICAL CHARACTERISTICS:

A. REFER TO PLUMBING ELECTRICAL WATER HEATER SCHEDULES FOR INFORMATION.

4. INSTALL COMBINATION TEMPERATURE AND PRESSURE RELIEF VALVES IN WATER PIPING FOR ELECTRIC DOMESTIC-WATER HEATERS WITHOUT STORAGE. EXTEND COMMERCIAL WATER HEATER RELIEF-VALVE OUTLET WITH DRAIN PIPING, SAME MATERIAL AS DOMESTIC WATER PIPING, IN CONTINUOUS DOWNWARD PITCH AND DISCHARGE BY POSITIVE AIR GAP INTO CLOSEST FLOOR DRAIN/SINK.

5. INSTALL WATER HEATER DRAIN PAN PIPING AS INDIRECT WASTE TO SPILL BY POSITIVE AIR GAP INTO OPEN DRAINS OR OVER FLOOR DRAINS/SINKS. INSTALL HOSE-END DRAINS VALVES AT LOW POINT IN WATER PIPING FOR ELECTRIC DOMESTIC-WATER HEATER THAT DO NOT HAVE TANK DRAINS.

6. INSTALL THERMOMETERS ON INLET AND OUTLET PIPING FOR ELECTRICAL DOMESTIC-WATER HEATERS.

7. ASSEMBLE AND INSTALL INLET AND OUTLET PIPING MANIFOLD KITS FOR MULTIPLE ELECTRIC DOMESTIC-WATER HEATERS. FABRICATE, MODIFY OR ARRANGE MANIFOLDS FOR BALANCED WATER FLOW THROUGH EACH ELECTRIC DOMESTIC-WATER HEATER. INCLUDE SHUTOFF VALVE AND THERMOMETER IN EACH DOMESTIC-WATER HEATER INLET AND OUTLET AND THROTTLING VALVE IN EACH ELECTRIC DOMESTIC-WATER HEATER OUTLET.

8. INSTALL PIPING-TYPE HEAT TRAPS ON INLET AND OUTLET PIPING OF ELECTRICAL DOMESTIC-WATER HEATER STORAGE TANKS WITHOUT INTEGRAL OR FITTING-TYPE HEAT TRAPS.

9. FILL ELECTRIC DOMESTIC-WATER HEATER WITH WATER.

M. CHARGE DOMESTIC-WATER COMPRESSION TANK WITH AIR.

1. DOMESTIC WATER HEATER ACCESSORIES:

A. DOMESTIC WATER COMPRESSION TANKS:

a. DESCRIPTION: STEEL, PRESSURE-RATED TANK CONSTRUCTED WITH WELDED JOINTS AND A FACTORY INSTALLED BUTYL-RUBBER DIAPHRAGM. INCLUDE AIR PRECHARGE TO MINIMUM SYSTEM-OPERATION PRESSURE TANK.

b. CONSTRUCTION:

a. TAPPING FACTORY FABRICATED STEEL, WELDED TO TANK BEFORE TESTING AND STORAGE.

b. LABELING: INCLUDE ASME B1.20.1 PIPE THREAD.

c. INTERIOR FINISH: COMPLY WITH NSF 61 ANNEX G BARRIER MATERIALS FOR POTABLE WATER TANK LININGS, INCLUDING EXTENDING FINISH INTO AND THROUGH TANK FITTING AND OULETS.

d. AIR-CHARGING VALVE: FACTORY INSTALLED.

c. CAPACITY AND CHARACTERISTICS:

A. DIMENSIONS NOT LESS THAN BASE OF DOMESTIC WATER HEATER AND INCLUDE DRAIN OULET NOT LESS THAN NPS 3/4 WITH ASME B1.20.1 PIPE THREADS OR ASME B1.20.7 GARDEN HOSE THREADS.

B. PIPING TYPE HEAT TRAPS: FIELD FABRICATED PIPING ARRANGEMENT ACCORDING TO ASHRAE/IESNA 90.1.

C. HEAT TRAP FITTINGS: ASHRAE 90.2.

D. MANIFOLD KITS: DOMESTIC WATER HEATER MANUFACTURER'S FACTORY FABRICATED INLET AND OUTLET PIPING FOR INSTALLATION. FOR MULTIPLE DOMESTIC WATER HEATER INSTALLATION, INCLUDE BALL SHUTOFF VALVES TO ISOLATE EACH DOMESTIC WATER HEATER AND CALIBRATE BALANCING VALVES TO PROVIDE BALANCED FLOW THROUGH EACH DOMESTIC WATER HEATER.

E. COMPLY WITH REQUIREMENTS FOR BALL SHUTOFF VALVES SPECIFIED IN SECTION 220523.12 "BALL VALVES FOR PLUMBING PIPING." SECTION 220523.13 "BUTTERFLY VALVES FOR PLUMBING PIPING."

a. COMPLY WITH REQUIREMENTS FOR BALANCING VALVES SPECIFIED IN SECTION 221119 "DOMESTIC WATER PIPING SPECIALTIES."

F. COMBINATION TEMPERATURE AND PRESSURE RELIEF VALVE: INCLUDE RELIEVING CAPACITY AT LEAST AS GREAT AS HEAT INPUT AND INCLUDE PRESSURE SETTING LESS THAN DOMESTIC WATER HEATER WORKING PRESSURE RATING. SELECT RELIEF VALVES WITH SENSING ELEMENT THAT EXTENDS INTO STORAGE TANK.

G. VACUUM RELIEF VALVES: ANSI Z21.22/CSA 4.4-M.

H. DOMESTIC WATER HEATER STANDS: MANUFACTURER'S FACTORY FABRICATED STEEL STAND FOR FLOOR MOUNTING, CAPABLE OF SUPPORTING DOMESTIC WATER HEATER AND WATER, PROVIDE DIMENSION THAT WILL SUPPORT BOTTOM OF DOMESTIC WATER HEATER A MINIMUM OF 18 INCHES ABOVE THE FLOOR.

I. DOMESTIC WATER HEATER MOUNTING BRACKETS: MANUFACTURER'S FACTORY FABRICATED STEEL BRACKET FOR WALL MOUNTING, CAPABLE OF SUPPORTING DOMESTIC WATER HEATER AND WATER.

1. DOMESTIC WATER HEATER INSTALLATION:

A. COMMERCIAL DOMESTIC WATER HEATER MOUNTING: INSTALL COMMERCIAL ELECTRICAL DOMESTIC WATER HEATER ON CONCRETE BASE. COMPLY WITH REQUIREMENTS FOR CONCRETE BASES SPECIFIED IN SECTION 033000 "CAST IN PLACE CONCRETE" OR SECTION 033053 "MISCELLANEOUS CAST IN PLACE CONCRETE.

a. EXCEPTION: OMIT CONCRETE BASE FOR COMICAL ELECTRIC DOMESTIC WATER HEATER IF THY INSTALLATION ON TANK BRACKET, SUSPENDED PLATFORM OR DIRECTLY ON FLOOR IS INDICATED.

b. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES.

c. ARRANGE WATER CONTROL AND DEVICES THAT REQUIRE SERVICING ARE ACCESSIBLE.

d. INSTALL DOWEL RODS TO CONNECT CONCRETE BASE TO CONCRETE FLOOR, UNLESS OTHERWISE INDICATED. INSTALL DOWEL RODS ON 18 INCH CENTERS AROUND THE FULL PERIMETER OF CONCRETE BASE.

B. INSTALL DOMESTIC WATER HEATER LEVEL AND PLUMB, ACCORDING TO LAYOUT DRAWINGS, ORIGINAL DESIGN AND REFERENCED STANDARDS. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES. ARRANGE UNITS SO CONTROLS AND DEVICES NEEDING SERVICE ARE ACCESSIBLE.

FIRE SPRINKLER COORDINATION NOTES

SECTION 21 05 00 - COMMON WORK RESULTS FOR FIRE SUPPRESSION

1. SCOPE OF WORK:

A. THE CONTRACTOR IS RESPONSIBLE FOR ALL WORK, MATERIALS, AND LABOR REQUIRED TO SATISFY A COMPLETE WORKING SYSTEM WHETHER SPECIFIED OR IMPLIED.

B. THE WORK IS TO BE PERFORMED IN STRICT COMPLIANCE WITH NFPA 13 (LATEST EDITION), ALL CODES AND OTHER NFPA REGULATIONS GOVERNING WORK OF THIS NATURE.

C. THE CONTRACTOR SHALL BEFORE SUBMITTING ANY PROPOSAL, EXAMINE THE PROPOSED SITE AND SHALL DETERMINE FOR HIMSELF THE CONDITIONS THAT MAY EFFECT THE WORK. NO ALLOWANCE SHALL BE MADE IF THE CONTRACTOR FAILS TO MAKE SUCH EXAMINATIONS.

D. ALL EQUIPMENT AND MATERIALS SHALL BE AS SPECIFIED OR "APPROVED EQUAL" BY THE ENGINEER OR ARCHITECT.

E. WHEN NEW SPRINKLERS ARE ADDED INTO THE EXISTING SPRINKLER SYSTEM, THE NEW SPRINKLER SHALL BE OF THE SAME TYPE, ORIFICE AND TEMPERATURE RATINGS UNLESS CONDITIONS REQUIRE A DIFFERENT TYPE SPRINKLER BE INSTALLED.

F. ISOLATE, DRAIN AND REFILL EXISTING PIPING SYSTEM AS REQUIRED TO ACCOMMODATE INSTALLATION OF NEW WORK.

G. ALL SPRINKLER LINES SHALL BE AVOIDING INTERFERENCE WITH LIGHTING FIXTURES, DUCTS, AND PIPING.

2. PERMITS:

A. THE CONTRACTOR SHALL SECURE ALL PERMITS OR APPLICATIONS AND PAY ANY AND ALL FEES.

3. SHOP DRAWINGS:

A. SUBMIT MATERIAL LIST AND SHOP DRAWINGS FOR MAJOR EQUIPMENT TO THE ARCHITECT/ENGINEER FOR APPROVAL.

4. SPRINKLER PIPING:

A. PIPE SHALL BE STEEL, MADE TO ANY OF THE THREE SPECIFICATIONS LISTED. WELDING WILL NOT BE PERMITTED IF THERE ARE ANY SPRINKLERS CONNECTED THERETO.

a. ASTM A53

b. ASTM A120

c. ASTM A135

B. PIPING MATERIAL MAY BE SCHEDULE 10 (THINWALL) PROVIDED JOINTS ARE MADE BY ROLL-GROOVE COUPLING. THREADING WILL NOT BE PERMITTED.

C. PIPING MATERIAL MAY BE SCHEDULE 40 WITH FLANGED, ROLL-GROOVED COUPLINGS, OR THREADED JOINTS.

5. PIPE SUPPORTS:

A. ALL HANGERS MUST BE AN APPROVED TYPE BY NFPA 13. NO SPRINKLER PIPING IS TO BE SUPPORTED FROM ANY MECHANICAL OR ELECTRICAL DEVICES.

B. ALL PIPE SHALL BE SUPPORTED FROM THE BUILDING STRUCTURE IN A NEAT AND WORKMANLIKE MANNER. VERTICAL RISERS SHALL BE SUPPORTED AT EACH FLOOR LEVEL WITH STEEL PIPE CLAMPS. THE USE OF WIRE OR STRAP METAL HANGER TO SUPPORT PIPES WILL NOT BE PERMITTED. HANGING PIPES FROM OTHER PIPES WILL NOT BE PERMITTED. PIPING SHALL BE CAREFULLY COORDINATED BEFORE INSTALLATION WITH OTHER SYSTEMS AND EQUIPMENT IN CHASES AND OTHER CONGESTED AREAS.

C. MAXIMUM DISTANCE BETWEEN PIPE SUPPORTS:

a. 12'-0" FOR 1 1/4" DIAMETER PIPE AND SMALLER

b. 15'-0" FOR 1 1/2" DIAMETER PIPE AND LARGER

6. WORKING PLANS:

A. ANY DRAWINGS INCLUDED AS PART OF THIS SET OF CONSTRUCTION DOCUMENTS ARE DEFINED AS PRELIMINARY SPRINKLER PLANS AS SPECIFIED IN SECTION 23.1. OF NFPA 13. FINAL WORKING DRAWINGS SHALL BE PREPARED BY THE SPRINKLER CONTRACTOR AND REVIEWED AND APPROVED BY THE ENGINEER.

7. TESTS:

A. ACCEPTANCE AND HYDROSTATIC TESTS TO BE PERFORMED IN ACCORDANCE WITH CHAPTER 25 OF NFPA 13.

8. MISCELLANEOUS:

A. FIELD VERIFY ALL FIGURES, CONDITIONS, AND DIMENSIONS AT THE JOB SITE.

B. ANY PRELIMINARY SPRINKLER PLANS ARE INTENDED TO BE DIAGRAMMATIC. THEY ARE NOT INTENDED TO SHOW EVERY ITEM IN ITS EXACT LOCATION.

9. GUARANTEED:

A. MATERIALS, EQUIPMENT AND INSTALLATION SHALL BE GUARANTEED FOR PERIOD OF ONE (1) YEAR FROM DATE OF ACCEPTANCE. DEFECTS WHICH APPEAR DURING THAT PERIOD SHALL BE CORRECTED AT THE SPRINKLER CONTRACTOR'S EXPENSE.

SECTION 23 05 93 - TESTING, ADJUSTING, AND BALANCING FOR HVAC EXECUTION

1. GENERAL PROCEDURES FOR TESTING AND BALANCING:

A. PERFORM TESTING AND BALANCING PROCEDURES ON EACH SYSTEM ACCORDING TO THE PROCEDURES CONTAINED IN AABC (ASSOCIATED AIR BALANCE COUNCIL) NATIONAL STANDARDS FOR TOTAL SYSTEM BALANCE; NEBB'S "PROCEDURAL STANDARDS FOR TESTING, ADJUSTING, AND BALANCING OF ENVIRONMENTAL SYSTEMS"; CUT INSULATION, PIPES, AND EQUIPMENT CABINETS FOR INSTALLATION OF TEST PROBES TO THE MINIMUM EXTENT NECESSARY FOR THE PROCEDURES.

B. MARK EQUIPMENT AND BALANCING DEVICES AND VALVE POSITION INDICATORS WITH PAINT OR OTHER SUITABLE, PERMANENT IDENTIFICATION MATERIAL TO SHOW FINAL SETTINGS.

C. TAKE AND REPORT TESTING AND BALANCING MEASUREMENTS IN INCH-POUND (IP) UNITS.

D. VERIFY FINAL SYSTEM CONDITIONS.

SECTION 22 05 13 - COMMON WORK REQUIREMENTS FOR PLUMBING EQUIPMENT

1. TEST MOTORS IN ACCORDANCE WITH NEMA MG 1, INCLUDING WINDING RESISTANCE, NO-LOAD SPEED AND CURRENT, LOCKED ROTOR CURRENT, INSULATION HIGH-POTENTIAL TEST, AND MECHANICAL ALIGNMENT TESTS. INSTALL SECURELY ON FIRM FOUNDATION. MOUNT BALL BEARING MOTORS WITH SHAFT IN ANY POSITION. INSTALL ENGRAVED PLASTIC NAMEPLATES. GROUND AND BOND MOTORS.

2. SINGLE-PHASE MOTORS: PERMANENT SPLIT-CAPACITOR TYPE, WHERE AVAILABLE; OTHERWISE, USE SPLIT-PHASE START/CAPACITOR RUN OR CAPACITOR START/CAPACITOR RUN MOTOR. TERMINAL LUGS TO MATCH BRANCH CIRCUIT CONDUCTOR QUANTITIES, SIZES AND MATERIALS.

3. THREE-PHASE MOTORS: NEMA MG 1, DESIGN B, PREMIUM-EFFICIENCY SQUIRREL-CAGE INDUCTION MOTOR, WITH WINDINGS TO ACCOMPLISH STARTING METHODS AND NUMBER OF SPEEDS INDICATED. SERVICE FACTOR: 1.15 UNLESS OTHERWISE INDICATED ON DRAWINGS. ENCLOSURE: MEET CONDITIONS OF INSTALLATION UNLESS SPECIFIC ENCLOSURE IS SPECIFIED OR INDICATED. DESIGN FOR CONTINUOUS OPERATION IN 40 DEGREES C ENVIRONMENT, WITH TEMPERATURE RISE IN ACCORDANCE WITH NEMA MG 1 LIMITS FOR INSULATION CLASS, SERVICE FACTOR, AND MOTOR ENCLOSURE TYPE. INSULATION SYSTEM: NEMA CLASS F. MOTOR FRAMES: NEMA STANDARD T-FRAMES OF STEEL, ALUMINUM, OR CAST IRON WITH END BRACKETS OF CAST IRON OR ALUMINUM WITH STEEL INSERTS. THERMISTOR SYSTEM (MOTOR FRAME SIZES 254T AND LARGER); THREE PTC THERMISTORS EMBEDDED IN MOTOR WINDINGS AND EPOXY ENCAPSULATED SOLID STATE CONTROL RELAY WITH WIRING TO TERMINAL BOX BEARINGS. GREASE LUBRICATED ANTI-FRICTION BALL BEARINGS WITH HOUSINGS EQUIPPED WITH PLUGGED PROVISION FOR RELUBRICATION, RATED FOR MINIMUM ABMA 9, L-10 LIFE OF 200,000 HOURS. CALCULATE BEARING LOAD WITH NEMA MINIMUM V-BELT PULLEY WITH BELT CENTER LINE AT END OF NEMA STANDARD SHAFTEXTENSION. STAMP BEARING SIZES ON NAMEPLATE. SOUND POWER LEVELS: CONFORM TO NEMA MG 1. TERMINAL LUGS TO MATCH BRANCH CIRCUIT CONDUCTOR QUANTITIES, SIZES AND MATERIALS.

SECTION 22 05 23.12 - BALL VALVES FOR PLUMBING PIPING

1. BRASS:

A. PRODUCTS: TWO-PIECE BRASS BALL VALVES WITH FULL PORT AND STAINLESS-STEEL TRIM.

B. MANUFACTURERS: NIBCO, KITZ CORPORATION, MILWAUKEE VALVE COMPANY.

C. DESCRIPTION: STANDARD: MSS SP-110, SWP RATING: 150 PSIG, CWP RATING: 600 PSIG, BODY DESIGN: TWO PIECE, BODY MATERIAL: FORGED BRASS, ENDS: THREADED, SEATS: PTFE, STEM: STAINLESS STEEL, BALL: STAINLESS STEEL, VENTED, PORT: FULL.

2. BRONZE:

A. PRODUCTS: TWO-PIECE LEAD FREE BRONZE BALL VALVES WITH FULL PORT AND STAINLESS-STEEL TRIM.

B. MANUFACTURERS: NIBCO, KITZ CORPORATION, WATTS.

C. DESCRIPTION: STANDARD: MSS SP-110, SWP RATING: 150 PSIG, CWP RATING: 600 PSIG, BODY DESIGN: TWO PIECE, BODY MATERIAL: FORGED BRONZE, ENDS: THREADED, SEATS: PTFE, STEM: STAINLESS STEEL, BALL: STAINLESS STEEL, VENTED, PORT: FULL.

3. SELECT VALVES WITH THE FOLLOWING END CONNECTIONS:

A. FOR COPPER TUBING, NPS 2 AND SMALLER: THREADED ENDS EXCEPT WHERE SOLDER-JOINT VALVE-END OPTION IS INDICATED IN VALVE SCHEDULES BELOW.

B. FOR STEEL PIPING, NPS 2 AND SMALLER: THREADED ENDS.

SECTION 22 05 23.14 - CHECK VALVES FOR PLUMBING PIPING

1. SOURCE LIMITATIONS FOR VALVES: OBTAIN EACH TYPE OF VALVE FROM SINGLE SOURCE FROM SINGLE MANUFACTURER.

2. COMPLY WITH ASME: COMPLY WITH AWWA C606 FOR GROOVED-END CONNECTIONS. NSF 61 ANNEX G AND NSF 372 FOR VALVE MATERIALS FOR POTABLE-WATER SERVICE.

3. BRONZE VALVES SHALL BE MADE WITH DEZINCIFICATION-RESISTANT MATERIALS. BRONZE VALVES MADE WITH COPPER ALLOY (BRASS) CONTAINING MORE THAN 15 PERCENT ZINC ARE NOT PERMITTED. VALVE PRESSURE-TEMPERATURE RATINGS: NOT LESS THAN INDICATED AND AS REQUIRED FOR SYSTEM PRESSURES AND TEMPERATURES. VALVE SIZES: SAME AS UPSTREAM PIPING UNLESS OTHERWISE INDICATED. VALVE BYPASS AND DRAIN CONNECTIONS: MSS SP-45.

4. BRONZE SWING CHECK VALVES

A. CLASS 125, BRONZE, SWING CHECK VALVES WITH BRONZE DISC.

B. MANUFACTURERS: NIBCO INC., WATTS.

C. DESCRIPTION: STANDARD: MSS SP-80, TYPE 3, CWP RATING: 300 PSIG, BODY DESIGN: HORIZONTAL FLOW, BODY MATERIAL: ASTM B 82, BRONZE-, ENDS: THREADED OR SOLDERED. SEE VALVE SCHEDULE ARTICLES, DISC: BRONZE.

SECTION 22 05 29 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

1. SINGLE PIPE HANGERS WILL BE MALLEABLE IRON OR CARBON STEEL, ADJUSTABLE SWIVEL, SPLIT RING FOR PIPE SIZES UP TO 1 INCHES AND CARBON STEEL, ADJUSTABLE CLEVIS HANGERS FOR PIPE SIZES 2 INCHES AND OVER.

2. PROVIDE COPPER HANGERS OR WONDER TAPE AT ALL HANGERS FOR DISSIMILAR METALS.

3. MULTIPLE OR TRAPEZE HANGERS WILL BE UNISTRUT STEEL CHANNELS WITH UNISTRUT CLAMPS AND ALL-THREAD HANGER RODS AND DOUBLE NUTS. WHERE REQUIRED, UNISTRUT CHANNELS MAY BE REPLACED WITH STRUCTURAL STEEL CHANNELS OR I-BEAMS TO MEET STRUCTURAL LOADING.

4. WALL SUPPORTS WILL BE UNISTRUT BOLTED TO THE WALL WITH UNISTRUT PIPE CLAMPS FOR PIPE SIZES UP TO 3 INCHES AND WELDED STEEL BRACKETS AND STEEL CLAMPS FOR PIPE SIZES GREATER THAN 4 INCHES.

5. VERTICAL PIPE WILL BE SUPPORTED WITH STEEL RISER CLAMPS FOR STEEL PIPE AND COPPER RISER CLAMPS FOR COPPER PIPE.

6. GALVANIZED STEEL PIPE INSULATION SHIELDS WILL BE PROVIDED FOR PIPE SIZES UP TO 2 INCH. PRE-FABRICATED PIPE INSERTS AND SHIELDS WILL BE USED FOR PIPE SIZES OVER 2 INCHES.

7. HORIZONTAL PIPE WILL BE SUPPORTED WITH THE MAXIMUM SPACING:

| PIPE SIZE (INCH) | COPPER PIPE (FT.) | STEEL PIPE (FT.) |
|------------------|-------------------|------------------|
| 1/2"             | 6'                | 6'               |
| 3/4" TO 1"       | 6'                | 8'               |
| 1-1/4" TO 1-1/2" | 6'                | 10'              |
| 2"               | 10'               | 10'              |
| 2-1/2" TO 4"     | 10'               | 10'              |
| 6" TO 8"         | 10'               | 10'              |

8. ALL PIPING WILL BE BRACED IN ACCORDANCE WITH NFPA 13, ASME B31.1, ASCO 705, AND ASCE 7-10 STANDARDS FOR THE APPROPRIATE SEISMIC HAZARD LEVEL.

9. INSERTS WILL BE USED TO SUPPORT PIPING FROM CONCRETE STRUCTURES. WHERE INSERTS ARE NOT USED DRILL-IN EXPANSION BOLTS WILL BE USED. LOW VELOCITY SHOT PIN FASTENERS WILL ALSO BE USED WHERE PERMITTED. APPROPRIATE DRILL-IN ANCHORS WILL BE USED ON TENSION SLABS.

SECTION 22 05 53 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

1. EQUIPMENT: PERMANENT LABEL (STENCIL, METAL TAG OR ENGRAVED PLASTIC) WITH UNIT TAG OR NAME AND AREA OR SPACE SERVED.

2. PIPING:

A. PIPING LABEL MANUFACTURERS: BRIMAR INDUSTRIES, INC., CRAFTMARK PIPE MARKERS., SETON IDENTIFICATION PRODUCTS.

B. GENERAL REQUIREMENTS FOR MANUFACTURED PIPE LABELS: PREPRINTED, COLOR-CODED, WITH LETTERING INDICATING SERVICE, AND SHOWING FLOW DIRECTION ACCORDING TO ASME A13.1. PROVIDE PIPE MARKERS EVERY 20 FEET. IDENTIFY SERVICE AND FLOW DIRECTION. INSTALL IN CLEAR VIEW AND ALIGN WITH AXIS OF PIPING.

C. PRETENSIONED PIPE LABELS: PRECOILED, SEMIRIGID PLASTIC FORMED TO PARTIALLY COVER CIRCUMFERENCE OF PIPE AND TO ATTACH TO PIPE WITHOUT FASTENERS OR ADHESIVE.

D. SELF-ADHESIVE PIPE LABELS: PRINTED PLASTIC WITH CONTACT-TYPE, PERMANENT-ADHESIVE BACKING.

E. PIPE LABEL CONTENTS: INCLUDE IDENTIFICATION OF PIPING SERVICE USING SAME DESIGNATIONS OR ABBREVIATIONS AS USED ON DRAWINGS. ALSO INCLUDE PIPE SIZE AND AN ARROW INDICATING FLOW DIRECTION.

F. FLOW-DIRECTION ARROWS: INTEGRAL WITH PIPING SYSTEM SERVICE LETTERING TO ACCOMMODATE BOTH DIRECTIONS OR AS SEPARATE UNIT ON EACH PIPE LABEL TO INDICATE FLOW DIRECTION, LETTERING SIZE: SIZE LETTERS ACCORDING TO ASME A13.1 FOR PIPING.

G. PIPE LABEL COLOR SCHEDULE:

a. DOMESTIC WATER PIPING:

• BACKGROUND: SAFETY GREEN

• LETTER COLORS: WHITE

b. SANITARY WASTE AND STORM DRAINAGE PIPING:

• BACKGROUND: SAFETY BLACK

• LETTER COLORS: WHITE

SECTION 22 07 19 - PLUMBING PIPING INSULATION

1. PIPE INSULATION WILL BE PROVIDED ON ALL COLD WATER, HOT WATER AND STEAM PIPING. THE INSULATION WILL BE GLASS FIBER WITH A K-FACTOR OF 0.24 AT 75°F, NON-COMBUSTIBLE, WITH A SERVICE TEMPERATURE RANGE OF -20°F TO 450°F.

2. AN ALL SERVICE JACKET WITH INTEGRAL VAPOR BARRIER CONSISTING OF WHITE KRAFT PAPER WITH FOIL FACE AND SEALED JOINTS WILL BE USED ON LOW TEMPERATURE SERVICES. FITTING COVERS WILL BE PVC ONE-PIECE, PRE-MOLDED TYPE.

3. ALL INSULATED EXTERIOR PIPING WILL BE COVERED WITH 0.006 INCH THICK CORRUGATED ALUMINUM JACKET. PIPE INSULATION THICKNESS WILL BE AS FOLLOWS:

| INSULATION THICKNESS (IN.) FOR NOMINAL PIPE SIZES |                       |             |              |              |        |           |
|---------------------------------------------------|-----------------------|-------------|--------------|--------------|--------|-----------|
| SYSTEM                                            | TEMPERATURE RANGE (F) | 1" AND LESS | 1-1/4" TO 2" | 2-1/2" TO 4" | 6"     | 8" AND UP |
| DOMESTIC WATER                                    | ALL                   | 1"          | 1"           | 1"           | 1-1/2" | 1-1/2"    |

SECTION 22 11 16 - DOMESTIC WATER PIPING

1. COPPER TUBE AND FITTINGS:

2. PIPING

A. UNDERGROUND: NOT IN SCOPE.

B. ABOVE GROUND: ASTM B-88 TYPE 1 SEAMLESS COPPER TUBING, HARD TEMPER, COLD DRAWN. PEX-a, ASTM F 876/877, SDR 9, CTS, ENGEL METHOD.

3. FITTINGS:

A. COPPER TUBING: ANSI B16.22 WROUGHT COPPER SOLDER SWEAT TYPE.

B. CAST-COPPER, SOLDER-JOINT FITTINGS: PRESSURE FITTINGS.

C. WROUGHT-COPPER, SOLDER-JOINT FITTINGS: WROUGHT-COPPER PRESSURE FITTINGS.

D. BRASS: ANSI B16.15 85% RED BRASS, CAST IRON PATTERN. SCREWED TYPE MAY BE USED WHERE CLOSE FITTING IS REQUIRED.

E. ADAPTERS WILL BE USED WHERE COPPER TUBING CONNECTS TO IRON PIPE SIZE BRASS PIPING.

F. PEX-a ENGINEERED POLYMER OR LEAD-FREE BRASS ASTM F 1960 COLD EXPANSION WITH PEX-a REINFORCING RING.

G. FLANGES: VICTALUC MECHANICAL GROOVED COUPLINGS.

H. COPPER UNIONS:

a. CAST-COPPER-ALLOY, HEXAGONAL, STOCK BODY, BALL-AND-SOCKET, METAL-TO-METAL SEAL IN EACH END.

b. COPPER PRESSURE-SEAL-JOINT FITTINGS:

a. MANUFACTURERS: VIEGA LLC.

b. FITTINGS FOR NPS 2 AND SMALLER: WROUGHT-COPPER FITTING WITH EPDM-RUBBER, O-RING SEAL IN EACH END.

c. FITTINGS FOR NPS 2-1/2 TO NPS 4: CAST-BRONZE OR WROUGHT-COPPER FITTING WITH EPDM-RUBBER, O-RING SEAL IN EACH END.

SECTION 22 11 19 - DOMESTIC WATER PIPING SPECIALTIES

1. DUAL-CHECK VALVE BACKFLOW PREVENTER

a. STANDARD: ASSE 1024.

b. OPERATION: CONTINUOUS PRESSURE APPLICATIONS.

c. SIZE: NPS 1/2 TO NPS 1-1/4.

d. BODY: BRONZE WITH UNION INLET.

OUTLET BOXES

A. ICEMAKER OUTLET BOXES

a. MOUNTED: RECESSED.

b. MATERIAL AND FINISH: (ENAMELED STEEL OR EPOXY PAINTED STEEL) (ENAMELED STEEL, EPOXY PAINTED STEEL OR PLASTIC) (PLASTIC) (STAINLESS STEEL) BOX AND FACEPLATE.

c. FAUCET: VALVED FITTING COMPLYING WITH ASME A112.18.1. INCLUDE NPS 1/2 OR SMALL COPPER TUBE OUTLET.

d. SUPPLY SHUTOFF FITTING: NPS 1/2 BALL VALVE AND NPS 1/2 COPPER WATER TUBING.

SECTION 22 13 16 - SANITARY WASTE AND VENT PIPING

1. MANUFACTURERS SHALL BE ISO 9001 CERTIFIED.

2. PIPE AND FITTINGS ABOVE GROUND: CISPI 310 OR ASTM C 1277, CAST IRON NO-HUB SOIL PIPE AND FITTINGS WITH STAINLESS STEEL NO-HUB COUPLINGS. ALL CAST IRON SOIL PIPE AND FITTINGS SHALL BE MARKED WITH THE COLLECTIVE TRADEMARK OF THE CAST IRON SOIL PIPE INSTITUTE AND BE LISTED BY NSF INTERNATIONAL®.

3. INDIRECT DRAIN: TYPE M, ASTM B-88, SEAMLESS HARD TEMPER, COLD DRAWN COPPER TUBING WITH ANSI B16.29 WROUGHT COPPER SOLDER JOINTS AND FITTINGS.

4. CAST IRON SOIL PIPE NO-HUB COUPLINGS: CONFORMING TO ASTM C 1540. HEAVY DUTY, 24 GAUGE, TYPE 304 STAINLESS STEEL, SHIELD AND CLAMP ASSEMBLY WITH ASTM C 564 NEOPRENE SEALING SLEEVE TORQUED TO A MINIMUM OF 100 INCH.LBS. COUPLING WILL BE FACTORY MUTUAL APPROVED PER STANDARD 1680, CLASS I. (HUSKY 4000 OR EQUIVALENT)

SECTION 22 33 00 - ELECTRIC DOMESTIC WATER HEATER

1. COMMERCIAL, LIGHT-DUTY, STORAGE, ELECTRICAL, DOMESTIC-WATER HEATERS:

A. STANDARD: UL 174.

B. STORAGE-TANK CONSTRUCTION: STEEL, VERTICAL ARRANGEMENT.

A. TAPPINGS: ASME B1.20.1 PIPE THREAD.

B. PRESSURE RATING: 150 PSIG

C. INTERIOR FINISH: COMPLY WITH NSF 61 ANNEX G BARRIER MATERIAL FOR POTABLE-WATER TANK LININGS, INCLUDING EXTENDING LINING MATERIAL INTO TAPPINGS.

C. FACTORY-INSTALLED STORAGE-TANK APPURTENANCES::

A. ANODE ROD: REPLACEABLE MAGNESIUM.

B. DIP TUBE: REQUIRED UNLESS COLD-WATER INTLET IS NEAR BOTTOM OF TANK.

C. DRAIN VALVE: ASSE 1005.

D. INSULATION: COMPLY WITH ASHRAE/IESNA 90.1.

E. JACKET: STEEL WITH ENAMELED FINISH.

F. HEAT TRAP/FITTINGS: INTLET TYPE IN COLD-WATER INLET AND OUTLET IN HOT-WATER OUTLET.

G. HEATING ELEMENTS: TWO, ELECTRIC, SCREW-IN IMMERSION TYPE, WIRED FOR SIMULTANEOUS OPERATION UNLESS OTHERWISE INDICATED. LIMITED TO 12 KW TOTAL.

H. TEMPERATURE CONTROL: ADJUSTABLE THERMOSTAT.

I. SAFETY CONTROL: HIGH-TEMPERATURE-LIMIT CUT-OFF DEVICE OR SYSTEM.

J. RELIEF VALVE: ASME RATED AND STAMPED FOR COMBINATION TEMPERATURE-AND-PRESSURE RELIEF VALVES, INCLUDING RELIEVING CAPACITY AT LEAST AS GREAT AS HEAT INPUT, AND INCLUDE PRESSURE SETTING LESS THAN DOMESTIC-WATER HEATER WORKING-PRESSURE RATING. SELECT RELIEF VALVE WITH SENSING ELEMENT THAT EXTENDS INTO STORAGE TANK.

D. CAPACITY AND CHARACTERISTICS:

A. REFER TO PLUMBING ELECTRICAL WATER HEATER SCHEDULES FOR INFORMATION.

E. ELECTRICAL CHARACTERISTICS:

A. REFER TO PLUMBING ELECTRICAL WATER HEATER SCHEDULES FOR INFORMATION.

G. INSTALL COMBINATION TEMPERATURE AND PRESSURE RELIEF VALVES IN WATER PIPING FOR ELECTRIC DOMESTIC-WATER HEATERS WITHOUT STORAGE. EXTEND COMMERCIAL WATER HEATER RELIEF-VALVE OUTLET WITH DRAIN PIPING. SAME MATERIAL AS DOMESTIC WATER PIPING, IN CONTINUOUS DOWNWARD PITCH AND DISCHARGE BY POSITIVE AIR GAP INTO CLOSEST FLOOR DRAIN/SINK.

H. INSTALL WATER HEATER DRAIN PAN PIPING: AS INDIRECT WASTE TO SPILL BY POSITIVE AIR GAP INTO OPEN DRAINS OR OVER FLOOR DRAIN SINKS. INSTALL HOSE-END DRAINS VALVES AT LOW POINT IN WATER PIPING FOR ELECTRIC DOMESTIC-WATER HEATER THAT DO NOT HAVE TANK DRAINS.

I. INSTALL THERMOMETERS ON INLET AND OUTLET PIPING FOR ELECTRICAL DOMESTIC-WATER HEATER(S).

J. ASSEMBLE AND INSTALL INLET AND OUTLET PIPING MANIFOLD KITS FOR MULTIPLE ELECTRIC DOMESTIC-WATER HEATERS. FABRICATE, MODIFY OR ARRANGE MANIFOLDS FOR BALANCED WATER FLOW THROUGH EACH ELECTRIC DOMESTIC-WATER HEATER. INCLUDE SHUTOFF VALVE AND THERMOMETER IN EACH DOMESTIC-WATER HEATER INLET AND OUTLET AND THROTTLING VALVE IN EACH ELECTRIC DOMESTIC-WATER HEATER OUTLET.

K. INSTALL PIPING-TYPE HEAT TRAPS ON INLET AND OUTLET PIPING OF ELECTRICAL DOMESTIC-WATER HEATER STORAGE TANKS WITHOUT INTEGRAL OR FITTING-TYPE HEAT TRAPS.

L. FILL ELECTRIC DOMESTIC-WATER HEATER WITH WATER.

M. CHARGE DOMESTIC-WATER COMPRESSION TANK WITH AIR.

1. DOMESTIC WATER HEATER ACCESSORIES:

A. DOMESTIC WATER COMPRESSION TANKS:

a. DESCRIPTION: STEEL, PRESSURE-RATED TANK CONSTRUCTED WITH WELDED JOINTS AND FACTORY INSTALLED BUTYL-RUBBER DIAPHRAGM. INCLUDE AIR PRECHARGE TO MINIMUM SYSTEM-OPERATION PRESSURE. TANK.

b. CONSTRUCTION:

• TAPPING FACTORY FABRICATED STEEL, WELDED TO TANK BEFORE TESTING AND LABELING. INCLUDE ASME B1.20.1 PIPE THREAD.

• INTERIOR FINISH: COMPLY WITH NSF 61 ANNEX G BARRIER MATERIALS FOR POTABLE WATER TANK LININGS, INCLUDING EXTENDING FINISH INTO AND THROUGH TANK FITTING AND OULETS.

• AIR-CHARGING VALVE: FACTORY INSTALLED.

c. CAPACITY AND CHARACTERISTICS:

• REFER TO FIXTURE SCHEDULE.

A. DIMENSIONS NOT LESS THAN BASE OF DOMESTIC WATER HEATER AND INCLUDE DRAIN OULET NOT LESS THAN NSP 3/4 WITH ASME B1.20.1 PIPE THREADS OR ASME B1.20.7 GARDEN HOSE THREADS.

B. PIPING TYPE HEAT TRAPS: FIELD FABRICATED PIPING ARRANGEMENT ACCORDING TO ASHRAE/IESNA 90.1.

C. HEAT TRAP FITTINGS: ASHRAE 90.2.

D. MANIFOLD KITS: DOMESTIC WATER HEATER MANUFACTURER'S FACTORY FABRICATED INLET AND OUTLET PIPING FOR INSTALLATION. FOR MULTIPLE DOMESTIC WATER HEATER INSTALLATION, INCLUDE BALL SHUTOFF VALVES TO ISOLATE EACH DOMESTIC WATER HEATER AND CALIBRATE BALANCING VALVES TO PROVIDE BALNCED FLOW THROUGH EACH DOMESTIC WATER HEATER.

E. COMPLY WITH REQUIREMENTS FOR BALL SHUTOFF VALVES SPECIFIED IN SECTION 220523.12 "BALL VALVES FOR PLUMBING PIPING." SECTION 220523.13 "BUTTERFLY VALVES FOR PLUMBING PIPING."

F. COMPLY WITH REQUIREMENTS FOR BALANCING VALVES SPECIFIED IN SECTION 221119 "DOMESTIC WATER PIPING SPECIALTIES."

F. COMBINATION TEMPERATURE AND PRESSURE RELIEF VALVE: INCLUDE RELIEVING CAPACITY AT LEAST AS GREAT AS HEAT INPUT AND INCLUDE PRESSURE SETTING LESS THAN DOMESTIC WATER HEATER WORKING PRESSURE RATING. SELECT RELIEF VALVES WITH SENSING ELEMENT THAT EXTENDS INTO STORAGE TANK.

G. VACUUM RELIEF VALVES: ANSI Z21.22/CSA 4.4-M.

H. DOMESTIC WATER HEATER STANDS: MANUFACTURER'S FACTORY FABRICATED STEEL STAND FOR FLOOR MOUNTING. CAPABLE OF SUPPORTING DOMESTIC WATER HEATER AND WATER. PROVIDE DIMENSION THAT WILL SUPPORT BOTTOM OF DOMESTIC WATER HEATER A MINIMUM OF 18 INCHES ABOVE THE FLOOR.

I. DOMESTIC WATER HEATER MOUNTING BRACKETS: MANUFACTURER'S FACTORY FABRICATED STEEL BRACKET FOR WALL MOUNTING. CAPABLE OF SUPPORTING DOMESTIC WATER HEATER AND WATER.

1. DOMESTIC WATER HEATER INSTALLATION:

A. COMMERCIAL DOMESTIC WATER HEATER MOUNTING: INSTALL COMMERCIAL ELECTRICAL DOMESTIC WATER HEATER ON CONCRETE BASE. COMPLY WITH REQUIREMENTS FOR CONCRETE BASES SPECIFIED IN SECTION 033000 "CAST IN PLACE CONCRETE." OR SECTION 033053 "MISCELLANEOUS CAST IN PLACE CONCRETE.

a. EXCEPTION: OMIT CONCRETE BASE FOR COMIERAL ELECTRIC DOMESTIC WATER HEATER IF INSTALLATION ON STAND, BRACKET, SUSPENDED PLATFORM OR DIRECTLY ON FLOOR IS INDICATED.

b. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES.

c. ARRANGE UNITS SO CONTROL AND DEVICES THAT REQUIRE SERVICING ARE ACCESSIBLE.

d. INSTALL DOWEL RODS TO CONNECT CONCRETE BASE TO CONCRETE FLOOR, UNLESS OTHERWISE INDICATED. INSTALL DOWEL RODS ON 18 INCH CENTERS AROUND THE FULL PERIMETER OF CONCRETE BASE.

B. INSTALL DOMESTIC WATER HEATER LEVEL AND PLUMB, ACCORDING TO LAYOUT DRAWINGS, ORIGINAL DESIGN AND REFERENCED STANDARDS. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES. ARRANGE UNITS SO CONTROLS AND DEVICES NEEDING SERVICE ARE ACCESSIBLE.

FIRE SPRINKLER COORDINATION NOTES

SECTION 21 05 00 - COMMON WORK RESULTS FOR FIRE SUPPRESSION

1. SCOPE OF WORK:

A. THE CONTRACTOR IS RESPONSIBLE FOR ALL WORK, MATERIALS, AND LABOR REQUIRED TO SATISFY A COMPLETE WORKING SYSTEM WHETHER SPECIFIED OR IMPLIED.

B. THE WORK IS TO BE PERFORMED IN STRICT COMPLIANCE WITH NFPA 13 (LATEST EDITION), ALL CODES AND OTHER NFPA REGULATIONS GOVERNING WORK OF THIS NATURE.

C. THE CONTRACTOR SHALL, BEFORE SUBMITTING ANY PROPOSAL, EXAMINE THE PROPOSED SITE AND SHALL DETERMINE FOR HIMSELF THE CONDITIONS THAT MAY EFFECT THE WORK. NO ALLOWANCE SHALL BE MADE IF THE CONTRACTOR FAILS TO MAKE SUCH EXAMINATIONS.

D. ALL EQUIPMENT AND MATERIALS SHALL BE AS SPECIFIED OR "APPROVED EQUAL" BY THE ENGINEER OR ARCHITECT.

E. WHEN NEW SPRINKLERS ARE ADDED INTO THE EXISTING SPRINKLER SYSTEM, THE NEW SPRINKLER SHALL BE OF THE SAME TYPE, ORIFICE, AND TEMPERATURE RATINGS UNLESS CONDITIONS REQUIRE A DIFFERENT TYPE SPRINKLER BE INSTALLED.

F. ISOLATE, DRAIN AND REFILL EXISTING PIPING SYSTEM AS REQUIRED TO ACCOMMODATE INSTALLATION OF NEW WORK.

G. ALL SPRINKLER LINES SHALL BE AVOIDING INTERFERENCE WITH LIGHTING FIXTURES, DUCTS, AND PIPING.

2. PERMITS:

A. THE CONTRACTOR SHALL SECURE ALL PERMITS OR APPLICATIONS AND PAY ANY AND ALL FEES.

3. SHOP DRAWINGS:

A. SUBMIT MATERIAL LIST AND SHOP DRAWINGS FOR MAJOR EQUIPMENT TO THE ARCHITECT/ENGINEER FOR APPROVAL.

SPRINKLER PIPING

A. PIPE SHALL BE STEEL, MADE TO ANY OF THE THREE SPECIFICATIONS LISTED. WELDING WILL NOT BE PERMITTED IF THERE ARE ANY SPRINKLERS CONNECTED THERETO.

a. ASTM A53

b. ASTM A120

c. ASTM A135

B. PIPING MATERIAL MAY BE SCHEDULE 10 (THINWALL) PROVIDED JOINTS ARE MADE BY ROLL-GROOVE COUPLING. THREADING WILL NOT BE PERMITTED.

C. PIPING MATERIAL MAY BE SCHEDULE 40 WITH FLANGED, ROLL-GROOVED COUPLINGS, OR THREADED JOINTS.

5. PIPE SUPPORTS:

A. ALL HANGERS MUST BE AN APPROVED TYPE BY NFPA 13. NO SPRINKLER PIPING IS TO BE SUPPORTED FROM ANY MECHANICAL, OR ELECTRICAL DEVICES.

B. ALL PIPE SHALL BE SUPPORTED FROM THE BUILDING STRUCTURE IN A NEAT AND WORKMANLIKE MANNER. VERTICAL RISERS SHALL BE SUPPORTED AT EACH FLOOR LEVEL WITH STEEL PIPE CLAMPS. THE USE OF WIRE OR STRAP METAL HANGER TO SUPPORT PIPES WILL NOT BE PERMITTED. HANGING PIPES FROM OTHER PIPES WILL NOT BE PERMITTED. PIPING SHALL BE CAREFULLY COORDINATED BEFORE INSTALLATION WITH OTHER SYSTEMS AND EQUIPMENT IN CHASES AND OTHER CONGESTED AREAS.

C. MAXIMUM DISTANCE BETWEEN PIPE SUPPORTS:

a. 12'-0" FOR 1 1/4" DIAMETER PIPE AND SMALLER

b. 15'-0" FOR 1 1/2" DIAMETER PIPE AND LARGER

6. WORKING PLANS:

A. ANY DRAWINGS INCLUDED AS PART OF THIS SET OF CONSTRUCTION DOCUMENTS ARE DEFINED AS PRELIMINARY SPRINKLER PLANS AS SPECIFIED IN SECTION 23.1. OF NFPA 13. FINAL WORKING DRAWINGS SHALL BE PREPARED BY THE SPRINKLER CONTRACTOR AND REVIEWED AND APPROVED BY THE ENGINEER.

7. TESTS:

A. ACCEPTANCE AND HYDROSTATIC TESTS TO BE PERFORMED IN ACCORDANCE WITH CHAPTER 25 OF NFPA 13.

8. MISCELLANEOUS:

A. FIELD VERIFY ALL FIGURES, CONDITIONS, AND DIMENSIONS AT THE JOB SITE.

B. ANY PRELIMINARY SPRINKLER PLANS ARE INTENDED TO BE DIAGRAMMATIC. THEY ARE NOT INTENDED TO SHOW EVERY ITEM IN ITS EXACT LOCATION.

9. GUARANTEE:

A. MATERIALS, EQUIPMENT AND INSTALLATION SHALL BE GUARANTEED FOR PERIOD OF ONE (1) YEAR FROM DATE OF ACCEPTANCE. DEFECTS WHICH APPEAR DURING THAT PERIOD SHALL BE CORRECTED AT THE SPRINKLER CONTRACTOR'S EXPENSE.



PROJECT #

23.1002

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SEAL



PROJECT

LUMINARY  
RICE VILLAGE  
2556 UNIVERSITY BLVD, SUITE A-103  
HOUSTON, TEXAS 77005

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Costruction Drawings

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ISSUE FOR PERMIT

03/13/23

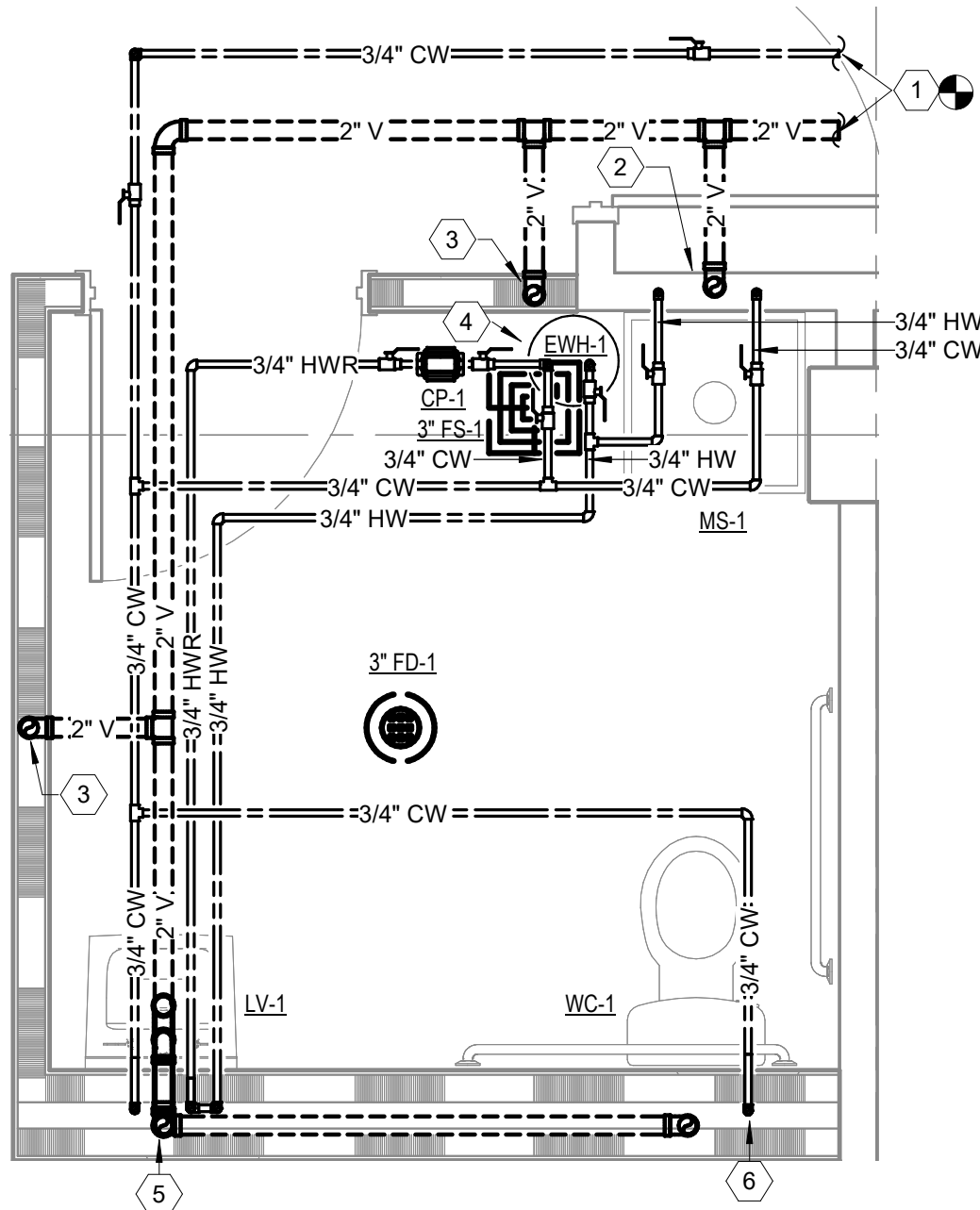
SHEET

PLUMBING  
FLOOR PLAN

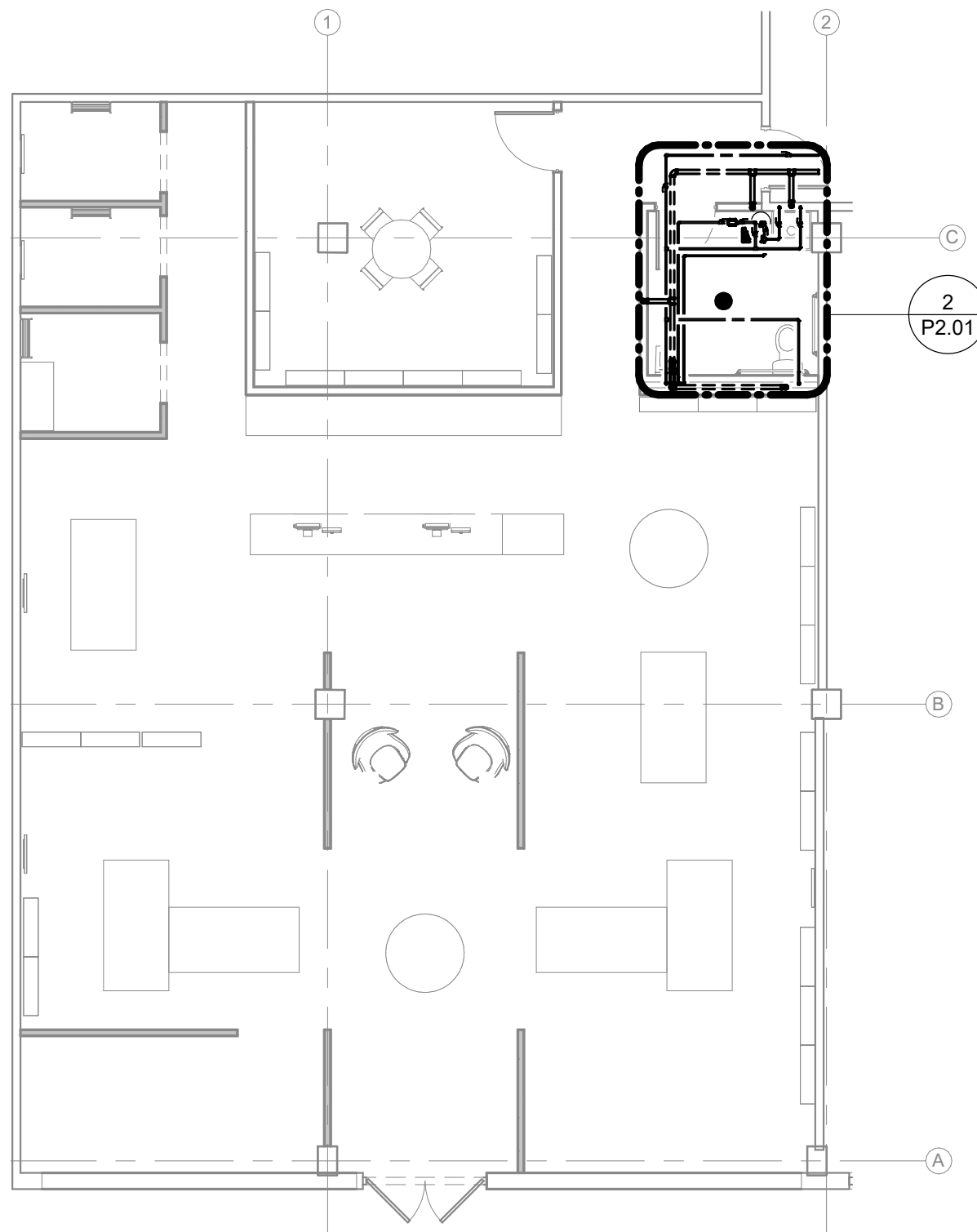
P2.01

## KEYNOTES

| KEYNOTE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | CONNECT NEW 3/4" COLD WATER AND 2" VENT TO EXISTING RESPECTIVE PIPING SYSTEM. CONTRACTOR SHALL FIELD VERIFY EXACT LOCATION, SIZE, AND ELEVATION OF EXISTING PIPING PRIOR TO COMMENCING NEW WORK.                                                                                                                                                                                                                                            |
| 2       | ROUTE 3/4" COLD WATER AND 3/4" HOT WATER DOWN WITHIN WALL TO FIXTURE WATER CONNECTION POINT. ROUTE 2" VENT UP WITHIN WALL INTO CEILING AND CONNECT TO VENT SYSTEM.                                                                                                                                                                                                                                                                          |
| 3       | ROUTE 2" VENT UP WITHIN WALL INTO CEILING AND CONNECT TO VENT SYSTEM.                                                                                                                                                                                                                                                                                                                                                                       |
| 4       | NEW ELECTRIC WATER HEATER IN CEILING SPACE. ROUTE 3/4" COLD WATER TO WATER HEATER AND ROUTE 3/4" HOT WATER FROM WATER HEATER TO SERVE PLUMBING FIXTURES. ROUTE T&P AND DRAIN PAN LINE TO INDIRECT WASTE AT FLOOR SINK BELOW. PROVIDE AIR GAP FOR INDIRECT WASTE. CONNECT 3/4" HOT WATER RETURN PIPE TO COLD WATER PIPE GOING TO WATER HEATER. INCLUDE RECIRCULATION PUMP TO HOT WATER RETURN LINE BEFORE CONNECTION INTO COLD WATER PIPING. |
| 5       | ROUTE 2" SANITARY UP THROUGH SLAB BELOW TO FIXTURE DRAIN CONNECTION POINT. ROUTE 2" VENT UP WITHIN WALL INTO CEILING AND CONNECT TO VENT SYSTEM. ROUTE 3/4" COLD WATER AND 3/4" HOT WATER DOWN WITHIN WALL TO FIXTURE WATER CONNECTION POINT. CONNECT NEW 3/4" HOT WATER RETURN TO HOT WATER PIPING SERVING FIXTURE AND ROUTE BACK UP WITHIN WALL TO RECIRCULATION PUMP LOCATION.                                                           |
| 6       | ROUTE 3/4" COLD WATER DOWN WITHIN WALL TO FIXTURE WATER CONNECTION POINT. ROUTE 2" VENT WITHIN WALL AND CONNECT TO VENT SYSTEM.                                                                                                                                                                                                                                                                                                             |



**2 PLUMBING ENLARGED PLAN**  
1/2" = 1'-0"



**1 PLUMBING PLAN**  
1/8" = 1'-0"

PROJECT #

23.1002

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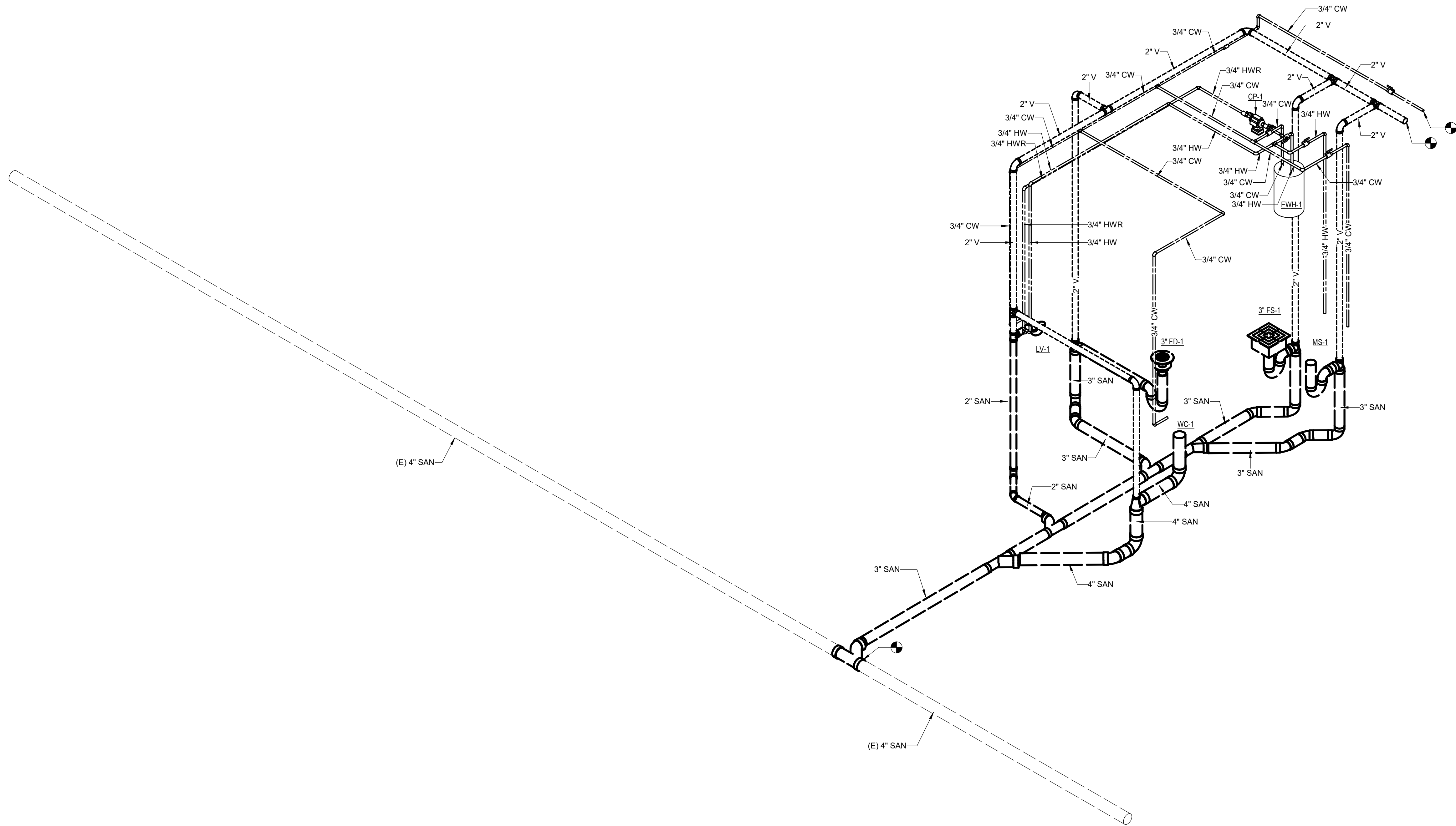
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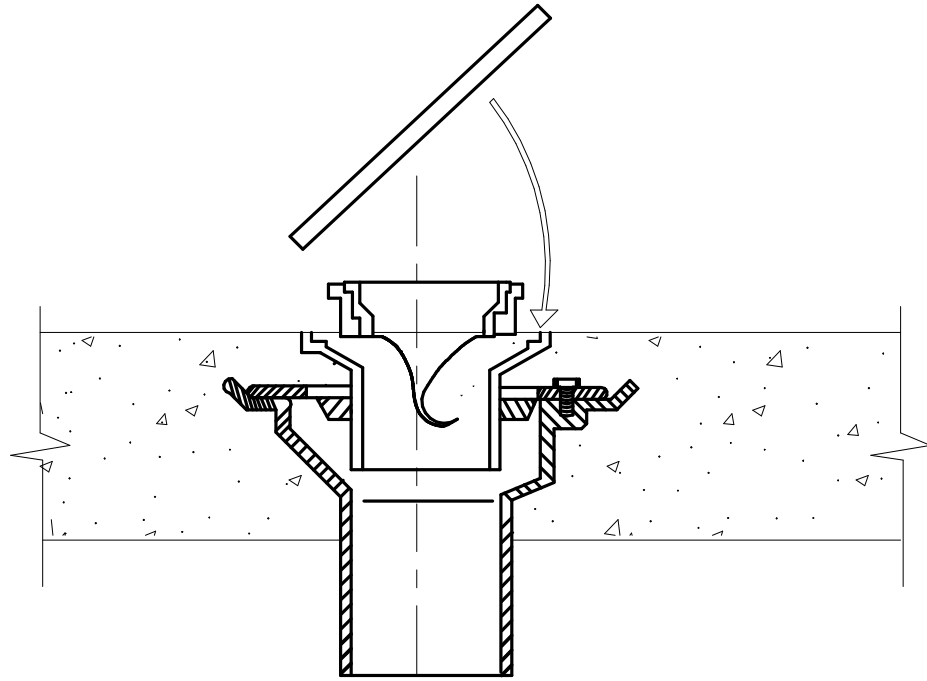
PLUMBING  
RISERS

P5.01



1 PLUMBING RISER DIAGRAM

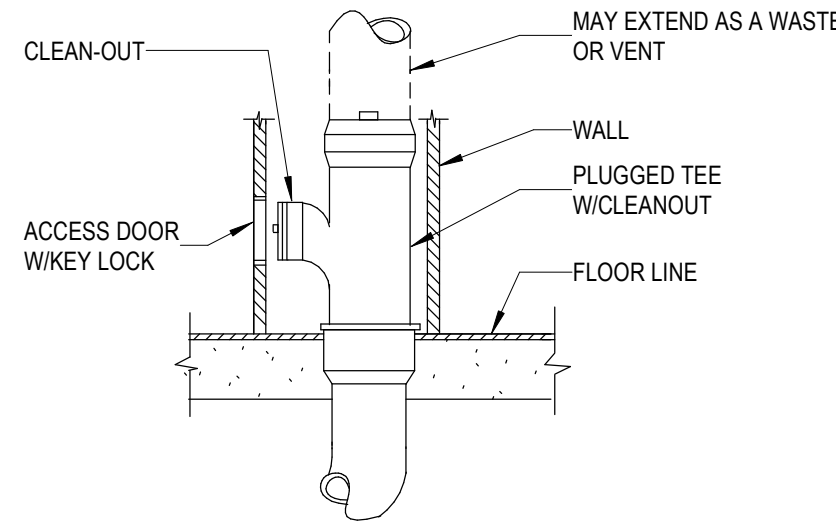
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1. TRAP PRIMER ALTERNATE DEVICES RECOGNIZED IN THIS LISTING IS INTENDED TO SERVE AS ALTERANTES TO THE CODE-REQUIRED TRAP PRIMER IN FLOOR DRAINS CONFORMING TO EITHER ASME/ANSI A112.6.3 OR CSA B79, AND MUST SERVE A FUNCTIONING TRAP.
2. THE PLUMBING SYSTEMS MUST BE INSTALLED IN ACCORDANCE WITH THE APPLICABLE CODE IN UPC 2015 WHEN INSTALLED.
3. THE FLOOR DRAIN MUST BE ACCESSIBLE AND INSTALLED ABOVE THE TRAP SEAL OF THE FLOOR DRAIN.
4. TRAP PRIMER ALTERNATES ARE MANUFACTURED UNDER A QUALITY CONTROL PROGRAM WITH YEARLY INSPECTIONS BY ICC-ES

## 6 COH TRAP SEAL DETAIL

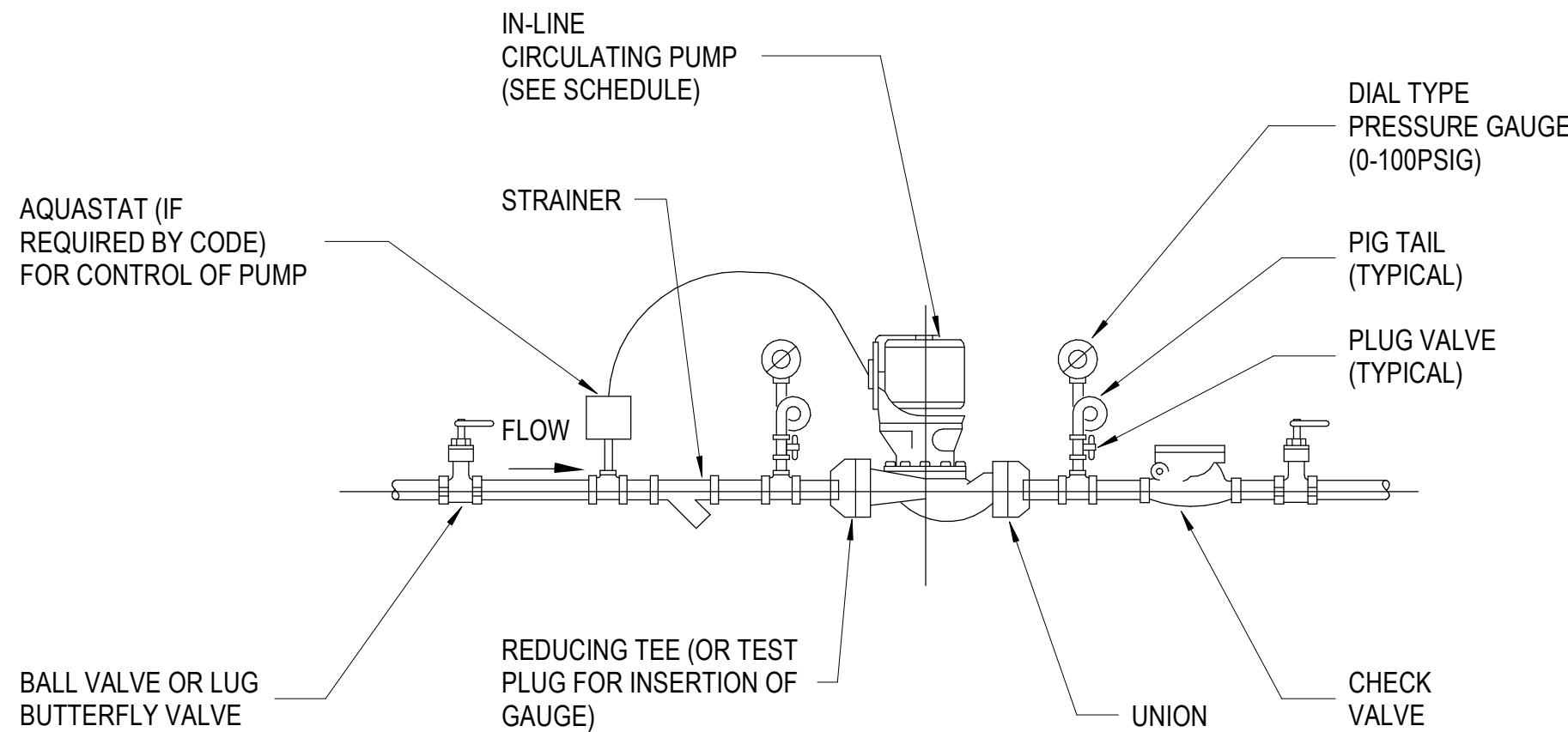
NOT TO SCALE



- NOTES:
1. BALANCE OF PIPING SAME AS CLEANOUT TO GRADE.

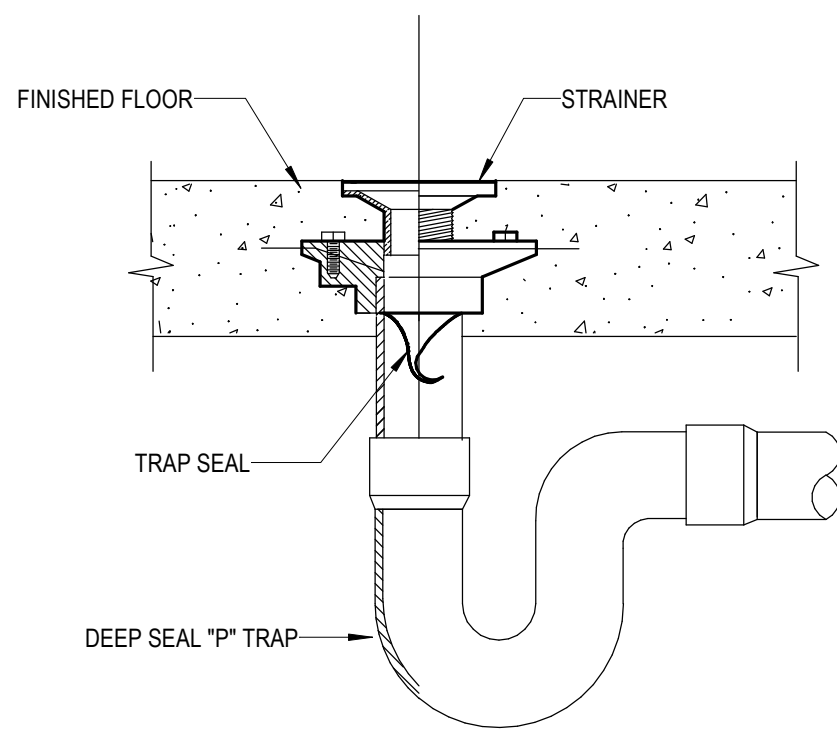
## 3 WALL CLEANOUT DETAIL

NOT TO SCALE



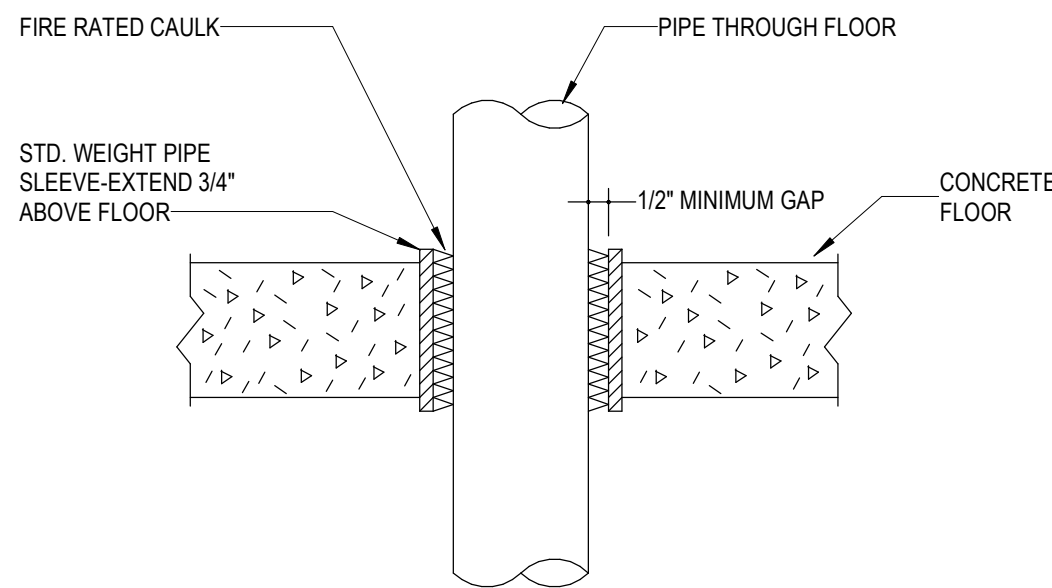
## 8 IN-LINE CIRCULATING PUMP DETAIL

NOT TO SCALE



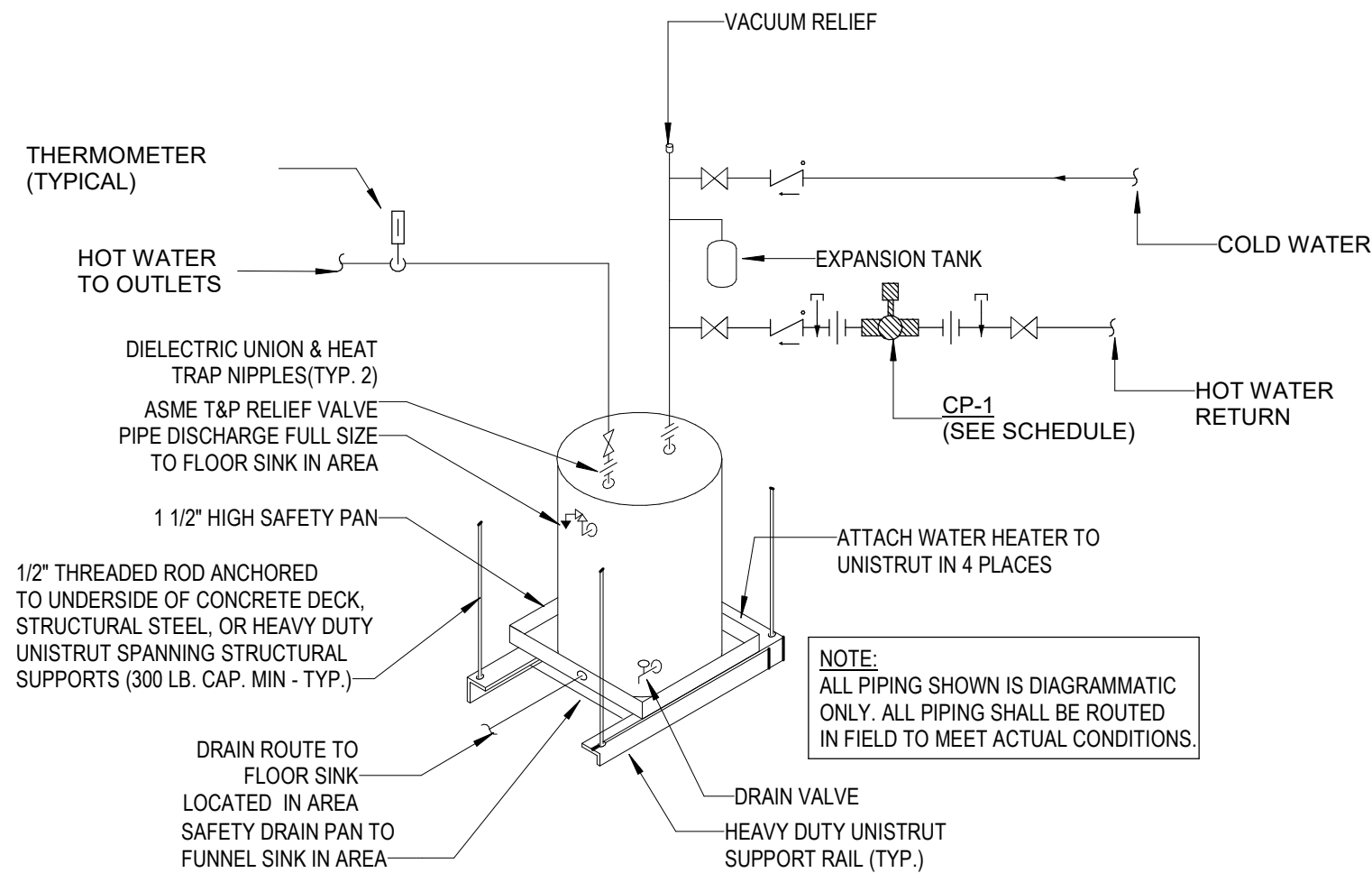
## 5 FLOOR DRAIN DETAIL

NOT TO SCALE



## 2 PIPE SLEEVE THROUGH FLOOR DETAIL

NOT TO SCALE

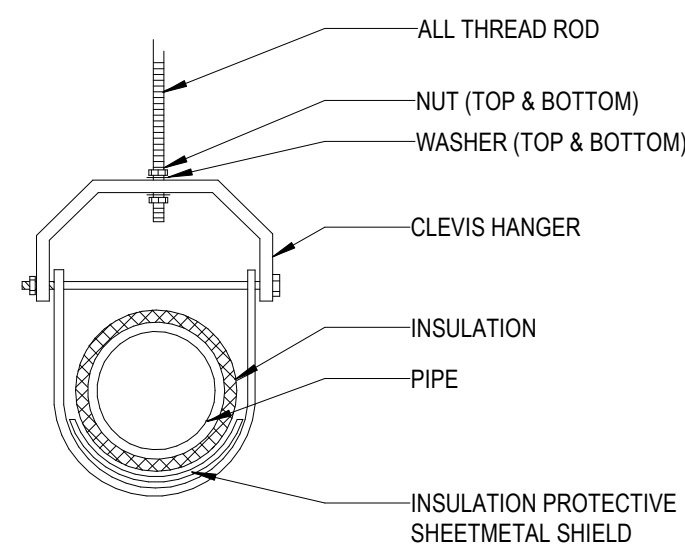


NOTES:

1. TEMPERATURE RISE.
2. HEATERS SHALL BE UL OR ETL LISTED AND SHALL MEET ALL ENERGY CONSERVATION CODE REQUIREMENTS OF THE AUTHORITY HAVING JURISDICTION.
3. REFER TO FLOORPLANS AND RISER DIAGRAMS FOR PIPE SIZING.
4. INSULATE ALL HOT WATER PIPING.
5. INSULATE ALL COLD WATER PIPING WITHIN 8'-0" OF THE WATER HEATER PER 2000 IECC.

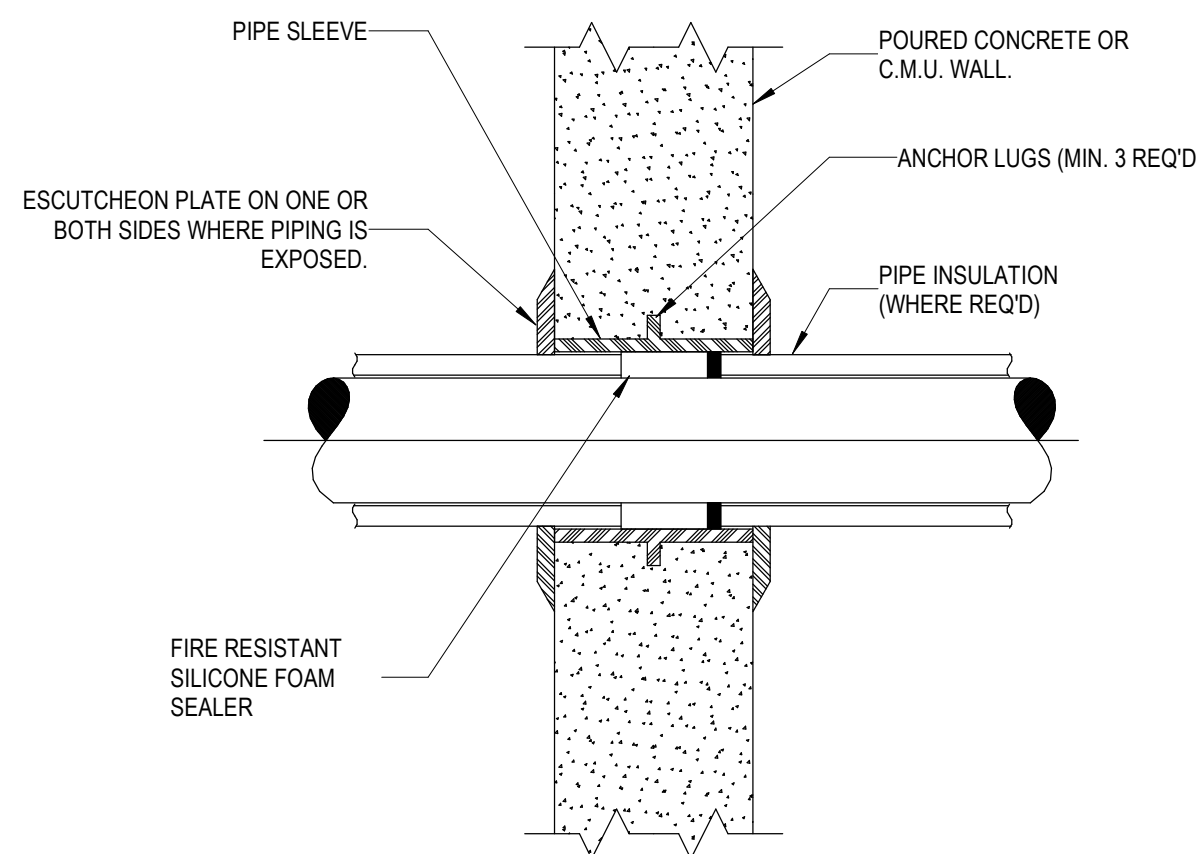
## 7 SUSPENDED WATER HEATER W/ CIRCULATING PUMP DETAIL

NOT TO SCALE



## 4 TYPICAL CLEVIS HANGER DETAIL

NOT TO SCALE



- NOTES:
1. FOR INTERIOR WALLS AND BELOW GRADE GROUND TO GROUND WALLS.

## 1 PIPE SLEEVE THROUGH WALL DETAIL

NOT TO SCALE





DESIGN TEAM

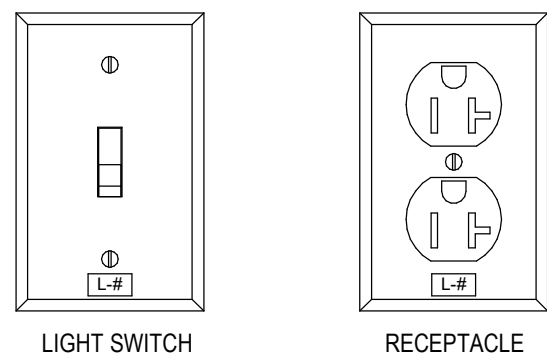
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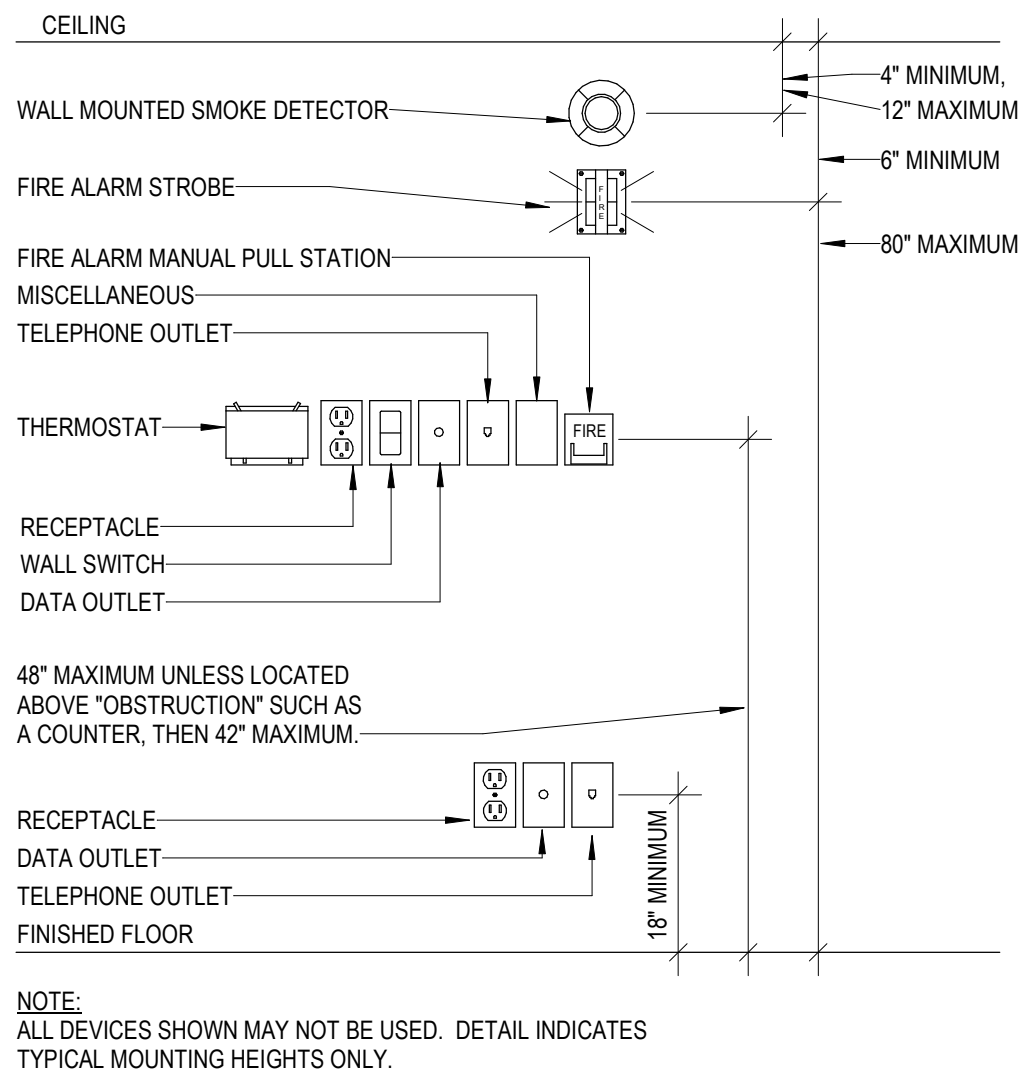
PROJECT



CIRCUIT LABEL  
L = PANEL DESIGNATION  
# = CIRCUIT DESIGNATION

### 3 DEVICE PLATES - CIRCUIT LABEL

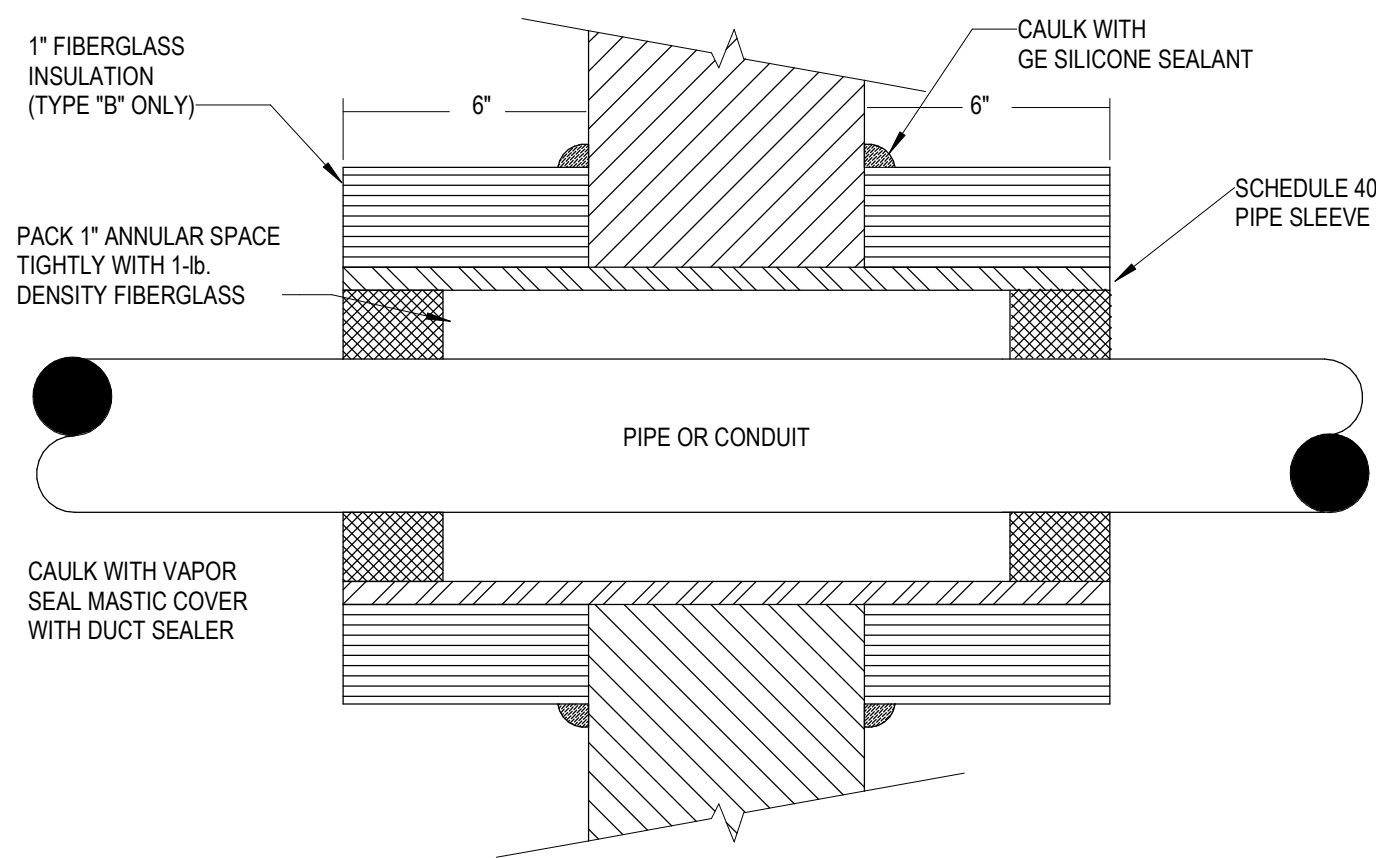
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NOTE:  
ALL DEVICES SHOWN MAY NOT BE USED. DETAIL INDICATES  
TYPICAL MOUNTING HEIGHTS ONLY.

### 2 MOUNTING HEIGHT DETAIL

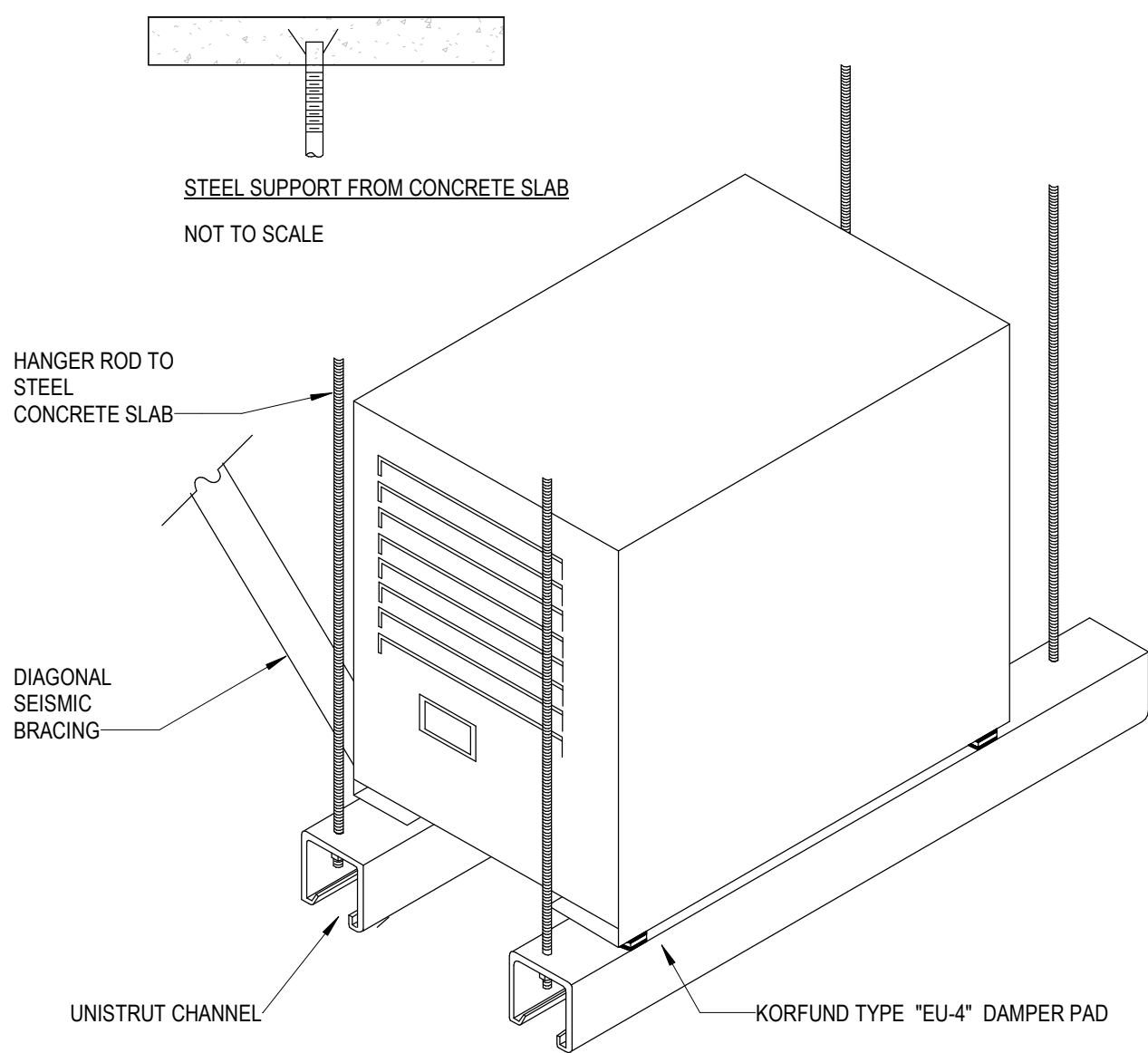
NOT TO SCALE



- NOTES:
1. TYPICAL TREATMENT FOR CONDUIT AND H.V.A.C. PIPING FOR NON-FIRE-RATED WALLS.
  2. SIZE SLEEVE TO ACCEPT CONDUIT.
  3. FOR FIRE WALLS PROVIDE U.L. APPROVED SEAL ASSEMBLY FOR OPENING SIZE AND WALL TYPE.

### 1 CONDUIT SEALANT THRU WALLS

NOT TO SCALE



### 4 CEILING HUNG TRANSFORMER DETAIL

NOT TO SCALE

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SHEET

ELECTRICAL  
DETAILS

E0.03

| FIRE ALARM NOTES |                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A                | FIRE ALARM SYSTEM DESIGN (DEVICES AND LAYOUT) ARE UNDER SEPARATE PERMIT BY THE FIRE ALARM CONTRACTOR.                                                                                                                                                                                                                           |
| B                | FIRE ALARM SYSTEM CONSTRUCTION DOCUMENTS FOR THE SCOPE OF WORK INDICATED IN THIS PROJECT SHALL BE SUBMITTED TO THE LOCAL AUTHORITY HAVING JURISDICTION FOR APPROVAL PRIOR TO COMMENCING FIRE ALARM WORK AND THE INSTALLATION MUST BE APPROVED BY THE LOCAL AUTHORITY HAVING JURISDICTION AFTER COMPLETION.                      |
| C                | AN EXISTING FIRE ALARM SYSTEM IS IN PLACE. REUSE EXISTING DEVICES IN GOOD CONDITION AND PROVIDE NEW DEVICES TO MATCH EXISTING WHERE NECESSARY. COORDINATE DEVICE LOCATIONS WITH ARCHITECTURAL DRAWINGS. SUBMIT SHOP DRAWINGS AND SEQUENCE OF OPERATIONS TO ENGINEER FOR REVIEW.                                                 |
| D                | THE FIRE ALARM SYSTEM MODIFICATIONS FOR THIS PROJECT SHALL BE DESIGNED BY A LICENSED FIRE ALARM CONTRACTOR AND BE IN ACCORDANCE WITH NFPA 72 & 101 AND THE LOCAL AUTHORITY HAVING JURISDICTION'S BUILDING CODE. CONTRACTOR IS RESPONSIBLE FOR SUBMISSION OF PLANS TO THE CITY FOR APPROVAL AND ALL ASSOCIATED FEES.             |
| E                | ALL 120V CIRCUITS REQUIRED FOR THE OPERATION OF THE FIRE ALARM SYSTEM SHALL BE INCLUDED. LOCATIONS OF ALL PANELS AND BOOSTERS SHALL BE COORDINATED WITH ARCHITECT. CONTRACTOR SHALL TEST THE SYSTEM IN THE PRESENCE OF LOCAL AUTHORITIES AND MAKE ALL REQUIRED MODIFICATIONS AND ADDITIONS TO THE DESIGN AT NO ADDITIONAL COST. |

| GENERAL NOTES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A             | REMOVE ALL ABANDONED CABLING AND CONDUIT IN SPACE. TERMINATE CONDUITS AT ELECTRICAL ROOM WITH JUNCTION BOX. TURN OFF UNUSED BREAKERS AND UPDATE PANEL DIRECTORY.                                                                                                                                                                                                                                                                                                                                                  |
| B             | REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION AND MOUNTING HEIGHTS OF ALL DEVICES. REFER TO IT REPRESENTATIVE FOR EXACT LOCATION OF IT DEVICES. REFER TO AV DRAWINGS FOR A DETAILED LAYOUT OF CONDUIT AND CABLE FOR AV DEVICES.                                                                                                                                                                                                                                                                              |
| C             | PROVIDE RECESSED CLOCK TYPE BOX FOR ALL WALL MOUNTED TV POWER AND AV DEVICES.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| D             | CONTRACTOR SHALL TRACE EXISTING CIRCUITS FOR ALL PANELS IN SCOPE AND REUSE THOSE THAT ARE NOT CURRENTLY BEING USED OR BECOME AVAILABLE DUE TO DEMO. CIRCUIT NUMBERS SHOWN ARE DIAGRAMMATIC AND MAY NEED TO BE MODIFIED FOR FIELD CONDITIONS. CONTRACTOR SHALL COORDINATE ALL WORK WITH BUILDING ENGINEER AND MAY NEED TO SCHEDULE WORK AFTER HOURS. CONTRACTOR SHALL CHECK ALL PANELS FOR ELECTRICAL LOAD. IF SUFFICIENT NUMBER OF CIRCUITS CANNOT BE LOCATED AND REUSED, ENGINEER SHALL BE NOTIFIED IMMEDIATELY. |
| E             | ALL 125-VOLT THROUGH 250-VOLT RECEPTACLES SUPPLIED BY SINGLE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 50 AMPERES OR LESS, AND ALL RECEPTACLES SUPPLIED BY THREE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 100 AMPERES OR LESS, INSTALLED IN THE LOCATIONS SPECIFIED IN 210.8(B) (1) THROUGH (B) (12) SHALL HAVE GROUND FAULT CIRCUIT-INTERRUPTER PROTECTION FOR PERSONNEL. 210.8(B)(2) KITCHENS OR AREAS WITH A SINK AND PROVISIONS FOR EITHER FOOD PREPARATION OR COOKING.        |

| KEYNOTES |                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KEYNOTE  | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                |
| 1        | PROVIDE FLOOR BOX. FLOOR BOX SHALL BE HUBBELL S1PFB, S1SP4X4, AND S1CFCBRS. PROVIDE 3/4" CW PULLSTRING FOR DATA UNDER FLOOR TO NEAREST WALL, STUBBED TO ACCESSIBLE CEILING, AND 3/4" C FOR POWER UNDER FLOOR TO NEAREST WALL. HOMERUN TO PANEL. COORDINATE EXACT LOCATION OF POWER/DATA DEVICES AND FINISH WITH ARCHITECT. X-RAY SLAB PRIOR TO SAW CUT. PATCH AS REQUIRED. |
| 2        | PROVIDE J-BOX FOR SIGNAGE. COORDINATE EXACT LOCATION WITH ARCHITECT. HOMERUN VIA TIME SWITCH 'T1'. COORDINATE EXACT SCHEDULE WITH BUILDING OWNER.                                                                                                                                                                                                                          |
| 3        | COORDINATE EXACT LOCATION OF NEW ELECTRICAL EQUIPMENT WITH BUILDING OWNER PRIOR TO INSTALLATION. MAINTAIN CLEARANCES AS REQUIRED BY NEC ART. 110.26.                                                                                                                                                                                                                       |
| 4        | PROVIDE NEW TIME SWITCH 'T1' INTERMATIC #ET2845C. COORDINATE EXACT LOCATION WITH BUILDING ENGINEER.                                                                                                                                                                                                                                                                        |
| 5        | DISCONNECT SHALL BE 240V/60A/2P/1N1.                                                                                                                                                                                                                                                                                                                                       |
| 6        | TIE TO LOCAL LIGHT SWITCH.                                                                                                                                                                                                                                                                                                                                                 |
| 7        | HOMERUN CP-1 VIA TIME CLOCK PROVIDED BY PLUMBING CONTRACTOR. COORDINATE EXACT SCHEDULE WITH OWNER.                                                                                                                                                                                                                                                                         |
| 8        | DISCONNECT SHALL BE 600V/30A/1P/1N1.                                                                                                                                                                                                                                                                                                                                       |

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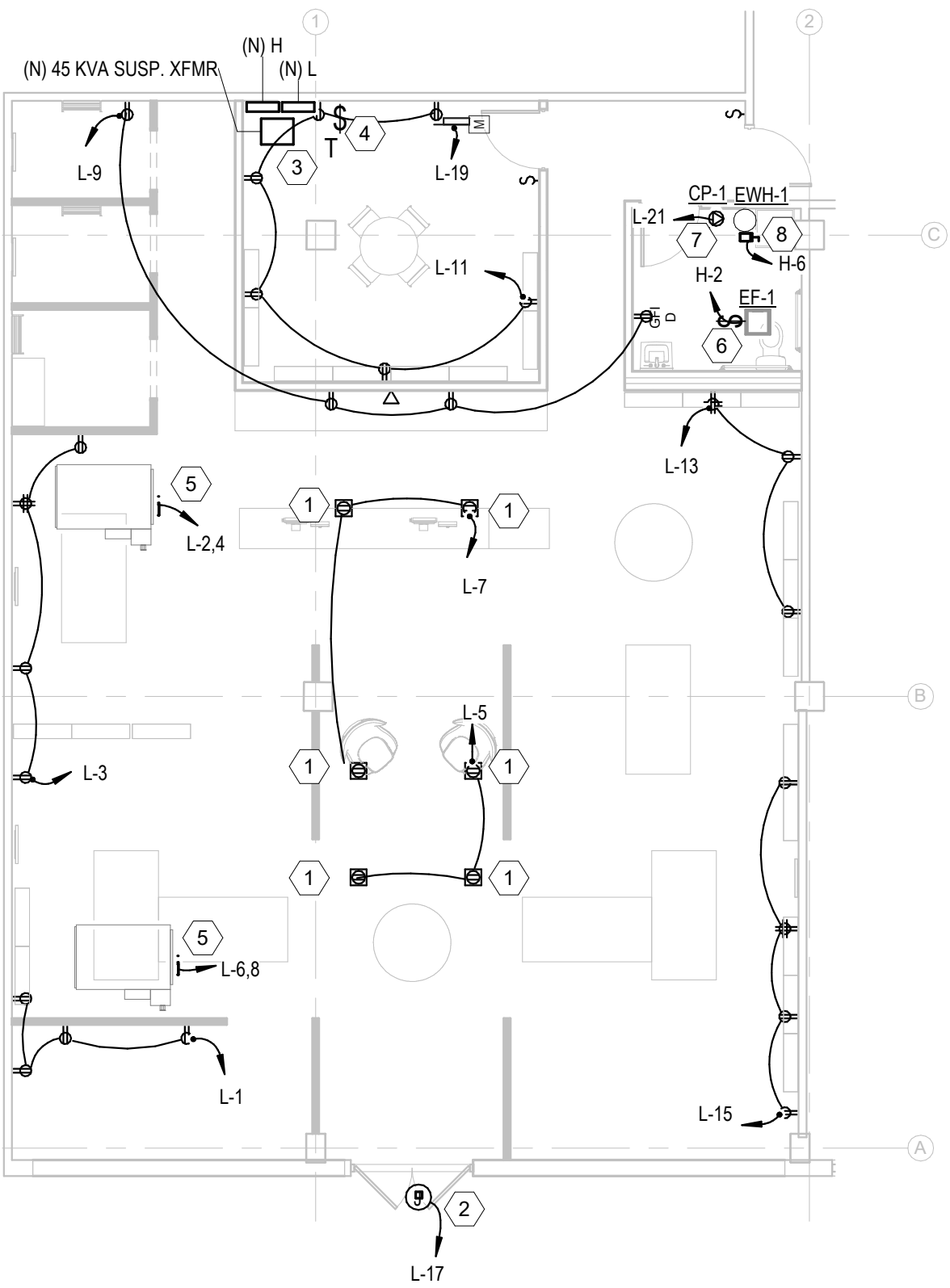
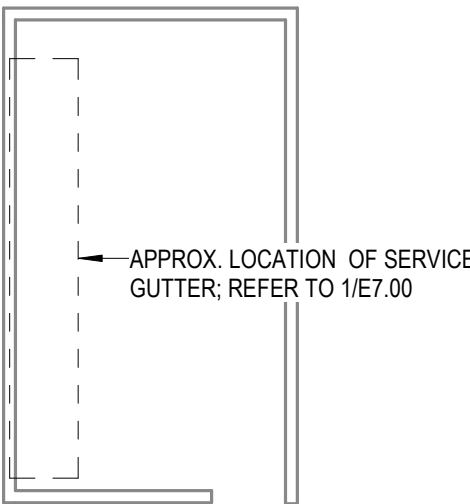
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PROJECT



1 POWER PLAN  
1/8" = 1'-0"

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SHEET  
POWER PLAN

INFINITY  
MEP+ CONSULTANTS  
TBPE Registration Number 18865  
10260 Westheimer Rd., Suite 400  
Houston, Texas 77042  
713.429.4949  
www.infinitymep.com



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | ALL ELECTRICAL EQUIPMENT EXISTING UNLESS OTHERWISE NOTED. CONTRACTOR SHALL PROVIDE CIRCUIT BREAKERS IN EXISTING PANELBOARDS PER PANEL SCHEDULES. CIRCUIT BREAKERS SHALL BE OF SAME MANUFACTURER AS EXISTING PANELS AND EQUAL A.I.C. RATING.                                                                                                                                                                                                                                                          |
| B | CONTRACTOR SHALL TRACE EXISTING CIRCUITS IN SCOPE AND REMOVE THOSE THAT ARE NOT CURRENTLY BEING USED OR BECOME AVAILABLE DUE TO DEMOLITION. CIRCUIT NUMBER SHOWN ARE DIAGRAMMATIC AND MAY NEED TO BE MODIFIED FOR FIELD CONDITIONS. CONTRACTOR SHALL COORDINATE ALL WORK WITH BUILDING ENGINEER AND MAY NEED TO SCHEDULE WORK AFTER HOURS. CONTRACTOR SHALL CHECK PANEL FOR ELECTRICAL LOAD. IF SUFFICIENT NUMBER OF CIRCUITS CAN NOT BE LOCATED AND REUSED, ENGINEER SHALL BE NOTIFIED IMMEDIATELY. |
| C | ALL NEW WIRES SHALL HAVE THINWYNTH INSULATION.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

NEW

EXISTING

WWW

SWITCHBOARD  
AMP BUS RATINGS

REMARKS

FULL SIZE NEUTRAL  
COPPER GROUND BUS  
1 NEMA ENCLOSURE

DEVICE TRIP CODES

T-M  
GFI  
SHUNT  
ELEC  
FUSE

THERMAL MAGNETIC  
GROUND FAULT INTERRUPT  
SHUNT TRIP  
ELECTRONIC TRIP  
FUSE

1600  
480  
277  
3  
4  
65000

AMP  
MLO  
VOLT P-P  
VOLT P-N  
PHASE  
WIRE  
INTERRUPTING RATING IN RMS  
SYMMETRICAL AMPS

TYPE

TRIP

AMPS

POLE

TYPE

TRIP

AMPS

POLE

TYPE

TRIP

AMPS

POLE

C B

T-M

400

3

C B

T-M

400

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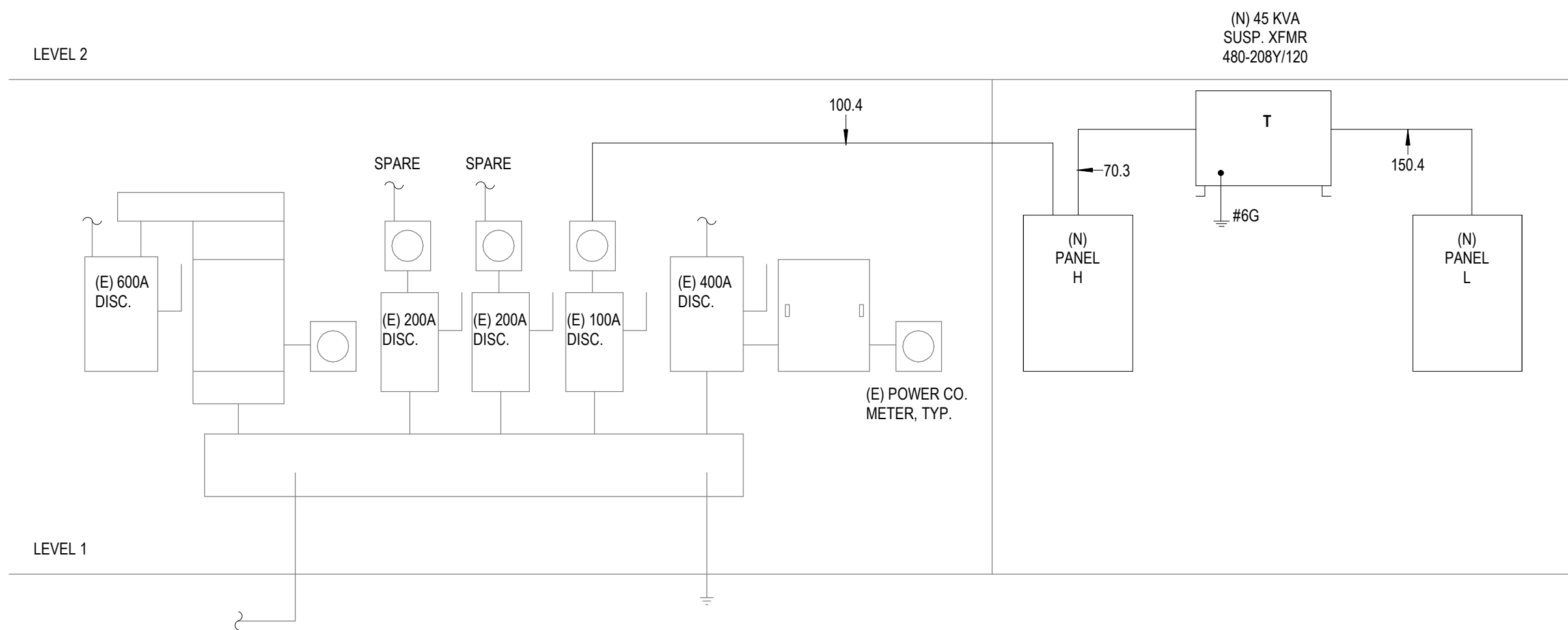
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LOAD ANALYSIS

|                              |              |              |                      |                      | 480<br>277<br>3<br>4 | VOLT P-P<br>VOLT P-N<br>PHASE<br>WIRE |                                                     |
|------------------------------|--------------|--------------|----------------------|----------------------|----------------------|---------------------------------------|-----------------------------------------------------|
| DESCRIPTION                  | LOAD         |              |                      |                      |                      |                                       | REMARKS ON NEC CALCULATION                          |
|                              | CON<br>(KVA) | NEC<br>(KVA) | PHASE<br>A<br>(AMPS) | PHASE<br>B<br>(AMPS) | PHASE<br>C<br>(AMPS) | NEUTRAL<br>(AMPS)                     |                                                     |
| ELECTRIC HEAT                |              |              |                      |                      |                      |                                       | LESS THAN A/C                                       |
| A/C REFRIGERATION            |              |              |                      |                      |                      |                                       | 100% OF CONNECTED                                   |
| MISCELLANEOUS MOTORS         | 0.5          | 0.5          | 0.6                  | 0.6                  | 0.6                  |                                       | 100% OF CONNECTED                                   |
| OUTDOOR LIGHTING             |              |              |                      |                      |                      |                                       | 125% OF CONNECTED                                   |
| INDOOR LIGHTING              | 2.0          | 2.5          | 3.0                  | 3.0                  | 3.0                  | 3.0                                   | 125% OF THE LARGER OF CONNECTED or NEC TABLE 220.12 |
| RECEPTACLES                  | 7.7          | 7.7          | 9.3                  | 9.3                  | 9.3                  | 9.3                                   | NEC TABLE 220.44                                    |
| EXISTING DEMAND              |              |              |                      |                      |                      |                                       | NOT APPLICABLE                                      |
| MISCELLANEOUS CONTINUOUS     |              |              |                      |                      |                      |                                       | 125% OF CONNECTED                                   |
| MISCELLANEOUS NON-CONTINUOUS | 435.0        | 435.0        | 523.3                | 523.3                | 523.3                | 523.3                                 | 100% OF CONNECTED                                   |
| KITCHEN EQUIPMENT            |              |              |                      |                      |                      |                                       | NO COMMERCIAL KITCHEN                               |
| 25% LARGEST MOTOR            | 0.1          | 0.1          | 0.2                  | 0.2                  | 0.2                  |                                       | 25% OF LARGEST MOTOR ADDED                          |
| *** TOTALS ***               | 445.4        | 445.9        | 536.3                | 536.3                | 536.3                | 535.5                                 |                                                     |
| *** CAPACITY ***             |              |              | 1600.0               | 1600.0               | 1600.0               | 1600.0                                |                                                     |
| *** SPARE CAPACITY ***       |              |              | 1063.7               | 1063.7               | 1063.7               | 1064.5                                |                                                     |

|                                                                                                                    |           |
|--------------------------------------------------------------------------------------------------------------------|-----------|
| REWIRED 277V LOAD                                                                                                  | 4.50 KVA  |
| REWIRED 120V LOAD                                                                                                  | 25.04 KVA |
| TOTAL REWIRED LOAD                                                                                                 | 29.54 KVA |
| LOAD @ 480Y/277V                                                                                                   | 35.55 A   |
| TOTAL LOAD LESS THAN 200A AND AREA OF WORK IS LESS THAN 5000SQFT. THEREFORE, LOAD IS ADEQUATE FOR ONE-STOP PERMIT. |           |



# 1 ELECTRICAL ONE-LINE DIAGRAM